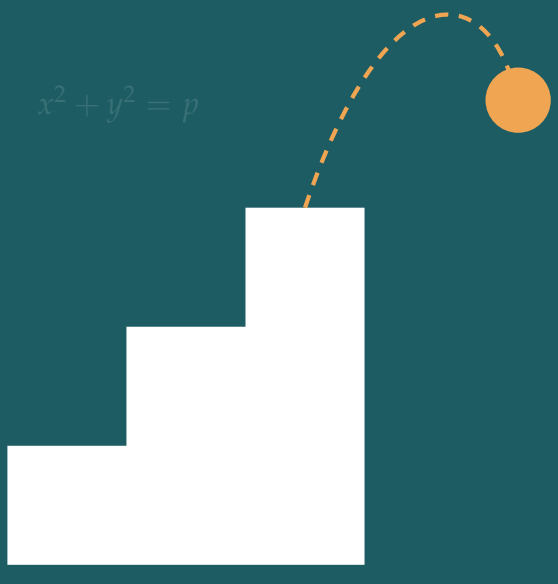
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

1989 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

1989 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

1989 Worked Solutions

Worked solutions to all 48 questions of the 1989 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 1989, Question 1

Pure • Areas of Composite Regions

> **STEP I 1989, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/89-s1-q1

The key idea

The region is built from semicircles. You can get it by starting with one big half-disc, then taking away and adding smaller half-discs. So all we need is each radius, and in particular the radius of the lower semicircle CDE, which comes from the lengths along the line AG. Changing to parabolas makes no difference to the method, because a parabolic arch has a simple area formula that does the job of $\frac{1}{2}\pi a^2$.

Solution

The semicircles

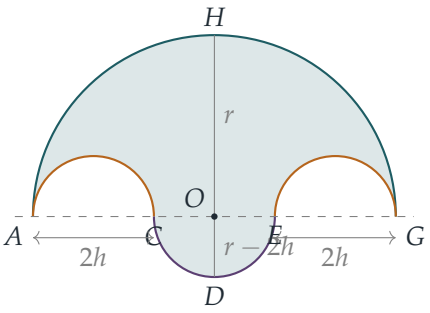
Measure along the line from A. Since ABC has radius h, $AC = 2h$. Similarly $EG = 2h$, and $AG = 2r$ because AHG has radius r. So

$$CE = 2r - 4h,$$

which is positive because $2h < r$. So the semicircle CDE has radius $r - 2h$. Its centre is the midpoint of CE, which is also the midpoint of AG by symmetry, so it shares its centre O with the big semicircle.

From the diagram, ABC and EFG lie on the same side of AG as H, inside the big semicircle, and CDE lies on the other side. The enclosed region is therefore

$$(\text{half-disc } AHG) \;-\; (\text{half-discs } ABC \text{ and } EFG) \;+\; (\text{half-disc } CDE).$$



A half-disc of radius a has area $\frac{1}{2}\pi a^2$, so the area enclosed is

$$\begin{aligned}\tfrac{1}{2}\pi r^2 - 2\cdot\tfrac{1}{2}\pi h^2 + \tfrac{1}{2}\pi(r-2h)^2 &= \tfrac{1}{2}\pi(r^2 - 2h^2 + r^2 - 4rh + 4h^2)\\ &= \tfrac{1}{2}\pi(2r^2 - 4rh + 2h^2)\\ &= \pi(r-h)^2.\end{aligned}$$

Ask yourself How long is HD?

The point H is the top of the big semicircle, so it lies on the perpendicular to AG through O, at distance r from O. The point D is the lowest point of CDE, so it lies on the same perpendicular, at distance $r-2h$ from O on the other side of the line. So

$$HD = r + (r-2h) = 2(r-h),$$

and a circle with diameter HD has radius $r-h$ and area $\pi(r-h)^2$. This is the area we found, as required.

The parabolas

Each semicircle of radius a is replaced by a parabolic arch on the same chord of length $2a$, with its vertex at height a above the midpoint of the chord. Put the origin at the midpoint of the chord. The parabola is symmetric about the axis through its vertex, so it has the form $y = a - kx^2$. It passes through $(\pm a, 0)$, so $k = 1/a$. The area between the arch and its chord is

$$\int_{-a}^{a}\left(a - \frac{x^2}{a}\right)\mathrm{d}x = 2a^2 - \frac{2a^3}{3a} = \frac{4a^2}{3}.$$

(This is Archimedes' result: two-thirds of the enclosing rectangle, $\frac{2}{3}\cdot 2a\cdot a$.)

Every arch still lies on the same side of AG as its semicircle did. We should also check that each small arch still lies under the big one. Both arches start from A with gradient 2, so this isn't obvious near A. At distance x from A, where $0 \le x \le 2h$, the big arch has height $x(2r-x)/r$ and the arch ABC has height $x(2h-x)/h$. Their difference is

$$\frac{x(2r-x)}{r} - \frac{x(2h-x)}{h} = x^2\left(\frac{1}{h} - \frac{1}{r}\right) \ge 0,$$

so ABC really does lie inside AHG, and EFG does too by symmetry. So the same add-and-subtract decomposition works, with $\frac{4}{3}a^2$ in place of $\frac{1}{2}\pi a^2$:

$$\begin{aligned}\tfrac{4}{3}r^2 - 2\cdot\tfrac{4}{3}h^2 + \tfrac{4}{3}(r-2h)^2 &= \tfrac{4}{3}(2r^2 - 4rh + 2h^2)\\ &= \tfrac{8}{3}(r-h)^2.\end{aligned}$$

So with parabolas the enclosed area is $\frac{8}{3}(r-h)^2$.

Checks

- Both answers are a constant times $(r-h)^2$. That has to happen, because the two calculations have exactly the same structure: only the factor for one arch changes, from $\frac{\pi}{2}$ to $\frac{4}{3}$. The ratio $\frac{8/3}{\pi} \approx 0.849$ is less than 1, as it should be, because a parabolic arch with the same chord and height lies inside the semicircle.
- Limiting case $h \to 0$: the small arches disappear and CDE becomes a full lower semicircle of radius r, giving a whole disc of area πr^2. The formula $\pi(r-h)^2$ agrees.
- Numerically, with $r = 3$ and $h = 1$ (the proportions of the paper's diagram), the areas are $4\pi \approx 12.566$ and $\frac{32}{3} \approx 10.667$. A shoelace computation on the traced boundary, for eight random pairs (r, h), matched both formulas to within the discretisation error.

Common slips

- Giving CDE radius $r-h$ or $r-4h$. Draw the lengths along AG: $CE = 2r - 4h$, so its radius is $r-2h$.
- Adding the small upper semicircles instead of subtracting them. They cut into the big half-disc, so their area comes off.

Beyond STEP

The figure in the first part is the *arbelos* ("shoemaker's knife"), a shape that Archimedes studied. Its classical form is a big semicircle with two smaller ones inside it, and its area also equals that of a circle on a well-chosen chord. Archimedes also found the $\frac{2}{3}$ rule for parabolic segments, by a limiting argument that comes close to integration. At university, the step from his "method of exhaustion" to the Riemann integral is a central part of a first analysis course. A good place to see it is Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I 1989, Question 2

Pure • Riemann Sums and Limits

STEP I 1989, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q2

The key idea

The strange expression is nothing more than a sum of rectangle areas in disguise. If we write down the total area of the n rectangles directly, we get a sum involving $k \ln k$; the factorial logarithms are just that same sum regrouped. Once the two are matched up, the limit is the integral, which the first part lets us evaluate.

Solution

The indefinite integral

Integrating by parts with $u = \ln x$ and $\frac{\mathrm{d}v}{\mathrm{d}x} = x$, so that $\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x}$ and $v = \frac{1}{2}x^2$,

$$\begin{aligned}\int x \ln x \,\mathrm{d}x &= \tfrac{1}{2}x^2 \ln x - \int \tfrac{1}{2}x^2 \cdot \frac{1}{x}\,\mathrm{d}x \\ &= \tfrac{1}{2}x^2 \ln x - \tfrac{1}{4}x^2 + C.\end{aligned}$$

The value of the integral

Since $\ln(1/x) = -\ln x$, the integrand is $-x \ln x$. The integrand isn't defined at $x = 0$, so we integrate over $[\varepsilon, 1]$ and let $\varepsilon \to 0^+$:

$$\int_\varepsilon^1 x \ln(1/x)\,\mathrm{d}x = \left[\tfrac{1}{4}x^2 - \tfrac{1}{2}x^2 \ln x\right]_\varepsilon^1 = \tfrac{1}{4} - \tfrac{1}{4}\varepsilon^2 + \tfrac{1}{2}\varepsilon\,(\varepsilon \ln \varepsilon).$$

As $\varepsilon \to 0$, $\varepsilon^2 \to 0$ and $\varepsilon \ln \varepsilon \to 0$ (the given fact), so

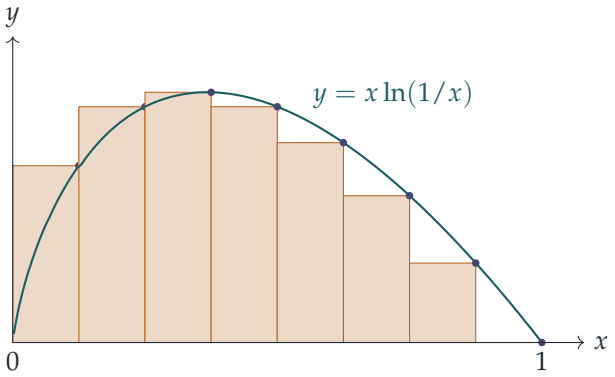
$$\int_0^1 x \ln(1/x)\,\mathrm{d}x = \frac{1}{4}.$$

The same fact shows that $f(x) = x \ln(1/x)$ tends to 0 as $x \to 0^+$, so setting $f(0) = 0$ makes f continuous on the whole of $[0, 1]$. The curve rises from 0 to a maximum of $\frac{1}{\mathrm{e}}$ at $x = \frac{1}{\mathrm{e}}$ (where $f'(x) = -\ln x - 1 = 0$) and falls back to 0 at $x = 1$.

The rectangles

Split $[0,1]$ into n strips of width $\frac{1}{n}$. The kth strip is $\left[\frac{k-1}{n}, \frac{k}{n}\right]$, and its rectangle has its top right-hand corner on the curve, so its height is

$$f\left(\frac{k}{n}\right) = \frac{k}{n}\ln\frac{n}{k} = \frac{k}{n}(\ln n - \ln k).$$



The total area of the rectangles is therefore

$$\begin{aligned} S_n = \sum_{k=1}^{n} \frac{1}{n}\cdot\frac{k}{n}(\ln n - \ln k) &= \frac{\ln n}{n^2}\sum_{k=1}^{n} k - \frac{1}{n^2}\sum_{k=1}^{n} k\ln k \\ &= \frac{\ln n}{n^2}\cdot\frac{n(n+1)}{2} - \frac{1}{n^2}\sum_{k=1}^{n} k\ln k \\ &= \frac{1}{2}\left(1+\frac{1}{n}\right)\ln n - \frac{1}{n^2}\sum_{k=1}^{n} k\ln k. \end{aligned}$$

The first term already matches the expression we want.

Ask yourself Where do the factorials come from?

For $0 \le m \le n-1$,

$$\ln\frac{n!}{m!} = \ln\big((m+1)(m+2)\cdots n\big) = \sum_{k=m+1}^{n} \ln k.$$

Now add these up for $m = 0, 1, \ldots, n-1$ and count how often each $\ln k$ appears. The term $\ln k$ is included exactly when $m+1 \le k$, that is for $m = 0, 1, \ldots, k-1$: that's k times. Hence

$$\ln\frac{n!}{0!} + \ln\frac{n!}{1!} + \cdots + \ln\frac{n!}{(n-1)!} = \sum_{k=1}^{n} k\ln k.$$

(For example, with $n = 3$: $\ln 6 + \ln 6 + \ln 3 = \ln 1 + 2\ln 2 + 3\ln 3$, both equal to $\ln 108$.) So the expression in the question is exactly S_n, the area of the n rectangles.

The limit

As $n \to \infty$ the rectangles fit the curve ever more closely, and their total area tends to the area under the curve, since f is continuous on $[0,1]$. Therefore

$$\frac{1}{2}\left(1+\frac{1}{n}\right)\ln n - \frac{1}{n^2}\left[\ln\frac{n!}{0!} + \ln\frac{n!}{1!} + \cdots + \ln\frac{n!}{(n-1)!}\right] = S_n \to \int_0^1 x\ln(1/x)\,\mathrm{d}x = \frac{1}{4},$$

as required.

If you want something firmer than "the rectangles fit ever more closely", here is a bound. On each strip, the rectangle and the area under the curve differ by at most $\frac{1}{n}$ times the amount by which f varies across that strip. Add these variations over all the strips: f climbs by $\frac{1}{\mathrm{e}}$ and comes back down by $\frac{1}{\mathrm{e}}$, so the total is at most $\frac{2}{\mathrm{e}}$. Hence $\left|S_n - \frac{1}{4}\right| \le \frac{2}{\mathrm{e}n} \to 0$.

Checks

- Differentiate the answer to the first part: $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{2}x^2\ln x - \frac{1}{4}x^2\right) = x\ln x + \frac{1}{2}x - \frac{1}{2}x = x\ln x$.
- With $n = 2$ the expression is $\frac{3}{4}\ln 2 - \frac{1}{4}(\ln 2 + \ln 2) = \frac{1}{4}\ln 2 \approx 0.173$, which is the area of the two rectangles, $\frac{1}{2}\cdot\frac{1}{2}\ln 2 + \frac{1}{2}\cdot 0$.
- Numerically, the expression is about 0.24559 for $n = 10$, 0.249937 for $n = 100$ and 0.2499992 for $n = 1000$, closing in on $\frac{1}{4}$.

Common slips

- Losing the sign: the integrand is $x\ln(1/x) = -x\ln x$, so the antiderivative from the first part must be negated, and the area is $+\frac{1}{4}$, not $-\frac{1}{4}$.
- Writing $\sum k = \frac{1}{2}n^2$ instead of $\frac{1}{2}n(n+1)$: the $\frac{1}{n}$ inside the bracket would then go missing.
- Forgetting that the integral is improper at 0: the lower limit needs the fact $x\ln x \to 0$, which the question gives you.

Beyond STEP

The rectangle sum is a Riemann sum, and the fact that it converges to the integral for every continuous function on a closed interval is one of the first big theorems of a university analysis course. The bound we used, with the total rise and fall of f, is the idea of a function of *bounded variation*. Turned round, this question also gives information about $\sum k\ln k$, a quantity closely related to the hyperfactorial $1^1\,2^2\cdots n^n$, whose growth can be found more precisely in the same spirit as Stirling's formula for $n!$. Abbott's *Understanding Analysis* develops the Riemann integral carefully.

WORKED SOLUTION

STEP I 1989, Question 3

Pure • Vectors and Scalar Product

STEP I 1989, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q3

The key idea

Put everything in terms of the two vectors $\mathbf{a}$ and $\mathbf{b}$, with O as origin. Then "perpendicular" becomes "scalar product zero", and because $\mathbf{a}\cdot\mathbf{a} = \mathbf{b}\cdot\mathbf{b} = 1$, that scalar product depends only on x, OY and the one unknown $\mathbf{a}\cdot\mathbf{b}$. Using $OX + OY = 1$ leaves a single equation linking x and $\mathbf{a}\cdot\mathbf{b}$. The bounds then come from how far x can range.

Solution

The condition

Take O as origin and write $k = \mathbf{a}\cdot\mathbf{b}$. Since C is one third of the way from A to B, and D two thirds of the way,

$$\overrightarrow{OC} = \mathbf{a} + \tfrac{1}{3}(\mathbf{b}-\mathbf{a}) = \tfrac{1}{3}(2\mathbf{a}+\mathbf{b}), \qquad \overrightarrow{OD} = \tfrac{1}{3}(\mathbf{a}+2\mathbf{b}).$$

Since X is on OA with $OX = x$ and $|\mathbf{a}| = 1$, we have $\overrightarrow{OX} = x\mathbf{a}$. Similarly $\overrightarrow{OY} = y\mathbf{b}$ with $y = OY = 1 - x$. So

$$3\overrightarrow{CY} = (3y-1)\mathbf{b} - 2\mathbf{a}, \qquad 3\overrightarrow{DX} = (3x-1)\mathbf{a} - 2\mathbf{b}.$$

The lines are perpendicular exactly when the scalar product is zero. Expanding, and using $\mathbf{a}\cdot\mathbf{a} = \mathbf{b}\cdot\mathbf{b} = 1$,

$$\begin{aligned} 9\,\overrightarrow{CY}\cdot\overrightarrow{DX} &= (3x-1)(3y-1)k - 2(3y-1) - 2(3x-1) + 4k \\ &= k\big[(3x-1)(3y-1)+4\big] - 2(3x+3y-2). \end{aligned}$$

Now use $x + y = 1$. Then $3x + 3y - 2 = 1$, and

$$(3x-1)(3y-1) = 9xy - 3(x+y) + 1 = 9xy - 2.$$

So the perpendicularity condition becomes $k(9xy+2) = 2$. With $y = 1 - x$, this is

$$\boxed{\mathbf{a}\cdot\mathbf{b} = \frac{2}{9x(1-x)+2}}$$

(the denominator is at least 2, so dividing by it is safe).

The bounds

X lies on the segment OA, which has length 1, so $0 \le x \le 1$. (Then $y = 1 - x$ is also in $[0, 1]$, so Y is automatically on OB.) On this interval

$$x(1-x) = \tfrac{1}{4} - \left(x - \tfrac{1}{2}\right)^2,$$

so $0 \le x(1-x) \le \frac{1}{4}$, with the maximum at $x = \frac{1}{2}$ and the minimum at $x = 0$ or 1. Hence

$$2 \le 9x(1-x) + 2 \le \tfrac{9}{4} + 2 = \tfrac{17}{4}.$$

Taking reciprocals reverses the inequalities (everything is positive), and multiplying by 2 gives

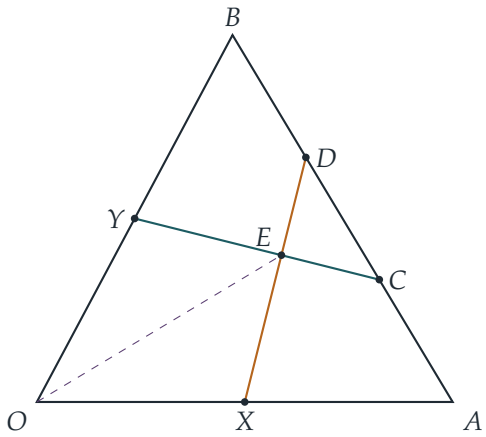
$$\frac{8}{17} \le \mathbf{a} \cdot \mathbf{b} \le 1,$$

as required. (The value 1 would need $\mathbf{a} = \mathbf{b}$, which is not a genuine triangle, so in practice $\mathbf{a} \cdot \mathbf{b} < 1$; the stated inequality still holds.)

The largest angle and OE

Ask yourself Why does the largest angle correspond to the *smallest* value of $\mathbf{a} \cdot \mathbf{b}$?

Since $|\mathbf{a}| = |\mathbf{b}| = 1$, $\mathbf{a} \cdot \mathbf{b} = \cos A\hat{O}B$, and cosine is decreasing on $[0, \pi]$. So the angle is largest when $\mathbf{a} \cdot \mathbf{b} = \frac{8}{17}$, that is, when $x = \frac{1}{2}$ (and so $y = \frac{1}{2}$). Then X and Y are the midpoints of OA and OB, and the angle is $\arccos \frac{8}{17} \approx 61.9°$.



Now the whole figure is symmetric under swapping $\mathbf{a}$ and $\mathbf{b}$: this swaps $A \leftrightarrow B$, $C \leftrightarrow D$ and $X \leftrightarrow Y$, so it swaps the lines CY and DX. Their intersection E is therefore unchanged by the swap, so it lies on the line of symmetry: $\overrightarrow{OE} = t(\mathbf{a} + \mathbf{b})$ for some t.

A general point of DX is

$$\overrightarrow{OX} + s\overrightarrow{XD} = \tfrac{1}{2}\mathbf{a} + s\left(\tfrac{1}{3}\mathbf{a} + \tfrac{2}{3}\mathbf{b} - \tfrac{1}{2}\mathbf{a}\right) = \left(\tfrac{1}{2} - \tfrac{s}{6}\right)\mathbf{a} + \tfrac{2s}{3}\mathbf{b}.$$

Since $\mathbf{a}$ and $\mathbf{b}$ are not parallel, we can compare coefficients: this is a multiple of $\mathbf{a}+\mathbf{b}$ when $\frac{1}{2} - \frac{s}{6} = \frac{2s}{3}$, giving $s = \frac{3}{5}$ and $t = \frac{2}{5}$. So $\overrightarrow{OE} = \frac{2}{5}(\mathbf{a}+\mathbf{b})$. Finally

$$|\mathbf{a}+\mathbf{b}|^2 = \mathbf{a}\cdot\mathbf{a} + 2\,\mathbf{a}\cdot\mathbf{b} + \mathbf{b}\cdot\mathbf{b} = 2 + \tfrac{16}{17} = \tfrac{50}{17},$$

so

$$OE = \frac{2}{5}\sqrt{\frac{50}{17}} = \frac{2}{5}\cdot\frac{5\sqrt{2}}{\sqrt{17}} = \frac{2\sqrt{2}}{\sqrt{17}} = \boxed{\frac{2\sqrt{34}}{17}} \approx 0.686.$$

Checks

- In coordinates, take $\mathbf{a} = (1,0)$ and $\mathbf{b} = (\frac{8}{17}, \frac{15}{17})$. Then $X = (\frac{1}{2}, 0)$, $D = (\frac{11}{17}, \frac{10}{17})$, so $\overrightarrow{XD} = (\frac{5}{34}, \frac{10}{17})$. Also $Y = (\frac{4}{17}, \frac{15}{34})$ and $C = (\frac{14}{17}, \frac{5}{17})$, so $\overrightarrow{CY} = (-\frac{10}{17}, \frac{5}{34})$. Their scalar product is $-\frac{50}{578} + \frac{50}{578} = 0$, as it should be.
- $E = \frac{2}{5}(\frac{25}{17}, \frac{15}{17}) = (\frac{10}{17}, \frac{6}{17})$, and $|OE| = \frac{\sqrt{136}}{17} = \frac{2\sqrt{34}}{17} \approx 0.6860$. This matches a direct numerical intersection of the two lines.
- The value $s = \frac{3}{5}$ lies in $[0,1]$, so E really is between X and D, inside the triangle.

Common slips

- Forgetting that $\mathbf{a}\cdot\mathbf{a} = 1$ (and so leaving $|\mathbf{a}|^2$ terms in the condition), or getting the trisection points the wrong way round.
- Thinking that the largest angle means the largest $\mathbf{a}\cdot\mathbf{b}$: the cosine decreases as the angle grows.
- Writing $\overrightarrow{OX} = x$ rather than $x\mathbf{a}$. The point X is a vector; x is only its length.

Beyond STEP

The method here, writing every point as a combination of two basis vectors and turning geometry into scalar products, is the start of working with inner products and Gram matrices. A configuration of vectors is fixed up to rotation by its table of scalar products, which here is just 1, 1 and $\mathbf{a}\cdot\mathbf{b}$. That is why the whole question reduces to one number. Strang's *Introduction to Linear Algebra* develops these ideas.

WORKED SOLUTION

STEP I 1989, Question 4

Pure • Ramsey Theory

STEP I 1989, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q4

This question has a full guided treatment in *Beyond STEP*, Section 9.2, *Ramsey's Theorem in Miniature*: https://beyondstep.keithmcnulty.org/ch09/ramsey.html

The key idea

Forget the geometry: the conditions on the points only guarantee that any three of them form a genuine triangle, so this is purely a question about colouring the 15 edges joining six points. Following the hint, look at one point. It has five edges and only two colours, so by the pigeonhole principle at least three of those edges share a colour. Then the edges among the three far ends either complete a one-coloured triangle with the first point or form one on their own.

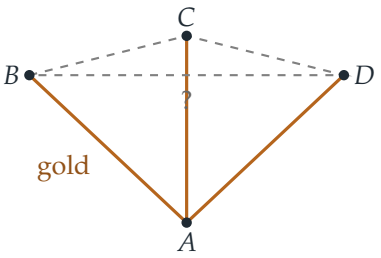
Solution

Six points

Since no three of the points are collinear, any three of them are the vertices of a genuine triangle, and its three sides are among the segments we have drawn. (The condition that no four are coplanar plays no part in the argument.)

Consider the five segments AB, AC, AD, AE, AF that come out of A. Each is gold or silver. If at most two were gold and at most two were silver, there would be at most four segments in total, which is false. So at least three of them have the same colour. By relabelling the points $B, \ldots, F$, and swapping the names of the two colours if necessary, we may assume that

$$AB,\ AC,\ AD \text{ are all gold.}$$



Now look at the three segments BC, CD and DB joining the far ends.

Ask yourself Why must one of the two cases below give a one-coloured triangle, whatever colours BC, CD and DB have?

There are only two possibilities.

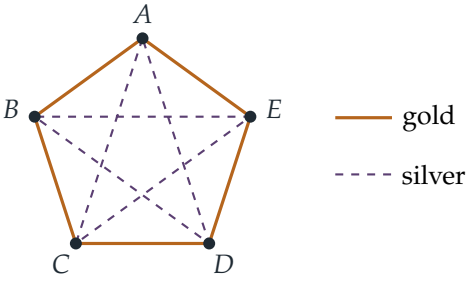
- *At least one of BC, CD, DB is gold.* Say it is BC (the other cases are the same with the letters changed). Then AB, AC and BC are all gold, so triangle ABC has edges of one colour.
- *None of BC, CD, DB is gold.* Then all three are silver, and triangle BCD has edges of one colour.

Either way there is a triangle whose edges are all the same colour, which is what we had to prove.

Five points

With five points, each point has only four edges, which can be split two and two, so the argument breaks down. To show the result really is false, we need one colouring with no one-coloured triangle.

Place the points A, B, C, D, E round a pentagon. Colour the five sides AB, BC, CD, DE, EA gold, and the five diagonals AC, BD, CE, DA, EB (the pentagram) silver.



Any triangle uses three of the five points. Among any three vertices of a pentagon, at least two are not next to each other (three mutually adjacent vertices would need a triangle inside the 5-cycle), so every triangle has a silver edge. Also, among any three

vertices at least two *are* next to each other: if no two of the three were adjacent, each gap between consecutive chosen vertices round the pentagon would contain at least one unchosen vertex, needing $3 + 3 = 6$ points. So every triangle also has a gold edge. Hence no triangle is all one colour.

Ask yourself A regular pentagon is flat, so four of its vertices are coplanar. Doesn't that break the rule about general positions?

It would, if the points were literally on a plane, but the colouring depends only on which pairs are joined, not on where the points are. Lift the vertices slightly off the plane by different amounts (for example, raise the kth vertex to height $0.3k^2$ for $k = 0, 1, 2, 3, 4$). Then no three are collinear and no four are coplanar, and seen from above the picture is still the pentagon and pentagram drawn here, with the same colouring and no one-coloured triangle.

Checks

There are only $2^{15} = 32768$ ways to colour the edges for six points, so a computer can try them all: every one has a one-coloured triangle, and in fact always at least two. For five points, checking all $\binom{5}{3} = 10$ triangles of the pentagon colouring confirms that each has both colours. A determinant check confirms that the lifted pentagon above has no four points coplanar.

Common slips

- Saying "by symmetry" without saying what is being assumed: be explicit that you choose three edges from A of the majority colour and call that colour gold.
- Only proving that A has three edges of one colour and stopping there; the second step, looking at BC, CD, DB, is the heart of the proof.
- For five points, giving a sketch without checking every triangle; a colouring such as "a star from A gold, everything else silver" fails.

Beyond STEP

This is the first case of Ramsey theory: the *Ramsey number* $R(3,3)$ is the smallest n such that every two-colouring of the edges of the complete graph K_n contains a one-coloured triangle, and this question shows that $R(3,3) = 6$. Ramsey's theorem says such a number exists for any sizes of one-coloured complete subgraph, but they are fiendishly hard to compute: even $R(5,5)$ is unknown. A lovely introduction to graph theory, including Ramsey's theorem, is Wilson, *Introduction to Graph Theory*.

WORKED SOLUTION

STEP I 1989, Question 5

Pure • Binomial Coefficient Identities

STEP I 1989, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q5

The key idea

The binomial expansion is an identity in x, so it stays true whatever number we put in. Choosing $x = 1$ adds up all the coefficients, and $x = -1$ adds them with alternating signs; combining the two separates the even-numbered from the odd-numbered ones. For the weighted sum in (ii), the factor r in front of $\binom{n}{r}$ is exactly what differentiating x^r produces, so we differentiate the expansion and then substitute.

Solution

For a positive integer n,

$$(1+x)^n = \sum_{r=0}^{n} \binom{n}{r} x^r = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n,$$

where $\binom{n}{r} = \dfrac{n!}{r!\,(n-r)!}$. This holds for every real x.

Part (i)

Write $E = \binom{n}{0} + \binom{n}{2} + \cdots + \binom{n}{n}$ for the sum of the even-numbered coefficients and $O = \binom{n}{1} + \binom{n}{3} + \cdots + \binom{n}{n-1}$ for the odd-numbered ones. Because n is even, the last term $\binom{n}{n}$ belongs to E and $\binom{n}{n-1}$ to O, so between them E and O contain every coefficient exactly once.

Putting $x = 1$ in the expansion gives

$$E + O = 2^n.$$

Putting $x = -1$, each even power $(-1)^r$ is $+1$ and each odd power is -1, while the left-hand side is $0^n = 0$ since $n \geq 1$. So

$$E - O = 0.$$

Hence $E = O$, and then $2E = 2^n$, so

$$E = O = 2^{n-1},$$

which is the required result.

Part (ii)

Differentiate both sides of the expansion with respect to x. The constant term vanishes and x^r becomes rx^{r-1}:

$$n(1+x)^{n-1} = \binom{n}{1} + 2\binom{n}{2}x + 3\binom{n}{3}x^2 + \cdots + n\binom{n}{n}x^{n-1}.$$

Now put $x = 1$:

$$n2^{n-1} = \binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n},$$

as required. This works for every positive integer n, odd or even.

Ask yourself Is there a way to see part (ii) without calculus?

Yes. Since $r\binom{n}{r} = r \cdot \dfrac{n!}{r!\,(n-r)!} = n \cdot \dfrac{(n-1)!}{(r-1)!\,(n-r)!} = n\binom{n-1}{r-1}$ for $1 \le r \le n$, the sum is $n\sum_{r=1}^{n}\binom{n-1}{r-1} = n \cdot 2^{n-1}$, by putting $x = 1$ in the expansion of $(1+x)^{n-1}$. In words: choosing a committee of any size from n people and then a chair from the committee is the same as choosing the chair first (n ways) and then any subset of the other $n-1$ people (2^{n-1} ways).

The final sum

Split the sum into two:

$$\sum_{r=0}^{n}\left(r + (-1)^r\right)\binom{n}{r} = \sum_{r=0}^{n} r\binom{n}{r} + \sum_{r=0}^{n}(-1)^r\binom{n}{r}.$$

The $r = 0$ term of the first sum is zero, so the first sum is exactly the sum in (ii), namely $n2^{n-1}$. The second sum is the expansion of $(1+x)^n$ at $x = -1$, namely $(1-1)^n = 0$ for every positive integer n. (This is the step $E - O = 0$ from (i); it needs only $n \ge 1$, not n even.) Therefore

$$\boxed{\sum_{r=0}^{n}\left(r + (-1)^r\right)\binom{n}{r} = n2^{n-1}.}$$

Checks

- For $n = 2$: the coefficients are $1, 2, 1$, so $E = 1 + 1 = 2 = O$ and $2^1 = 2$. The sum in (ii) is $2 + 2 \cdot 1 = 4 = 2 \cdot 2^1$.

- For $n = 3$ the final sum is $(0+1)\cdot 1 + (1-1)\cdot 3 + (2+1)\cdot 3 + (3-1)\cdot 1 = 1+0+9+2 = 12 = 3\cdot 2^2$.
- We checked all three identities exactly for every n from 1 to 29; for instance, $n = 6$ gives $192 = 6\cdot 2^5$.

Common slips

- Forgetting that $\sum(-1)^r\binom{n}{r} = 0$ holds for odd n too, and so thinking the final answer depends on the parity of n. Only at $n = 0$ does it fail, and n is positive here.
- Substituting $x = 1$ before differentiating in (ii). Substitution must come last: once x is a number, there is nothing left to differentiate.

Beyond STEP

Treating a polynomial as a store of coefficients, and then substituting, differentiating or multiplying to extract information, is the idea of a *generating function*, one of the most powerful tools in combinatorics and probability. Substituting complex roots of unity instead of ± 1 picks out every third or every fourth coefficient in the same way, and differentiating a probability generating function gives the mean of a distribution, just as it gave $\sum r\binom{n}{r}$ here. Grimmett and Welsh, *Probability: An Introduction*, shows generating functions at work.

WORKED SOLUTION

STEP I 1989, Question 6

Pure • Normals and Differential Equations

STEP I 1989, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q6

The key idea

The normal at P has gradient $-1/\mathrm{f}'(x)$, so finding Q is just a matter of putting $X = 0$ into its equation. Then PQ^2 is a horizontal distance squared plus a vertical one, and the given length turns into an equation for $\mathrm{f}'(x)$ alone. The only subtlety is the square root at the end: the sign we choose is decided by the minimum point.

Solution

The y-coordinate of Q

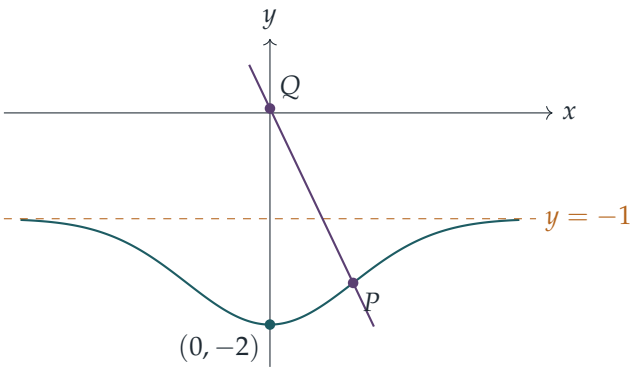
At $P = (x, \mathrm{f}(x))$ the tangent has gradient $\mathrm{f}'(x)$, so (provided $\mathrm{f}'(x) \neq 0$) the normal has gradient $-1/\mathrm{f}'(x)$. Using (X, Y) for the running coordinates, the normal is

$$Y - \mathrm{f}(x) = -\frac{1}{\mathrm{f}'(x)}\,(X - x).$$

Putting $X = 0$, the normal meets the y-axis at

$$Y = \mathrm{f}(x) + \frac{x}{\mathrm{f}'(x)}, \qquad \text{so} \qquad Q = \left(0,\ \mathrm{f}(x) + \frac{x}{\mathrm{f}'(x)}\right).$$

(If $\mathrm{f}'(x) = 0$ the normal is vertical and, unless $x = 0$, it never meets the y-axis.)



The differential equation

P and Q differ by x horizontally and by $x/\mathrm{f}'(x)$ vertically, so

$$PQ^2 = x^2 + \frac{x^2}{\left(\mathrm{f}'(x)\right)^2}.$$

Setting this equal to $\mathrm{e}^{x^2} + x^2$, the x^2 terms cancel and we get

$$\frac{x^2}{\left(\mathrm{f}'(x)\right)^2} = \mathrm{e}^{x^2}, \qquad \text{that is} \qquad \boxed{\left(\mathrm{f}'(x)\right)^2 = x^2\,\mathrm{e}^{-x^2}.}$$

Equivalently $\mathrm{f}'(x) = \pm x\,\mathrm{e}^{-x^2/2}$.

The curve with a minimum at $(0, -2)$

Ask yourself Which sign do we take, and could the sign change part-way along the curve?

Since f' is continuous (the curve has a minimum point, so we are assuming it is differentiable) and $\mathrm{f}'(x)^2 = x^2\mathrm{e}^{-x^2}$ vanishes only at $x = 0$, f' cannot change sign on $x > 0$ or on $x < 0$: it would have to pass through zero. For $(0, -2)$ to be a minimum, f must be decreasing just to the left of 0 and increasing just to the right, so $\mathrm{f}'(x) < 0$ for $x < 0$ and $\mathrm{f}'(x) > 0$ for $x > 0$. That is exactly the sign of x, so on both sides

$$\mathrm{f}'(x) = x\,\mathrm{e}^{-x^2/2}.$$

(The minus sign throughout would give a maximum at $x = 0$; mixing the signs would give f' of constant sign, i.e. a point of inflection.)

Integrating, since $\frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^{-x^2/2}\right) = -x\,\mathrm{e}^{-x^2/2}$,

$$\mathrm{f}(x) = -\mathrm{e}^{-x^2/2} + c.$$

The condition $\mathrm{f}(0) = -2$ gives $-1 + c = -2$, so $c = -1$ and the curve is

$$\boxed{y = -1 - \mathrm{e}^{-x^2/2}.}$$

It has its minimum value -2 at $x = 0$ and rises towards the asymptote $y = -1$ on both sides, as in the sketch.

Checks

- At $x = 1$: $\mathrm{f}(1) = -1 - \mathrm{e}^{-1/2} \approx -1.6065$ and $\mathrm{f}'(1) = \mathrm{e}^{-1/2} \approx 0.6065$, so $Q \approx (0,\, 0.0422)$ and $PQ \approx \sqrt{1 + 1.6487^2} \approx 1.9283$, which equals $\sqrt{\mathrm{e} + 1}$, as required.
- The computer recomputed Q by intersecting the normal line with the y-axis at 2000 random points and found PQ agreeing with $\sqrt{\mathrm{e}^{x^2} + x^2}$ to within 10^{-15} (relative).

Common slips

- Using the tangent's gradient $\mathrm{f}'(x)$ instead of the normal's $-1/\mathrm{f}'(x)$, which gives Q at $\mathrm{f}(x) - x\mathrm{f}'(x)$.
- Taking the square root as $\mathrm{f}'(x) = x\mathrm{e}^{-x^2/2}$ without saying why the minus sign is rejected.

Beyond STEP

Here a geometric condition on every point of a curve became a first-order differential equation, but one in which f' appears squared, so it does not determine a unique solution until extra information picks the branch. Such equations, not solved for y', are where uniqueness theorems for ODEs start to need care, and they lead on to envelopes and singular solutions. Boyce and DiPrima, *Elementary Differential Equations*, covers first-order equations and their existence and uniqueness theory.

WORKED SOLUTION

STEP I 1989, Question 7

Pure • Curve Sketching and Optimisation

STEP I 1989, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q7

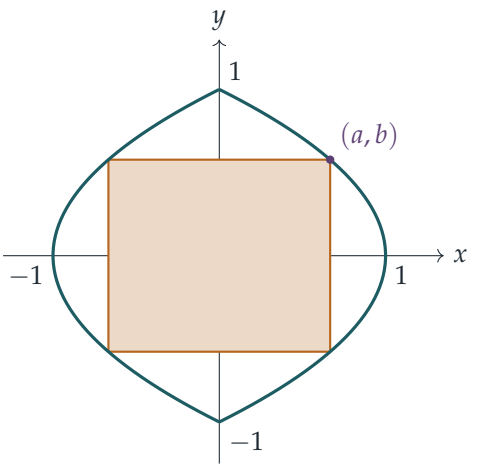
The key idea

The curve is symmetric in both axes, so the best rectangle should be centred on the origin with one corner on the curve in the first quadrant. That turns a two-variable search into maximising a function of one variable. For the general curve the same idea works once we solve for $|y|$ as a power of $1 - |x|$.

Solution

The sketch

Since $y^2 \geq 0$ we need $1 - |x| \geq 0$, so $-1 \leq x \leq 1$. For $x \geq 0$ the curve is $x = 1 - y^2$, a parabola opening to the left with vertex $(1, 0)$; for $x \leq 0$ it is its mirror image $x = y^2 - 1$. The two arcs meet at $(0, \pm 1)$, where they make corners (the slopes $\mathrm{d}y/\mathrm{d}x = \mp\frac{1}{2}$ and $\pm\frac{1}{2}$ do not match), and they cross the x-axis vertically at $(\pm 1, 0)$. The curve is a closed loop enclosing the region $|x| + y^2 \leq 1$, shaped like a rounded diamond.



The largest rectangle

Suppose a rectangle $[x_1, x_2] \times [y_1, y_2]$ lies inside the region $|x| + y^2 \leq 1$. Put $A = \max(|x_1|, |x_2|)$ and $B = \max(|y_1|, |y_2|)$. The corners are (x_i, y_j) with i, j chosen independently, so one corner has $|x| = A$ and $|y| = B$ at the same time. That corner lies in the region, so $A + B^2 \leq 1$. Then the rectangle $[-A, A] \times [-B, B]$ also lies in the region, because for $|x| \leq A$, $|y| \leq B$,

$$|x| + y^2 \leq A + B^2 \leq 1.$$

Its width $2A \geq x_2 - x_1$ and height $2B \geq y_2 - y_1$, so it is at least as big. Hence we may take the rectangle centred at the origin, with corners $(\pm a, \pm b)$, and pushing the corner out until it touches the curve, $a = 1 - b^2$ with $0 \leq b \leq 1$.

? **Ask yourself** Why is it enough to look at rectangles centred at the origin?

That is exactly the argument above: replacing each side by the larger of its two distances from the axis keeps the rectangle inside the curve (by symmetry) and never shrinks it.

The area is

$$S(b) = 4ab = 4b(1 - b^2) = 4b - 4b^3, \qquad S'(b) = 4 - 12b^2.$$

$S'(b) = 0$ at $b = 1/\sqrt{3}$, with $S' > 0$ before and $S' < 0$ after, and $S(0) = S(1) = 0$, so this is the maximum. There $a = 1 - \frac{1}{3} = \frac{2}{3}$ and

$$S_{\max} = 4 \cdot \frac{2}{3} \cdot \frac{1}{\sqrt{3}} = \frac{8}{3\sqrt{3}} = \frac{8}{\sqrt{27}},$$

as required.

The general curve

Now $y^{2m} = (1 - |x|)^n$. The left side is non-negative, and since n is odd, $(1 - |x|)^n$ has the same sign as $1 - |x|$. So again $|x| \leq 1$, and taking the (positive) $2m$-th root,

$$|y| = (1 - |x|)^k, \qquad k = \frac{n}{2m} > 0.$$

(If n were even, every x would be allowed and the curve would run off to infinity, so there would be no largest rectangle: this is why n must be odd.) The curve again is a closed loop through $(\pm 1, 0)$ and $(0, \pm 1)$, symmetric in both axes, enclosing $|y| \leq (1 - |x|)^k$. Since $(1 - |x|)^k$ decreases as $|x|$ increases, the same argument as before shows we may take a rectangle with corners $(\pm a, \pm b)$, $b = (1 - a)^k$, $0 \leq a \leq 1$. Its area is

$$S(a) = 4a(1 - a)^k.$$

Differentiating,

$$S'(a) = 4(1 - a)^{k-1}\big[(1 - a) - ka\big] = 4(1 - a)^{k-1}\big[1 - (1 + k)a\big],$$

which is positive for $0 < a < \frac{1}{1+k}$ and negative for $\frac{1}{1+k} < a < 1$. As $S(0) = S(1) = 0$, the maximum is at $a = \frac{1}{1+k}$, where $1 - a = \frac{k}{1+k}$:

$$S_{\max} = \frac{4}{1+k}\left(\frac{k}{1+k}\right)^k = \frac{4k^k}{(1+k)^{k+1}}.$$

With $k = \frac{n}{2m}$ and $1 + k = \frac{n+2m}{2m}$, the powers of $2m$ collect as $(2m)^{-k} \cdot (2m)^{k+1} = 2m$, so

$$S_{\max} = \frac{8m\, n^{n/(2m)}}{(n+2m)^{(n+2m)/(2m)}} = 8m\left(\frac{n^n}{(n+2m)^{n+2m}}\right)^{1/(2m)}.$$

Checks

- $m = n = 1$ gives $8\left(\frac{1}{27}\right)^{1/2} = 8/\sqrt{27} \approx 1.5396$, agreeing with the first part.
- A fine search over $a \in [0,1]$ for $(m,n) = (1,3), (2,1), (2,3), (3,5)$ matches the formula; for instance $(m,n) = (2,1)$ gives $16/5^{5/4} \approx 2.1400$. Random off-centre rectangles never beat it.
- As $m \to \infty$ with n fixed, $k \to 0$ and the curve fills out the square $|x| \le 1$, $|y| \le 1$; the formula tends to 4, the area of that square.

Common slips

- Forgetting the factor 4: the corner (a, b) gives a rectangle of width $2a$ and height $2b$.
- Ignoring why n odd matters: with n even there is no bounded curve.

Beyond STEP

Maximising $a(1-a)^k$ is a weighted AM–GM problem in disguise: writing $a \cdot \left(\frac{1-a}{k}\right)^k$ as a product of $1 + k$ factors with constant sum gives the maximum without calculus. The curves $|x|^p + |y|^q = 1$ are cousins of the unit balls of the ℓ^p norms, which appear in first-year analysis and linear algebra. For more on inequalities of this kind see Siklos, *Advanced Problems in Mathematics*, and for the analysis behind optimisation, Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I 1989, Question 8

Pure • Multiple Angle Formulae

STEP I 1989, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q8

The key idea

De Moivre's theorem turns $\cos n\theta$ into the real part of $(\cos\theta + \mathrm{i}\sin\theta)^n$, and replacing $\sin^2\theta$ by $1 - \cos^2\theta$ leaves a polynomial in $\cos\theta$. For the final part, the sextic looks like a combination of the two polynomials just found, so we look for one: it turns out to be half of $\cos 6\theta - \cos 4\theta$, and solving $\cos 6\theta = \cos 4\theta$ is easy.

Solution

Throughout, write $c = \cos\theta$ and $s = \sin\theta$, so $s^2 = 1 - c^2$.

Part (i)

By de Moivre's theorem, $\cos 4\theta + \mathrm{i}\sin 4\theta = (c + \mathrm{i}s)^4$. Expanding by the binomial theorem and keeping the real terms (those with even powers of $\mathrm{i}s$),

$$\begin{aligned}\cos 4\theta &= c^4 - \tbinom{4}{2}c^2s^2 + s^4 \\ &= c^4 - 6c^2(1 - c^2) + (1 - c^2)^2 \\ &= c^4 - 6c^2 + 6c^4 + 1 - 2c^2 + c^4 \\ &= 8c^4 - 8c^2 + 1,\end{aligned}$$

which is the required identity.

Part (ii)

In the same way, the real part of $(c + \mathrm{i}s)^6$ gives

$$\cos 6\theta = c^6 - 15c^4s^2 + 15c^2s^4 - s^6,$$

using $\binom{6}{2} = \binom{6}{4} = 15$ and $\mathrm{i}^2 = -1$, $\mathrm{i}^4 = 1$, $\mathrm{i}^6 = -1$. Put $u = c^2$, so $s^2 = 1 - u$:

$$\begin{aligned}\cos 6\theta &= u^3 - 15u^2(1-u) + 15u(1-u)^2 - (1-u)^3 \\ &= u^3 - 15u^2 + 15u^3 + 15u - 30u^2 + 15u^3 - 1 + 3u - 3u^2 + u^3 \\ &= 32u^3 - 48u^2 + 18u - 1 \\ &= 32c^6 - 48c^4 + 18c^2 - 1,\end{aligned}$$

as required.

The equation

Call the left-hand side $p(x) = 16x^6 - 28x^4 + 13x^2 - 1$.

> **Ask yourself** How can the two identities help with a polynomial that isn't either of them?

Compare coefficients. The leading coefficient 16 is half of 32, and the constant -1 is half of -2, which suggests subtracting (i) from (ii):

$$\cos 6\theta - \cos 4\theta = 32c^6 - 56c^4 + 26c^2 - 2 = 2\,p(c).$$

So if $x = \cos\theta$, then

$$p(x) = 0 \iff \cos 6\theta = \cos 4\theta.$$

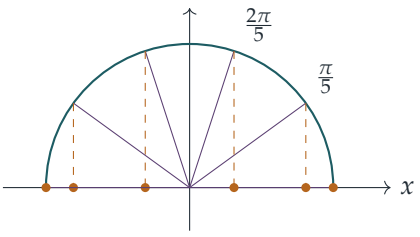
Now $\cos A = \cos B$ exactly when $A = \pm B + 2k\pi$ for some integer k. Here this gives

$$6\theta = 4\theta + 2k\pi \;\Rightarrow\; \theta = k\pi, \qquad \text{or} \qquad 6\theta = -4\theta + 2k\pi \;\Rightarrow\; \theta = \frac{k\pi}{5}.$$

The first family is contained in the second, so the solutions are $\theta = k\pi/5$, and the corresponding values of $x = \cos\theta$ are

$$\cos 0 = 1, \quad \cos\tfrac{\pi}{5}, \quad \cos\tfrac{2\pi}{5}, \quad \cos\tfrac{3\pi}{5} = -\cos\tfrac{2\pi}{5}, \quad \cos\tfrac{4\pi}{5} = -\cos\tfrac{\pi}{5}, \quad \cos\pi = -1.$$

Other values of k repeat these. They are six *distinct* numbers (cosine is strictly decreasing on $[0, \pi]$), and p has degree 6, so these are all the roots of p, real or complex. In particular, the substitution $x = \cos\theta$ only reaches $-1 \le x \le 1$, but we haven't missed any roots outside that interval, since there are none left to find.



These are not yet acceptable answers, since no numerical approximations are allowed and we should give the cosines exactly. Because $x = 1$ is a root, so is $x = -1$ (p is even), and $x^2 - 1$ is a factor. Dividing out (or comparing coefficients),

$$p(x) = (x^2 - 1)(16x^4 - 12x^2 + 1).$$

The quartic factor is a quadratic in x^2:

$$x^2 = \frac{12 \pm \sqrt{144 - 64}}{32} = \frac{12 \pm 4\sqrt{5}}{32} = \frac{3 \pm \sqrt{5}}{8}.$$

Notice that

$$\left(\frac{1+\sqrt{5}}{4}\right)^2 = \frac{6+2\sqrt{5}}{16} = \frac{3+\sqrt{5}}{8}, \qquad \left(\frac{\sqrt{5}-1}{4}\right)^2 = \frac{6-2\sqrt{5}}{16} = \frac{3-\sqrt{5}}{8}.$$

Both values of x^2 are positive, so each gives two real values of x. The real roots of the equation are therefore the six numbers

$$x = \pm 1, \qquad x = \pm\frac{1+\sqrt{5}}{4}, \qquad x = \pm\frac{\sqrt{5}-1}{4}.$$

As a bonus, matching sizes with the list above ($\cos\frac{\pi}{5} > \cos\frac{2\pi}{5} > 0$) gives the well-known values $\cos\frac{\pi}{5} = \frac{1+\sqrt{5}}{4}$ and $\cos\frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$.

Checks

- At $\theta = 0$ both identities give $1 = 1$; at $\theta = \frac{\pi}{2}$, (i) gives $1 = 1$ and (ii) gives $-1 = -1$.
- The coefficients of p add to $16 - 28 + 13 - 1 = 0$, confirming the root $x = 1$.
- Numerically, $\frac{1+\sqrt{5}}{4} \approx 0.80902 = \cos 36^\circ$ and $\frac{\sqrt{5}-1}{4} \approx 0.30902 = \cos 72^\circ$, and p vanishes at both (to rounding error).

Common slips

- Forgetting the solutions from $6\theta = -4\theta + 2k\pi$, and finding only $x = \pm 1$.
- Leaving answers as $\cos\frac{\pi}{5}$ and so on, or as decimals: the question wants exact surd forms.
- Not explaining why there are no roots with $|x| > 1$: the degree count settles it.

Beyond STEP

The polynomial expressing $\cos n\theta$ in terms of $\cos\theta$ is the Chebyshev polynomial T_n, so (i) and (ii) say $T_4(x) = 8x^4 - 8x^2 + 1$ and $T_6(x) = 32x^6 - 48x^4 + 18x^2 - 1$. Chebyshev polynomials are central in numerical analysis, where their roots give the best points for polynomial interpolation. The appearance of $\sqrt{5}$ in $\cos\frac{\pi}{5}$ reflects the geometry of the regular pentagon and the golden ratio. Needham's *Visual Complex Analysis* is a lovely place to see de Moivre's theorem and roots of unity geometrically.

WORKED SOLUTION

STEP I 1989, Question 9

Pure • Transformations of Graphs

STEP I 1989, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q9

The key idea

Once the basic cubic is sketched, every other graph is the same curve seen through a change of coordinates. A substitution $X = ax + b$, $Y = cy + d$ is just a stretch (or reflection) in each direction followed by a translation, and it carries turning points to turning points. So all the work is in finding the two turning points of the original cubic and tracking where they go.

Solution

The basic graph

Write $y = \frac{1}{8}(x^3 - 12x) = \frac{1}{8}x(x^2 - 12)$. Then

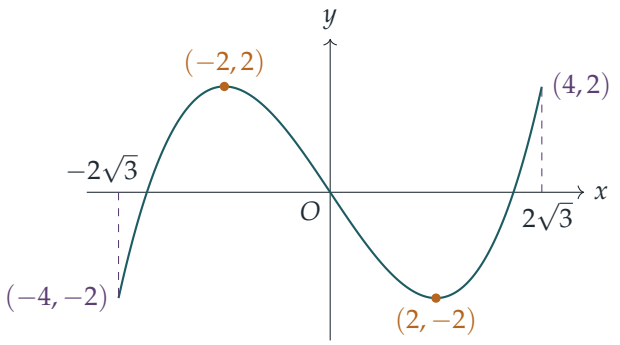
$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3x^2 - 12}{8} = \frac{3(x-2)(x+2)}{8}, \qquad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{6x}{8} = \frac{3x}{4}.$$

The gradient is zero at $x = \pm 2$.

- At $x = -2$: $y = \frac{1}{8}(-8 + 24) = 2$, and $y'' = -\frac{3}{2} < 0$, so $(-2, 2)$ is a local maximum.
- At $x = 2$: $y = \frac{1}{8}(8 - 24) = -2$, and $y'' = \frac{3}{2} > 0$, so $(2, -2)$ is a local minimum.

Other useful features:

- The curve meets the axes where $x(x^2 - 12) = 0$, i.e. at $x = 0$ and $x = \pm 2\sqrt{3} \approx \pm 3.46$.
- The function is odd, so the graph has rotational symmetry of order 2 about the origin, which is also the point of inflection; the gradient there is $-\frac{3}{2}$.
- At the ends, $y(4) = \frac{1}{8}(64 - 48) = 2$ and $y(-4) = -2$. So the curve climbs back to the height of the maximum exactly at $x = 4$.



Why turning points go to turning points

Ask yourself When we change coordinates, how do we know the new turning points are just the images of the old ones?

If $X = ax + b$ and $Y = cy + d$ with $a, c \neq 0$, then by the chain rule

$$\frac{\mathrm{d}Y}{\mathrm{d}X} = \frac{\mathrm{d}Y/\mathrm{d}x}{\mathrm{d}X/\mathrm{d}x} = \frac{c}{a}\frac{\mathrm{d}y}{\mathrm{d}x},$$

which is zero exactly where $\mathrm{d}y/\mathrm{d}x$ is zero. So the new turning points are the images of $(-2, 2)$ and $(2, -2)$. (If $c/a < 0$ the picture is reflected and the maximum and minimum swap roles.) Geometrically:

- $X = \frac{1}{2}x$ squashes the graph horizontally by a factor of 2;
- $Y = \frac{1}{2}y$ squashes it vertically by a factor of 2;
- adding 1 to X or Y then shifts it 1 unit right or up.

The four transformed graphs

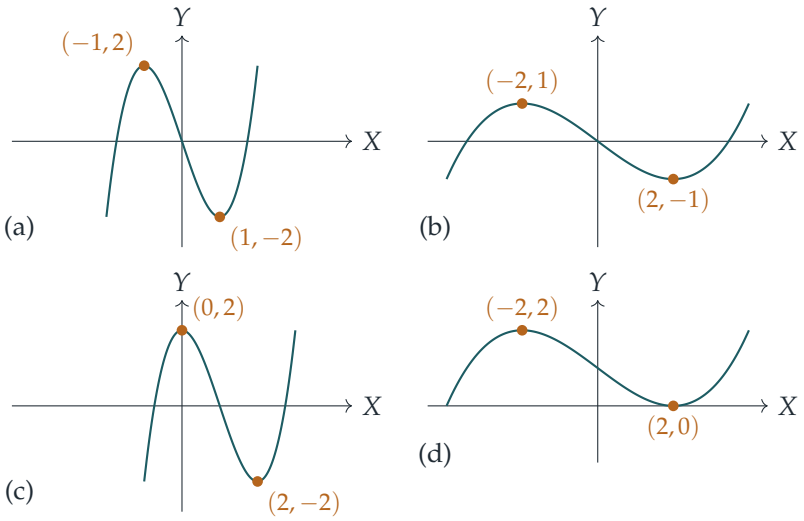
Applying each map to $(-2, 2)$, $(2, -2)$ and to the range $-4 \leq x \leq 4$:

(a) $X = \frac{1}{2}x$, $Y = y$: maximum $(-1, 2)$, minimum $(1, -2)$, for $-2 \leq X \leq 2$. The curve is $Y = X^3 - 3X$.

(b) $X = x$, $Y = \frac{1}{2}y$: maximum $(-2, 1)$, minimum $(2, -1)$, for $-4 \leq X \leq 4$. The curve is $16Y = X^3 - 12X$.

(c) $X = \frac{1}{2}x + 1$, $Y = y$: maximum $(0, 2)$, minimum $(2, -2)$, for $-1 \leq X \leq 3$. This is graph (a) moved 1 unit right, with inflection at $(1, 0)$.

(d) $X = x$, $Y = \frac{1}{2}y + 1$: maximum $(-2, 2)$, minimum $(2, 0)$, for $-4 \leq X \leq 4$. This is graph (b) moved 1 unit up, with inflection at $(0, 1)$; it touches the X-axis at its minimum.



The final part

We need $a, c \neq 0$ (otherwise the graph collapses to a line), and then the turning points of the new graph are

$$(-2a + b,\ 2c + d) \quad \text{and} \quad (2a + b,\ -2c + d).$$

These must be $(0,0)$ and $(1,1)$ in some order, which gives two linear systems.

Case 1: $(-2,2) \mapsto (0,0)$ and $(2,-2) \mapsto (1,1)$. Then $-2a + b = 0$ and $2a + b = 1$, so $b = \frac{1}{2}$, $a = \frac{1}{4}$; and $2c + d = 0$, $-2c + d = 1$, so $d = \frac{1}{2}$, $c = -\frac{1}{4}$.

Case 2: $(-2,2) \mapsto (1,1)$ and $(2,-2) \mapsto (0,0)$. In the same way $a = -\frac{1}{4}$, $b = \frac{1}{2}$, $c = \frac{1}{4}$, $d = \frac{1}{2}$.

So one answer is

$$a = \tfrac{1}{4}, \quad b = \tfrac{1}{2}, \quad c = -\tfrac{1}{4}, \quad d = \tfrac{1}{2},$$

and the other is $a = -\frac{1}{4}$, $b = \frac{1}{2}$, $c = \frac{1}{4}$, $d = \frac{1}{2}$.

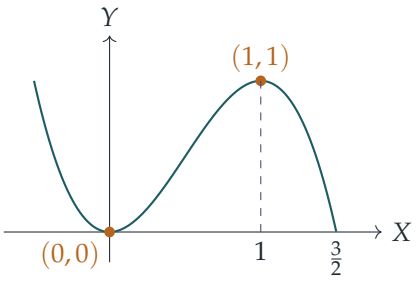
It is worth finding the new curve. In Case 1, $x = 4X - 2$ and $y = 2 - 4Y$, so

$$\begin{aligned} 8(2 - 4Y) &= (4X - 2)^3 - 12(4X - 2) \\ &= 64X^3 - 96X^2 + 48X - 8 - 48X + 24 \\ &= 64X^3 - 96X^2 + 16, \end{aligned}$$

which simplifies to

$$Y = 3X^2 - 2X^3 = X^2(3 - 2X).$$

This has $Y' = 6X(1 - X)$, so its turning points are indeed a minimum at $(0,0)$ and a maximum at $(1,1)$. Case 2 gives exactly the same curve: replacing (x,y) by $(-x,-y)$ doesn't change the original graph, because the cubic is odd, and that's the only difference between the two cases.



For $-4 \leq x \leq 4$ this runs from $(-\frac{1}{2}, 1)$ to $(\frac{3}{2}, 0)$.

Checks

- Substituting $X = 0$ and $X = 1$ into $Y = 3X^2 - 2X^3$ gives 0 and 1, and $Y' = 6X(1 - X)$ vanishes at both.
- Each transformed graph keeps the original feature that the curve returns to the height of a turning point at an end of the range: in (d), for instance, $Y = 2$ at $X = 4$ and $Y = 0$ at $X = -4$.
- Numerically, at $x = 1$ we have $y = -\frac{11}{8}$; in Case 1 this gives $X = 0.75$, $Y = 0.84375$, and $3(0.75)^2 - 2(0.75)^3 = 0.84375$.

Common slips

- Getting the direction of a stretch wrong: $X = \frac{1}{2}x$ makes the graph *narrower*, not wider, because the point at $x = 2$ moves to $X = 1$.
- Forgetting to transform the domain as well: graph (c) runs over $-1 \leq X \leq 3$, not $-4 \leq X \leq 4$.
- Giving only one solution to the final part without noticing the other ordering of the turning points (either is enough, but $c < 0$ in the first case is easy to miss).

Beyond STEP

The maps $X = ax + b$, $Y = cy + d$ are affine transformations, and the final part shows that any cubic with two turning points can be moved by one of them onto the standard curve $Y = 3X^2 - 2X^3$; this is the "smoothstep" function used in computer graphics. Classifying curves up to a group of transformations like this is a theme that runs through geometry and linear algebra. For the matrix side of the story, see Strang, *Introduction to Linear Algebra*.

WORKED SOLUTION

STEP I 1989, Question 10

Mechanics • Velocity-Dependent Resistance

STEP I 1989, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q10

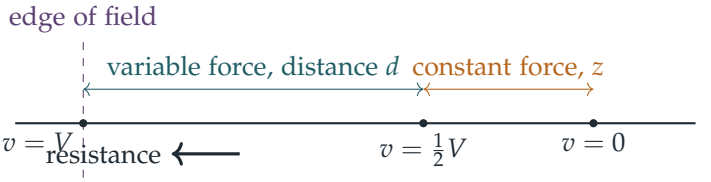
The key idea

The resistance depends on the speed v, not on time, and we want a *distance*. That is the signal to write the acceleration as $v\dfrac{\mathrm{d}v}{\mathrm{d}x}$, which turns Newton's second law into a separable equation linking v and x directly. The second part is a constant-deceleration problem, so the only work is to find the force at the moment of the change and let it vary by $\pm 10\%$.

Solution

The model

We treat the spaceship as a particle of mass M moving along a straight line. Let x be the distance travelled since entering the field, and v the speed, with $v = V$ at $x = 0$. The only force is the resistance, which acts backwards, so the speed decreases throughout.



Distance to halve the speed

Taking the direction of motion as positive and writing the acceleration as $v\dfrac{\mathrm{d}v}{\mathrm{d}x}$, Newton's second law gives

$$Mv\frac{\mathrm{d}v}{\mathrm{d}x} = -\frac{M\omega\,(v^2+V^2)}{v} \qquad \Longrightarrow \qquad \frac{v^2}{v^2+V^2}\,\mathrm{d}v = -\omega\,\mathrm{d}x.$$

Ask yourself How do we integrate $\dfrac{v^2}{v^2+V^2}$?

Write the numerator as $(v^2+V^2)-V^2$, so that

$$\frac{v^2}{v^2+V^2} = 1 - \frac{V^2}{v^2+V^2}, \qquad \int \frac{v^2}{v^2+V^2}\,\mathrm{d}v = v - V\arctan\frac{v}{V} + C,$$

using the standard integral $\int \frac{\mathrm{d}v}{v^2+V^2} = \frac{1}{V}\arctan\frac{v}{V}$.

Let d be the distance travelled when the speed has fallen to $\frac{1}{2}V$. Integrating from $(x,v)=(0,V)$ to $(d,\frac{1}{2}V)$,

$$\begin{aligned}\omega d &= \int_{V/2}^{V} \frac{v^2}{v^2+V^2}\,\mathrm{d}v = \Big[v - V\arctan\frac{v}{V}\Big]_{V/2}^{V} \\ &= \frac{V}{2} - V\Big(\arctan 1 - \arctan\tfrac{1}{2}\Big) = V\Big(\frac{1}{2} - \frac{\pi}{4} + \arctan\frac{1}{2}\Big).\end{aligned}$$

So the required distance is

$$d = \frac{V}{\omega}\Big(\frac{1}{2} - \frac{\pi}{4} + \arctan\frac{1}{2}\Big) = \frac{V}{\omega}\Big(\frac{1}{2} - \arctan\frac{1}{3}\Big) \approx 0.178\,\frac{V}{\omega}.$$

The second form uses $\arctan 1 - \arctan\frac{1}{2} = \arctan\dfrac{1-\frac{1}{2}}{1+\frac{1}{2}} = \arctan\frac{1}{3}$, from the tangent subtraction formula (both angles lie in $(0,\frac{\pi}{2})$, so there is no ambiguity). Either form is a complete answer.

The constant force

Immediately before the change, $v=\frac{1}{2}V$, so the resistive force has magnitude

$$\frac{M\omega\left(\frac{1}{4}V^2+V^2\right)}{\frac{1}{2}V} = \frac{5}{2}M\omega V.$$

The new constant force F is within 10% of this, so

$$\frac{9}{10}\cdot\frac{5}{2}M\omega V \le F \le \frac{11}{10}\cdot\frac{5}{2}M\omega V, \qquad \text{i.e.} \qquad \frac{9}{4}M\omega V \le F \le \frac{11}{4}M\omega V.$$

With a constant force the deceleration $a = F/M$ is constant, so $v^2 = u^2 - 2as$ with $u=\frac{1}{2}V$ and final speed 0 gives

$$z = \frac{(V/2)^2}{2a} = \frac{MV^2}{8F}.$$

This decreases as F increases, so the largest force gives the smallest z and vice versa:

$$\frac{MV^2}{8\cdot\frac{11}{4}M\omega V} \le z \le \frac{MV^2}{8\cdot\frac{9}{4}M\omega V}, \qquad \text{that is} \qquad \boxed{\frac{V}{22\,\omega} \le z \le \frac{V}{18\,\omega}}$$

(with strict inequalities if "within 10%" is read strictly). If the force were exactly unchanged, $z = V/(20\omega)$, in the middle of this range.

Checks

- Dimensions: ω has units s^{-1} (since $M\omega v$ must be a force), so V/ω is a length, as both answers require.
- The coefficient $\frac{1}{2} - \arctan\frac{1}{3} \approx 0.1782$ is positive and less than $\frac{1}{2}$. It should be: the ship travels at speeds between $\frac{1}{2}V$ and V, and the deceleration is at least $2\omega V$, so the trip lasts at most $1/(4\omega)$ and covers less than $V/(4\omega)$.
- Stepping $\dot{v} = -\omega(v^2 + V^2)/v$ numerically with $V = \omega = 1$ gives 0.17825 for the distance at which $v = \frac{1}{2}$, and stepping the constant-force phase gives $z = 0.0455$, 0.0500, 0.0556 for forces of 110%, 100% and 90%, matching $\frac{1}{22}$, $\frac{1}{20}$, $\frac{1}{18}$.

Common slips

- Using $\dfrac{\mathrm{d}v}{\mathrm{d}t}$ instead of $v\dfrac{\mathrm{d}v}{\mathrm{d}x}$: this gives the time, not the distance, and an extra integration is then needed.
- Pairing the bounds the wrong way round: a *larger* force gives a *shorter* stopping distance, so 1.1 times the force goes with the lower limit $V/(22\omega)$.

Beyond STEP

Writing $a = v\,\mathrm{d}v/\mathrm{d}x$ is the first glimpse of a bigger idea: for a one-dimensional force depending only on position or speed, one can trade time for another variable and reduce the order of the equation, and when the force depends only on position this leads to energy conservation and the phase-plane picture of motion. The second part is a small exercise in *sensitivity*: how an uncertainty in the data (here $\pm 10\%$ in the force) propagates to the answer (roughly $\mp 10\%$ in z). Kleppner and Kolenkow, *An Introduction to Mechanics*, treats velocity-dependent forces in this spirit.

WORKED SOLUTION

STEP I 1989, Question 11

Mechanics • Projectile Range Optimisation

STEP I 1989, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q11

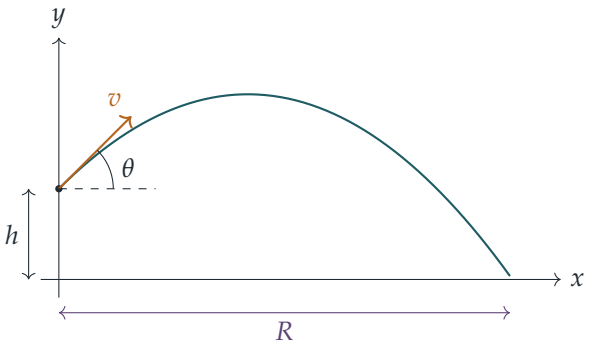
The key idea

The time of flight comes from a quadratic in t: the shot lands when its height returns to zero. Taking the positive root and multiplying by the horizontal speed gives R; the rest is tidying the square root into the required shape. For the second part, "does a small decrease in θ increase R?" is a question about the sign of $\mathrm{d}R/\mathrm{d}\theta$ at $\theta = 45°$, and the symmetry $\sin 45° = \cos 45°$ makes that derivative collapse very neatly.

Solution

The model

We treat the shot as a particle moving under gravity alone, released from the point $(0, h)$ with speed v at angle θ above the horizontal, where $0 < \theta < \frac{\pi}{2}$ and $h > 0$. Take x horizontal in the direction of the throw and y vertically upwards from ground level.



The range

After time t,

$$x = vt\cos\theta, \qquad y = h + vt\sin\theta - \tfrac{1}{2}gt^2.$$

The shot strikes the ground when $y = 0$, that is when

$$\tfrac{1}{2}gt^2 - (v\sin\theta)\,t - h = 0 \quad\Longrightarrow\quad t = \frac{v\sin\theta \pm \sqrt{v^2\sin^2\theta + 2gh}}{g}.$$

Since $h > 0$ the product of the roots, $-2h/g$, is negative, so exactly one root is positive; the minus sign gives the negative one (a time before release), so we take the plus sign. Then

$$\begin{aligned} R = v\cos\theta\, t &= \frac{v\cos\theta}{g}\left(v\sin\theta + \sqrt{v^2\sin^2\theta + 2gh}\right) \\ &= \frac{v^2\sin\theta\cos\theta}{g}\left(1 + \sqrt{1 + \frac{2gh}{v^2\sin^2\theta}}\right) \qquad \text{(taking out } v\sin\theta > 0\text{)} \\ &= \frac{v^2}{2g}\sin 2\theta\left(1 + \sqrt{1 + \frac{2hg}{v^2\sin^2\theta}}\right), \end{aligned}$$

using $2\sin\theta\cos\theta = \sin 2\theta$. This is the required result.

Ask yourself Why is it legitimate to pull $v\sin\theta$ inside the square root as $v^2\sin^2\theta$?

Because $\sqrt{a^2 b} = a\sqrt{b}$ only when $a \geq 0$. Here $0 < \theta < \frac{\pi}{2}$, so $v\sin\theta > 0$ and the step is valid. (For a downward throw it would fail, and the formula as given would be wrong.)

Is 45° the best angle?

For differentiating, the unsimplified form is easier. Write

$$R(\theta) = \frac{v}{g}\cos\theta\,(v\sin\theta + S), \qquad S = \sqrt{v^2\sin^2\theta + 2gh},$$

so that $\dfrac{\mathrm{d}S}{\mathrm{d}\theta} = \dfrac{v^2\sin\theta\cos\theta}{S}$. By the product rule,

$$\frac{\mathrm{d}R}{\mathrm{d}\theta} = \frac{v}{g}\left[-\sin\theta\,(v\sin\theta + S) + \cos\theta\left(v\cos\theta + \frac{v^2\sin\theta\cos\theta}{S}\right)\right].$$

At $\theta = 45^\circ$ put $\sin\theta = \cos\theta = c = \frac{1}{\sqrt{2}}$. The terms $-vc^2$ and $+vc^2$ cancel, leaving

$$\begin{aligned} \left.\frac{\mathrm{d}R}{\mathrm{d}\theta}\right|_{\theta=45^\circ} &= \frac{vc}{g}\left(-S + \frac{v^2c^2}{S}\right) = \frac{vc}{gS}\left(v^2c^2 - S^2\right) \\ &= \frac{vc}{gS}\left(\tfrac{1}{2}v^2 - \tfrac{1}{2}v^2 - 2gh\right) = -\frac{2vhc}{S} = -\frac{\sqrt{2}\,vh}{\sqrt{v^2 + 4gh}}, \end{aligned}$$

where at the last step $S = \sqrt{\frac{1}{2}v^2 + 2gh} = \frac{1}{\sqrt{2}}\sqrt{v^2 + 4gh}$.

Since $v > 0$ and $h > 0$, this derivative is strictly negative. So R is a strictly decreasing function of θ near $45°$, and for a sufficiently small decrease $\delta\theta > 0$ we have $R(45° - \delta\theta) \approx R(45°) + \frac{\sqrt{2}\,vh}{\sqrt{v^2+4gh}}\,\delta\theta > R(45°)$. **Yes: a small decrease in θ will increase R.**

Physically: released from a height, the shot stays in the air a little longer than on level ground, and that extra time is better spent travelling horizontally, so a flatter launch pays off. Only when $h = 0$ does the derivative vanish and $45°$ become optimal.

Checks

- With $h = 0$ the formula gives $R = \frac{v^2}{g}\sin 2\theta$, the familiar level-ground range, and the derivative at $45°$ is 0, as it should be.
- Units: v^2/g is a length, and $2hg/v^2$ is dimensionless.
- Numerically, with $g = 9.8$, $v = 13\,\mathrm{m\,s^{-1}}$ and $h = 2\,\mathrm{m}$: $R(45°) \approx 19.05\,\mathrm{m}$ but $R(40°) \approx 19.10\,\mathrm{m}$. Stepping the motion forward in time reproduces the formula for R, and a finite-difference derivative at $45°$ agrees with $-\sqrt{2}\,vh/\sqrt{v^2+4gh}$ for many random v and h.

Common slips

- Taking the wrong root of the quadratic for t, or forgetting to justify discarding the negative one.
- Reasoning that "$\sin 2\theta$ is maximal at $45°$, so R is too". The bracket also depends on θ (it *decreases* as θ increases), so that argument is incomplete.

Beyond STEP

Setting $\mathrm{d}R/\mathrm{d}\theta = 0$ in general shows that the best launch angle from height h satisfies $\tan\theta = v/\sqrt{v^2+2gh}$, which is a little under $45°$ (about $42°$ for the numbers above); real shot-putters release lower still, because they can push harder at flatter angles. Optimising over a family of trajectories like this leads to the *envelope* of the family, the "parabola of safety", which marks out every point the shot can reach. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops projectile motion and its variations carefully.

WORKED SOLUTION

STEP I 1989, Question 12

Mechanics • Elastic Strings and Energy

STEP I 1989, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q12

The key idea

Everything rests on the geometry of a regular tetrahedron: how far its centre of mass is from a vertex, from a face, and from the axis through a vertex. Once we have those three lengths, the first part is just Hooke's law. For the second part, symmetry keeps the motion a pure vertical translation. The tetrahedron starts and finishes at rest, so we don't need the equation of motion at all: an energy balance between gravitational and elastic potential energy does the job.

Solution

Geometry of the tetrahedron

We model the rods as uniform rigid rods and the strings as light elastic strings obeying Hooke's law. Each rod has its mass at its midpoint. Every vertex belongs to exactly three of the six rods, so the average of the six midpoints equals the average of the four vertices. The centre of mass G is therefore the centroid of the tetrahedron.

The face BCD is equilateral with side $2a$, so its centre N is at distance

$$\rho = \frac{2a}{\sqrt{3}}$$

from each of B, C, D. The height of the tetrahedron is $AN = \sqrt{(2a)^2 - \rho^2} = \sqrt{\frac{8}{3}}\,a = \frac{2\sqrt{6}}{3}a$. The centroid lies on AN, a quarter of the way up from the face. So

$$GN = r = \frac{\sqrt{6}}{6}\,a, \qquad GA = GB = GC = GD = R = \frac{\sqrt{6}}{2}\,a.$$

(As a check, $R^2 = \rho^2 + r^2 = \frac{4}{3}a^2 + \frac{1}{6}a^2 = \frac{3}{2}a^2$.)

The tension

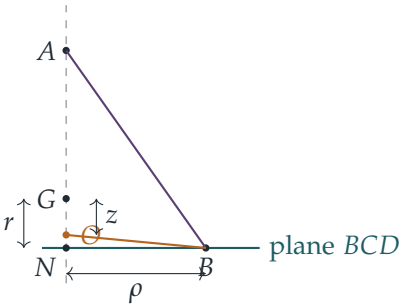
With X at G, each string has length $R = \frac{\sqrt{6}}{2}a$ and extension $R - a = \frac{\sqrt{6}-2}{2}a > 0$, so all four are taut. By Hooke's law each tension is

$$T = \frac{\lambda(R-a)}{a} = \frac{\sqrt{6}-2}{2}\lambda \approx 0.225\,\lambda.$$

After XA breaks

Before the break, the four equal tensions pull the vertices along the four directions towards G. The unit vectors from G to the vertices sum to zero, because G is the centroid, so the strings exert no net force and the table supports the full weight Mg.

When XA breaks, the three remaining strings pull with total force T vertically upwards, which is minus the missing pull of XA. Horizontally the three pulls cancel by the threefold symmetry about the vertical line AN. Each lies in a vertical plane through that line, so they give no twisting couple, and their tilting couples cancel by symmetry. The tetrahedron therefore moves as a pure vertical translation, with AN staying on the vertical through O. It leaves the table because $T > Mg$, which we confirm at the end.



Let z be the height of G above O. Then B is at horizontal distance ρ from the vertical through O and at height $z - r$ relative to O, so each of the three strings has length

$$L(z) = \sqrt{\rho^2 + (z-r)^2}.$$

At the start $z = 0$ and $L = \sqrt{\rho^2 + r^2} = R$. At the highest point, BCD is level with O, so $z = r$ and $L = \rho = \frac{2}{\sqrt{3}}a$. For $0 \le z \le r$, L stays between ρ and R, both greater than a, so the strings stay taut throughout.

> **Ask yourself** How can we find the maximum height without solving an awkward non-linear equation of motion?

The table is smooth and the strings are light, so mechanical energy is conserved. The normal reaction does no work, since the tetrahedron leaves the table at once. The tetrahedron is at rest both at the start and at its highest point, so kinetic energy drops

out. The elastic energy $3 \cdot \dfrac{\lambda(L-a)^2}{2a}$ lost by the three strings must equal the gravitational energy Mgr gained:

$$Mg\,r = \frac{3\lambda}{2a}\Big[(R-a)^2 - (\rho-a)^2\Big].$$

Factorising the bracket, and using $R^2 - \rho^2 = \frac{3}{2}a^2 - \frac{4}{3}a^2 = \frac{1}{6}a^2$,

$$\begin{aligned}(R-a)^2 - (\rho-a)^2 &= R^2 - \rho^2 - 2a(R-\rho)\\ &= a^2\Big(\tfrac{1}{6} - \sqrt{6} + \tfrac{4}{\sqrt{3}}\Big).\end{aligned}$$

Since $\dfrac{3}{2ar} = \dfrac{3}{2a}\cdot\dfrac{6}{\sqrt{6}\,a} = \dfrac{3\sqrt{6}}{2a^2}$, this gives

$$\begin{aligned}\frac{Mg}{\lambda} &= \frac{3\sqrt{6}}{2}\Big(\tfrac{1}{6} - \sqrt{6} + \tfrac{4}{\sqrt{3}}\Big) = \frac{\sqrt{6}}{4} - 9 + 6\sqrt{2}\\ &\approx 0.6124 - 9 + 8.4853 = 0.0977,\end{aligned}$$

which is 0.098 to 2 significant figures, as required.

Finally, $Mg \approx 0.098\lambda < 0.225\lambda \approx T$, so the net upward force $T - Mg$ just after the break is positive and the tetrahedron does lift off, consistent with the question.

Checks

- Setting $a = 1$, we placed the four vertices in coordinates, confirmed all six edges have length 2 and all four vertices are $\frac{\sqrt{6}}{2} \approx 1.2247$ from the centroid, and recovered $T \approx 0.2247\lambda$.
- Taking $Mg/\lambda = 6\sqrt{2} + \frac{\sqrt{6}}{4} - 9 \approx 0.09765$, we stepped the equation of motion $M\ddot{z} = 3T(L)\,\frac{r-z}{L} - Mg$ numerically from rest at $z = 0$. The velocity first returns to zero at $z \approx 0.40825$, which matches $r = \frac{\sqrt{6}}{6}$.

Common slips

- Taking R to be the distance from the centroid to a face. The centroid is $\frac{3}{4}$ of the way down from a vertex, so $R = 3r$.
- Forgetting that the strings are still stretched at the top. Each string's energy there is $\frac{\lambda(\rho-a)^2}{2a}$, not zero.
- Using the extension $L - a$ with L measured from O to the centre N of the face, rather than from O to a vertex.

Beyond STEP

The centroid trick used here (six rod midpoints average to the four vertices) is a small case of the general fact that the centre of mass of a system is the mass-weighted average

of the centres of mass of its parts. Treating the motion as a pure translation relies on symmetry arguments, which become systematic in rigid-body dynamics, where a body's motion splits into translation of the centre of mass and rotation about it. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both ideas carefully.

WORKED SOLUTION

STEP I 1989, Question 13

Mechanics • Ladder Equilibrium with Friction

STEP I 1989, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

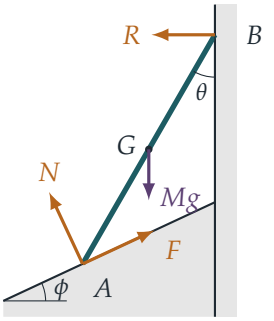
step.maths.org/questions/89-s1-q13

The key idea

This is a three-force-plus-friction equilibrium problem, and the trick is to choose good directions. Taking moments about the foot of the ladder removes both N and F and gives the wall reaction at once. Then, instead of resolving horizontally and vertically, we resolve *perpendicular to and along the slope*. Each of those equations contains only one of the unknowns N and F, so the given formulas drop straight out.

Solution

We model the ladder as a uniform rod AB of length $2a$ with its centre of mass G at the midpoint. The foot A is on the rough slope, and the top B is against the smooth wall, so the wall's reaction R is horizontal, pointing away from the wall. At A there is a normal reaction N perpendicular to the slope and a friction force F along it. The ladder would tend to slip with its foot sliding down the slope, away from the wall, so we take F as acting up the slope towards the wall. (The formula for F turns out positive, which confirms the choice.)



The wall reaction

Taking moments about A removes N and F. The ladder makes angle θ with the vertical, so B is a height $2a\cos\theta$ above A and G is a horizontal distance $a\sin\theta$ from A. The horizontal force R at B has moment $R \cdot 2a\cos\theta$, and the weight has moment $Mg \cdot a\sin\theta$ in the opposite sense, so

$$2aR\cos\theta = Mga\sin\theta \qquad \Longrightarrow \qquad R = \tfrac{1}{2}Mg\tan\theta.$$

Resolving perpendicular to and along the slope

> **Ask yourself** Which directions should we resolve in, so that each equation contains only one unknown?

Resolve perpendicular to the slope and along it. F has no component perpendicular to the slope, and N has no component along it. We need the components of the other two forces.

- The weight Mg has component $Mg\cos\phi$ into the slope and $Mg\sin\phi$ down the slope.
- The slope rises towards the wall, so its upward normal leans away from the wall, and the horizontal force R (also pointing away from the wall) has component $R\sin\phi$ along that normal, i.e. *out* of the slope. Along the slope, R points away from the wall, which is downhill, with component $R\cos\phi$.

Perpendicular to the slope:

$$N + R\sin\phi = Mg\cos\phi,$$
$$\text{so} \quad N = Mg\cos\phi - \tfrac{1}{2}Mg\tan\theta\sin\phi = Mg\left(\cos\phi - \frac{\tan\theta\sin\phi}{2}\right).$$

Along the slope (up the slope positive):

$$F = Mg\sin\phi + R\cos\phi \quad \Longrightarrow \quad F = Mg\left(\sin\phi + \frac{\tan\theta\cos\phi}{2}\right).$$

These are the two required results.

The largest angle

For the foot not to slip we need the contact to hold, $N > 0$, and friction to be limited: $F \le \mu N$ with $\mu = 2$. (Here $F > 0$, so this is the only friction condition.) With $\phi = 45^\circ$ we have $\sin\phi = \cos\phi = \frac{1}{\sqrt{2}}$, and writing $t = \tan\theta$,

$$F = \frac{Mg}{\sqrt{2}}\left(1 + \frac{t}{2}\right), \qquad N = \frac{Mg}{\sqrt{2}}\left(1 - \frac{t}{2}\right).$$

So $N > 0$ requires $t < 2$, and then $F \le 2N$ becomes

$$1 + \frac{t}{2} \le 2 - t \quad \Longleftrightarrow \quad \frac{3t}{2} \le 1 \quad \Longleftrightarrow \quad \tan\theta \le \frac{2}{3}.$$

This is stricter than $t < 2$, so it is the binding condition. Since $\tan\theta$ is increasing on $0 < \theta < 90°$, the largest angle for which the ladder can rest in equilibrium is

$$\theta = \arctan\tfrac{2}{3} \approx 33.7°,$$

and at that angle friction is limiting.

Checks

- With $\phi = 0$ (level ground) the formulas give $N = Mg$ and $F = \frac{1}{2}Mg\tan\theta = R$, the familiar results for a ladder on a horizontal floor.
- With $\theta = 0$ (ladder vertical) the wall does nothing, and $N = Mg\cos\phi$, $F = Mg\sin\phi$, just as for a particle resting on the slope.
- Numerically, solving the horizontal and vertical force equations directly for random θ and ϕ agrees with both formulas to 10^{-13}, and with $\phi = 45°$ the ratio F/N passes through 2 at $\theta = 33.690°$.

Common slips

- Getting the sign of the $R\sin\phi$ term wrong. Because the slope rises towards the wall, R partly pulls the foot *off* the slope, which reduces N.
- Resolving horizontally and vertically, then making errors solving the resulting simultaneous equations. It works, but resolving along and perpendicular to the slope is much quicker.

Beyond STEP

Notice that the steeper the slope, the less the ladder can lean, because the wall reaction reduces N while adding to F. Problems like this are the start of statics, where we choose axes and moment points to decouple the unknowns. The same idea underlies the analysis of trusses and frameworks, and of stability more generally. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body equilibrium and friction carefully.

WORKED SOLUTION

STEP I 1989, Question 14

Probability and Statistics • Conditional Distributions

STEP I 1989, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q14

The key idea

Everything rests on one set of probabilities: the chance that a seed lands in each league-wide strip. A seed grows into a tree exactly when it lands in the parkland or the farmland, so the number of trees is binomial. The distance Y at which a tree grows is just L *conditioned* on landing on land. Its density is f cut down to the land strips and scaled up so that the total area is 1. The final part is a hypothesis test: is 23 trees on the farmland surprisingly many for a binomial count?

Solution

The strip probabilities

The cumulative distribution function of L is, for $0 \le x \le 4$,

$$\mathrm{F}(x) = \int_0^x \left(\tfrac{1}{2} - \tfrac{1}{8}t\right)\mathrm{d}t = \frac{x}{2} - \frac{x^2}{16},$$

so $\mathrm{F}(1) = \frac{7}{16}$, $\mathrm{F}(2) = \frac{3}{4}$, $\mathrm{F}(3) = \frac{15}{16}$ and $\mathrm{F}(4) = 1$. Hence

$$\mathrm{P}(\text{parkland}) = \mathrm{P}(0 \le L \le 1) = \tfrac{7}{16}, \qquad \mathrm{P}(\text{lake}) = \tfrac{3}{4} - \tfrac{7}{16} = \tfrac{5}{16},$$
$$\mathrm{P}(\text{farmland}) = \mathrm{P}(2 \le L \le 3) = \tfrac{15}{16} - \tfrac{3}{4} = \tfrac{3}{16}, \qquad \mathrm{P}(\text{sea}) = \tfrac{1}{16}.$$

(The boundaries have probability zero, since L is continuous, so it doesn't matter which strip they belong to.) A seed grows into a tree with probability

$$p = \tfrac{7}{16} + \tfrac{3}{16} = \tfrac{5}{8}.$$

Let T be the number of trees in a year. Since the 96 seeds are blown independently, each growing with probability $\frac{5}{8}$, we have $T \sim \mathrm{B}\left(96, \frac{5}{8}\right)$, and the mean number of new fever trees each year is

$$\mathrm{E}(T) = 96 \times \tfrac{5}{8} = 60.$$

Part (i)

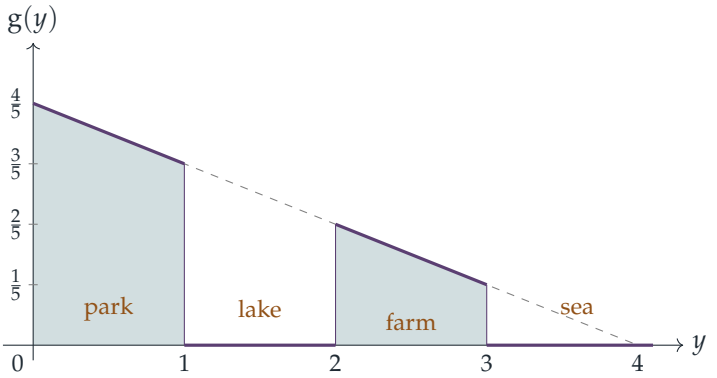
Y has the distribution of L given that the seed lands on land. Write $A = [0,1] \cup [2,3]$ for the land. For $y \in A$,

$$\mathrm{P}(y \le Y \le y+\delta y) = \frac{\mathrm{P}(y \le L \le y+\delta y)}{\mathrm{P}(L \in A)} \approx \frac{\mathrm{f}(y)\,\delta y}{5/8},$$

and Y never takes a value outside A. So the density of Y is

$$\mathrm{g}(y) = \begin{cases} \frac{8}{5}\left(\frac{1}{2} - \frac{1}{8}y\right) = \dfrac{4-y}{5} & \text{for } 0 \le y \le 1 \text{ or } 2 \le y \le 3, \\ 0 & \text{otherwise.} \end{cases}$$

As a check, its area is $\frac{8}{5}\left(\frac{7}{16} + \frac{3}{16}\right) = 1$. The graph is two pieces of the same falling line: from $\frac{4}{5}$ down to $\frac{3}{5}$ over the parkland, nothing over the lake, then from $\frac{2}{5}$ down to $\frac{1}{5}$ over the farmland.



The dashed line is $\frac{8}{5}\mathrm{f}$, shown for comparison; the density of Y follows it only over land.

For the mean, $\int y(4-y)\,\mathrm{d}y = 2y^2 - \frac{1}{3}y^3$, so

$$\begin{aligned} \mathrm{E}(Y) &= \frac{1}{5}\left[2y^2 - \tfrac{1}{3}y^3\right]_0^1 + \frac{1}{5}\left[2y^2 - \tfrac{1}{3}y^3\right]_2^3 \\ &= \frac{1}{5}\left(\frac{5}{3}\right) + \frac{1}{5}\left(9 - \frac{16}{3}\right) = \frac{1}{5}\cdot\frac{5}{3} + \frac{1}{5}\cdot\frac{11}{3} = \frac{16}{15}. \end{aligned}$$

So a new tree grows, on average, $\frac{16}{15}$ leagues south of the castle. That's just past the parkland, in the lake, which is fine: a mean need not be a value the variable can take.

Part (ii)

Let X be the number of trees that grow in the farmland in a year. If the tradition is followed, 96 seeds are released, and each lands in the farmland independently with probability $\frac{3}{16}$, so

$$X \sim \mathrm{B}\left(96, \tfrac{3}{16}\right), \qquad \mathrm{E}(X) = 18, \qquad \mathrm{Var}(X) = 96 \cdot \tfrac{3}{16} \cdot \tfrac{13}{16} = \tfrac{117}{8} \approx 14.6,$$

giving a standard deviation of about 3.82.

Ask yourself What exactly are we testing, and in which direction?

The null hypothesis is that the tradition was followed: 96 seeds, so $X \sim \mathrm{B}\left(96, \frac{3}{16}\right)$. The natural suspicion behind *more* farmland trees than usual is that more than 96 seeds were released, so we use a one-tailed test and ask how likely 23 *or more* is under the null hypothesis.

Since $np = 18$ and $n(1-p) = 78$ are both large, we use the normal approximation $X \approx \mathrm{N}(18,\, 14.625)$ with a continuity correction:

$$\mathrm{P}(X \geq 23) \approx \mathrm{P}\left(Z \geq \frac{22.5 - 18}{3.824}\right) = \mathrm{P}(Z \geq 1.18) \approx 0.12.$$

(The exact binomial value is 0.121.) So under normal conditions, 23 or more farmland trees happens about one year in eight. That's well short of significance at the 5% level (or even 10%). The observation is quite consistent with the tradition having been followed, and the king is *not* justified in his suspicions on this evidence alone.

For comparison, even at the 5% level the count would have to be at least 25 (exact $\mathrm{P}(X \geq 25) \approx 0.049$) before the king would have reasonable grounds. If he also had the parkland count, he could test the total number of trees, $T \sim \mathrm{B}\left(96, \frac{5}{8}\right)$, which would be much better evidence about how many seeds were released.

Checks

- The four strip probabilities $\frac{7}{16}$, $\frac{5}{16}$, $\frac{3}{16}$, $\frac{1}{16}$ add to 1, and the density of Y has total area 1.
- A midpoint-rule integration gives $\mathrm{E}(Y) = 1.06667$, matching $\frac{16}{15}$. A simulation of 400 000 seeds (using $L = 4 - 4\sqrt{1-U}$ with U uniform on $[0,1]$) found 62.5% landing on land and a mean landing distance of 1.069 among them.
- Summing the binomial probabilities exactly gives $\mathrm{P}(X \geq 23) = 0.1213$, close to the normal estimate of 0.120.

Common slips

- Using f itself as the density of Y. It has area only $\frac{5}{8}$ over the land, so it must be divided by $\frac{5}{8}$.
- Computing $\mathrm{E}(L\,;\, L \in A) = \int_A y\,\mathrm{f}(y)\,\mathrm{d}y = \frac{2}{3}$ and forgetting to divide by $\mathrm{P}(L \in A)$.
- Testing $\mathrm{P}(X = 23)$ rather than the tail $\mathrm{P}(X \geq 23)$. Any single value of X is fairly unlikely.

Beyond STEP

Part (i) is an example of a *conditional distribution*: restricting a random variable to an event and renormalising. The same operation underlies Bayes' theorem for densities and the truncated distributions used throughout statistics. Part (ii) is a first taste of hypothesis testing and p-values, which first-year courses make precise with the Neyman–Pearson framework. Blitzstein and Hwang, *Introduction to Probability*, treats conditional distributions with plenty of worked stories like this one.

WORKED SOLUTION

STEP I 1989, Question 15

Probability and Statistics • Comparing Distributions

STEP I 1989, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q15

The key idea

At $12\,\mathrm{km\,h^{-1}}$ every kilometre takes exactly 5 minutes, so each journey time is a fixed number of minutes plus a simple multiple of one random variable: uniform, Poisson or binomial. Every question about time then becomes a question about that one variable. The middle part is Bayes' theorem with three equally likely routes.

Solution

Journey times

Let T_A, T_B, T_D be the journey times in minutes by the three routes. Since $12\,\mathrm{km\,h^{-1}}$ is 1 km every 5 minutes:

- Angle Avenue: $T_A = 25 + A$, with $A \sim \mathrm{U}(0, 10)$.
- Bend Boulevard: $T_B = 20 + 3B$, with $B \sim \mathrm{Po}(4)$.
- Detour Drive: let X be the number of detours. Assuming the five road works act independently, each with probability $\frac{4}{5}$, we have $X \sim \mathrm{B}(5, \frac{4}{5})$, the distance is $2 + X$ km and $T_D = 10 + 5X$.

At least 27 minutes

Angle Avenue. $T_A \geq 27 \iff A \geq 2$, so

$$\mathrm{P}(T_A \geq 27) = \frac{10 - 2}{10} = \frac{4}{5}.$$

Bend Boulevard. $T_B \geq 27 \iff 3B \geq 7 \iff B \geq 3$, since B is a whole number. So

$$\mathrm{P}(T_B \geq 27) = 1 - \mathrm{e}^{-4}\left(1 + 4 + \frac{4^2}{2!}\right) = 1 - 13\mathrm{e}^{-4} \approx 0.7619.$$

Detour Drive. $T_D \geq 27 \iff 5X \geq 17 \iff X \geq 4$. So

$$\mathrm{P}(T_D \geq 27) = 5\left(\frac{4}{5}\right)^4 \frac{1}{5} + \left(\frac{4}{5}\right)^5 = \left(\frac{4}{5}\right)^4 \left(1 + \frac{4}{5}\right) = \frac{256}{625} \cdot \frac{9}{5} = \frac{2304}{3125} = 0.73728.$$

Which route, given a long journey?

Let R be the route chosen, with $\mathrm{P}(R = A) = \mathrm{P}(R = B) = \mathrm{P}(R = D) = \frac{1}{3}$, and let L be the event that the journey lasts at least 27 minutes. By Bayes' theorem the factors of $\frac{1}{3}$ cancel:

$$\begin{aligned}\mathrm{P}(R = B \mid L) &= \frac{\mathrm{P}(L \mid B)}{\mathrm{P}(L \mid A) + \mathrm{P}(L \mid B) + \mathrm{P}(L \mid D)} = \frac{1 - 13\mathrm{e}^{-4}}{\frac{4}{5} + 1 - 13\mathrm{e}^{-4} + \frac{2304}{3125}} \\ &= \frac{1 - 13\mathrm{e}^{-4}}{\frac{7929}{3125} - 13\mathrm{e}^{-4}} \approx \frac{0.7619}{2.2992} \approx 0.331.\end{aligned}$$

This is barely above $\frac{1}{3}$, which makes sense: the three routes are almost equally likely to give a journey of 27 minutes or more.

Part (i)

Using $\mathrm{E}(A) = 5$, $\mathrm{E}(B) = 4$ and $\mathrm{E}(X) = 5 \times \frac{4}{5} = 4$, and linearity of expectation,

$$\mathrm{E}(T_A) = 25 + 5 = 30, \qquad \mathrm{E}(T_B) = 20 + 3 \times 4 = 32, \qquad \mathrm{E}(T_D) = 10 + 5 \times 4 = 30.$$

? **Ask yourself** Two routes tie. Is that a mistake?

No. Angle Avenue and Detour Drive both average exactly 30 minutes, and Bend Boulevard averages 32. So to make the average as small as possible I should use *either* Angle Avenue or Detour Drive: they are equally good, and both beat Bend Boulevard.

Part (ii)

Now compare $\mathrm{P}(T < 32)$ for each route.

- $T_A < 32 \iff A < 7$, so $\mathrm{P}(T_A < 32) = \frac{7}{10} = 0.7$.
- $T_B < 32 \iff 3B < 12 \iff B \leq 3$, so

$$\mathrm{P}(T_B < 32) = \mathrm{e}^{-4}\left(1 + 4 + 8 + \frac{32}{3}\right) = \frac{71}{3}\mathrm{e}^{-4} \approx 0.4335.$$

- $T_D < 32 \iff 5X < 22 \iff X \leq 4$, so

$$\mathrm{P}(T_D < 32) = 1 - \left(\frac{4}{5}\right)^5 = 1 - \frac{1024}{3125} = \frac{2101}{3125} \approx 0.6723.$$

Since $0.7 > 0.6723 > 0.4335$, I should always use **Angle Avenue**, which gets me there in under 32 minutes on 70% of days.

Checks

- A simulation of 400 000 journeys gave $\mathrm{P}(T \geq 27) \approx 0.801,\ 0.761,\ 0.740$, mean times $\approx 30.0,\ 32.0,\ 30.0$ minutes, $\mathrm{P}(T < 32) \approx 0.700,\ 0.435,\ 0.670$, and $\mathrm{P}(R = B \mid L) \approx 0.331$. All agree with the exact values.
- Boundary sense-check: on Detour Drive with 3 detours the time is 25 minutes and with 4 it is 30, so "at least 27" really does mean "at least 4 detours".

Common slips

- Treating the integer conditions loosely: $3B \geq 7$ means $B \geq 3$, not $B \geq 2$, and $3B < 12$ means $B \leq 3$, not $B \leq 4$.
- Forgetting the 2 km base distance on Detour Drive, or the five-minute-per-km riding time on the other routes.
- Claiming a single winner in (i). The two means are exactly equal.

Beyond STEP

Parts (i) and (ii) show that "best" depends on what we are optimising: the mean is just one summary of a distribution, and two routes with the same mean can have very different spreads and tail probabilities. In decision theory this is the choice of a *loss function*. Grimmett and Welsh, *Probability: An Introduction*, covers the distributions here and Bayes' theorem. Blitzstein and Hwang, *Introduction to Probability*, has many problems of this kind.

WORKED SOLUTION

STEP I 1989, Question 16

Probability and Statistics • Zero-Sum Games

STEP I 1989, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q16

This question has a full guided treatment in *Beyond STEP*, Section 12.14, *Mixed Strategies*: https://beyondstep.keithmcnulty.org/ch12/mixed-strategies.html

The key idea

Everything follows from one small table: B's winnings for each of the nine pairs of guesses. Once that is written down, each expected value is just a weighted sum of table entries. For the last part, grouping the sum by A's probabilities a_j turns "whatever A does" into a condition on three coefficients, and making all three non-negative is a small system of linear inequalities.

Solution

Write $w(i,j)$ for B's winnings (in pounds) in a round where A guesses i and B guesses j. Reading off the rules:

- if $|i-j|=2$, the one who said 1 pays the one who said 3 two pounds, so $w(1,3)=2$ and $w(3,1)=-2$;
- if $|i-j|=1$, the one who said the larger number pays one pound, so $w(2,1)=w(3,2)=1$ and $w(1,2)=w(2,3)=-1$;
- $w(i,i)=0$.

	B says 1	B says 2	B says 3
A says 1	0	-1	2
A says 2	1	0	-1
A says 3	-2	1	0

Notice that $w(j,i)=-w(i,j)$: the rules treat the players identically, so swapping the guesses swaps who pays.

Part (i)

If A always says N, B should look along row N of the table and always pick the largest entry.

- $N = 1$: B always says 3 and wins £2 every round.
- $N = 2$: B always says 1 and wins £1 every round.
- $N = 3$: B always says 2 and wins £1 every round.

In each case this choice is the unique best one, and every other reply wins nothing or loses.

Part (ii)

In a given round A's guess and B's guess are independent (A randomises on his own, and B decides secretly), so the probability that A says i and B says j is $\frac{1}{3}b_j$. Hence B's expected winnings are

$$\begin{aligned} \mathrm{E}(W) &= \sum_{j=1}^{3} b_j \cdot \tfrac{1}{3}\big(w(1,j) + w(2,j) + w(3,j)\big) \\ &= \tfrac{1}{3}\big(b_1(0+1-2) + b_2(-1+0+1) + b_3(2-1+0)\big) \\ &= \tfrac{1}{3}\,(b_3 - b_1). \end{aligned}$$

Since $b_3 \le 1$ and $b_1 \ge 0$, this is at most $\frac{1}{3}$, with equality exactly when $b_3 = 1$ (which forces $b_1 = b_2 = 0$). So B should *always guess* 3, and then expects to win £$\frac{1}{3}$ per round. Against a uniformly random A, the guess 3 wins £2, loses £1 or draws, each with probability $\frac{1}{3}$.

Part (iii)

With A guessing i with probability a_i and B guessing j with probability b_j, independently,

$$\mathrm{E}(W) = \sum_{i=1}^{3}\sum_{j=1}^{3} a_i b_j\, w(i,j).$$

Collecting the terms for each a_i (one row of the table at a time):

$$\mathrm{E}(W) = \underbrace{(2b_3 - b_2)}_{X}\, a_1 + \underbrace{(b_1 - b_3)}_{Y}\, a_2 + \underbrace{(b_2 - 2b_1)}_{Z}\, a_3.$$

We want $X \ge 0$, $Y \ge 0$ and $Z \ge 0$, that is

$$b_2 \le 2b_3, \qquad b_3 \le b_1, \qquad 2b_1 \le b_2.$$

Chaining these, $b_2 \le 2b_3 \le 2b_1 \le b_2$, so all of them must be equalities: $b_3 = b_1$ and $b_2 = 2b_1$. Together with $b_1 + b_2 + b_3 = 1$ this gives

$$b_1 = \tfrac{1}{4}, \qquad b_2 = \tfrac{1}{2}, \qquad b_3 = \tfrac{1}{4},$$

and then $X = Y = Z = 0$, which are certainly non-negative. So such a choice exists (and in fact it is the only one).

Ask yourself Could B do better and make X, Y and Z all strictly positive?

No. For any b_j,

$$b_1X + b_2Y + b_3Z = b_1(2b_3 - b_2) + b_2(b_1 - b_3) + b_3(b_2 - 2b_1) = 0,$$

which is just the statement that B playing against a copy of himself breaks even (the table is antisymmetric). A weighted sum of X, Y, Z with non-negative weights summing to 1 can't be zero if all three are positive. That is why the inequalities above could only be satisfied with equality.

Now the deduction. With $b = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ we have

$$\mathrm{E}(W) = 0 \cdot a_1 + 0 \cdot a_2 + 0 \cdot a_3 = 0$$

for *every* choice of a_1, a_2, a_3; more generally, whenever $X, Y, Z \geq 0$, $\mathrm{E}(W) = Xa_1 + Ya_2 + Za_3 \geq 0$ because each $a_i \geq 0$. B's winnings on different rounds are independent and identically distributed (both players' guesses are independent from round to round), each with mean at least 0. By the law of large numbers, B's average winnings per round over many rounds tend to this mean, so in the long run B loses no money, whatever A does. □

Checks

- Part (ii) is the special case $a_1 = a_2 = a_3 = \frac{1}{3}$ of part (iii): $\frac{1}{3}(X + Y + Z) = \frac{1}{3}(b_3 - b_1)$, as found.
- Part (i) is the case where a puts all its weight on N: for $N = 1$, $\mathrm{E}(W) = X = 2b_3 - b_2$, maximised by $b_3 = 1$, giving 2.
- A simulation of 2×10^5 rounds with B using $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ against A using $(0.6, 0.1, 0.3)$ gave average winnings of about -0.004 per round, consistent with 0.

Common slips

- Getting the sign convention backwards: in a difference of 1 it is the *larger* guess that pays, but in a difference of 2 it is the *smaller*. Build the table first and check it is antisymmetric.
- Concluding from part (ii) that B should mix his guesses. Against a fixed mixed strategy, the best reply is always a pure one (here, always 3); mixing only pays when B must guard against *every* strategy of A, as in part (iii).

Beyond STEP

This is a two-player zero-sum game, and part (iii) is a small case of von Neumann's minimax theorem: B has a mixed strategy guaranteeing him at least the value of the game whatever A does, and because the game is symmetric that value is 0. Finding such strategies is a linear programming problem, which connects probability with optimisation and economics. For the probability side, see Grimmett and Welsh, *Probability: An Introduction*; for games themselves, a good first read is Binmore, *Playing for Real*.

WORKED SOLUTION

STEP II 1989, Question 1

Pure • Trigonometric Solution of Cubics

STEP II 1989, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q1

The key idea

The identity says that $4z^3 - 3z$ is just $\cos 3\theta$ in disguise when $z = \cos\theta$. So if we can squeeze the cubic into the shape $4z^3 - 3z = k$ with $|k| \leq 1$, solving it is the same as solving $\cos 3\theta = k$, which has three easy solutions. The two substitutions do two separate jobs: the shift $y = x + a$ removes the squared term, and the scaling $z = by$ fixes the ratio of the remaining coefficients to $4 : 3$.

Solution

The identity

Using the addition formulae and $\sin^2\theta = 1 - \cos^2\theta$,

$$\begin{aligned}\cos 3\theta &= \cos 2\theta\cos\theta - \sin 2\theta\sin\theta \\ &= (2\cos^2\theta - 1)\cos\theta - 2\sin^2\theta\cos\theta \\ &= 2\cos^3\theta - \cos\theta - 2(1 - \cos^2\theta)\cos\theta \\ &= 4\cos^3\theta - 3\cos\theta,\end{aligned}$$

as required.

Reducing the cubic

With $y = x + a$ we have $x = y - a$, and the coefficient of y^2 in $24(y-a)^3 - 72(y-a)^2 + \cdots$ is $-72a - 72$. To remove it we need $a = -1$, that is $x = y + 1$. Then

$$\begin{aligned}&24(y+1)^3 - 72(y+1)^2 + 66(y+1) - 19 \\ &\quad = (24y^3 + 72y^2 + 72y + 24) - (72y^2 + 144y + 72) \\ &\qquad + (66y + 66) - 19 \\ &\qquad\qquad = 24y^3 - 6y - 1,\end{aligned}$$

so $(*)$ becomes

$$24y^3 - 6y = 1.$$

Now put $y = z/b$:

$$\frac{24}{b^3}z^3 - \frac{6}{b}z = 1.$$

Ask yourself How do we choose b so that the left-hand side is a multiple of $4z^3 - 3z$?

We need the coefficients of z^3 and z in the ratio $4 : -3$, so

$$\frac{24/b^3}{6/b} = \frac{4}{3} \quad \Longrightarrow \quad \frac{4}{b^2} = \frac{4}{3} \quad \Longrightarrow \quad b^2 = 3.$$

Take $b = \sqrt{3}$. Then the equation is $\dfrac{8}{\sqrt{3}}z^3 - \dfrac{6}{\sqrt{3}}z = 1$, and multiplying by $\frac{\sqrt{3}}{2}$,

$$4z^3 - 3z = \frac{\sqrt{3}}{2}.$$

So $a = -1$, $b = \sqrt{3}$ and $k = \frac{\sqrt{3}}{2}$, with $z = \sqrt{3}\,(x - 1)$. (Taking $b = -\sqrt{3}$ instead gives $k = -\frac{\sqrt{3}}{2}$ and the same roots in the end.)

The roots

Since $|k| \leq 1$ we can write $z = \cos\theta$, and by the identity the equation becomes

$$\cos 3\theta = \frac{\sqrt{3}}{2} = \cos 30^\circ.$$

So $3\theta = 30^\circ,\ 390^\circ,\ 750^\circ$, giving $\theta = 10^\circ,\ 130^\circ,\ 250^\circ$. These give three different values of $\cos\theta$, and a cubic has at most three roots, so they are all of them:

$$z = \cos 10^\circ, \quad \cos 130^\circ, \quad \cos 250^\circ.$$

Undoing the substitutions, $x = 1 + \dfrac{z}{\sqrt{3}}$, so

$$x_1 = 1 + \frac{\cos 10^\circ}{\sqrt{3}} = 1.5686\ldots \approx 1.57,$$
$$x_2 = 1 + \frac{\cos 130^\circ}{\sqrt{3}} = 0.6289\ldots \approx 0.629,$$
$$x_3 = 1 + \frac{\cos 250^\circ}{\sqrt{3}} = 0.8025\ldots \approx 0.803.$$

Not every cubic works

For a general monic cubic $x^3 + \alpha x^2 + \beta x + \gamma = 0$, the shift is forced: the coefficient of y^2 after $x = y - a$ is $\alpha - 3a$, so $a = \frac{\alpha}{3}$ and we reach $y^3 + py + q = 0$ with $p = \beta - \frac{\alpha^2}{3}$. Putting $y = z/b$ and multiplying by $4b^3$ gives

$$4z^3 + 4pb^2z + 4qb^3 = 0,$$

and to match $4z^3 - 3z$ we need $4pb^2 = -3$, that is $b^2 = -\dfrac{3}{4p}$. This has a real solution only if $p < 0$.

A counterexample is

$$x^3 + 3x - 4 = 0.$$

Here $\alpha = 0$, so $a = 0$ and $p = 3 > 0$; we would need $b^2 = -\frac{1}{4}$, which no real b satisfies. So the equation cannot be put in the form $4z^3 - 3z = k$ at all. (It does have a root: it factorises as $(x-1)(x^2 + x + 4) = 0$, and the quadratic has discriminant $-15 < 0$, so $x = 1$ is its only real root. The method fails, not the equation.)

Even when $p < 0$ the method can fail at the next step. For $x^3 - 3x - 4 = 0$, the substitution $x = 2z$ gives $4z^3 - 3z = 2$; but $|\cos 3\theta| \leq 1$, so there is no real θ with $\cos 3\theta = 2$.

Checks

- The three roots should add to $\frac{72}{24} = 3$ and multiply to $\frac{19}{24}$. Indeed the cos terms sum to zero ($\cos 10^\circ + \cos 130^\circ + \cos 250^\circ = 0$), and numerically $1.5686 \times 0.6289 \times 0.8025 = 0.7917 \approx \frac{19}{24}$.
- Substituting $x = 1.5686$ into $24x^3 - 72x^2 + 66x - 19$ gives zero to the accuracy of a calculator; the same holds for the other two roots.

Common slips

- Stopping at $\theta = 10^\circ$: the other two roots come from $3\theta = 390^\circ$ and 750°.
- Getting the sign of a wrong: $y = x + a$ with $x = y + 1$ means $a = -1$, not $+1$.
- For the counterexample, choosing a cubic with $p < 0$ and $|k| \leq 1$, which the method handles perfectly well. Check $b^2 > 0$ and $|k| \leq 1$ before claiming failure.

Beyond STEP

The method works exactly when the cubic has three real roots, the "casus irreducibilis" in which Cardano's formula forces you through complex cube roots even though every answer is real. When $|k| > 1$ the same idea works with cosh in place of cos, since $\cosh 3u = 4\cosh^3 u - 3\cosh u$; for $x^3 - 3x - 4 = 0$ this gives the real root $x = 2\cosh\left(\frac{1}{3}\operatorname{arcosh} 2\right) \approx 2.196$. The deeper question of which polynomial equations can be solved by radicals at all leads to Galois theory; Allenby, *Rings, Fields and Groups*, is a good place to start, and Needham, *Visual Complex Analysis*, shows the geometry of the cube roots.

WORKED SOLUTION

STEP II 1989, Question 2

Pure • Power Series Coefficients

STEP II 1989, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q2

The key idea

A trigonometric identity that links a function at x to the same function at $2x$ turns into a link between power-series coefficients. Replacing x by $2x$ multiplies the coefficient of x^n by 2^n, so once both sides are written as power series, we can compare coefficients and read the relation straight off. The useful trick is to multiply through by x first, so that every term is a function with a genuine power series at 0.

Solution

Throughout, x is small and non-zero, so every series below converges (all the functions involved are well behaved for $0 < |x| < \frac{\pi}{4}$, say). Two convergent power series that agree on an interval around 0 have equal coefficients, so we may compare coefficients of x^n.

The relation between a_{n-1} and b_n

First we check $(*)$:

$$\cot x - \tan x = \frac{\cos^2 x - \sin^2 x}{\sin x \cos x} = \frac{\cos 2x}{\frac{1}{2}\sin 2x} = 2\cot 2x.$$

Multiply by x:

$$x\cot x - x\tan x = (2x)\cot(2x).$$

Now write each term as a series.

- $x\cot x = 1 + \sum_{n\geq 1} b_n x^n$.
- $x\tan x = \sum_{m\geq 0} a_m x^{m+1} = \sum_{n\geq 1} a_{n-1}x^n$, putting $n = m+1$.
- Since $2x$ is small whenever x is, we may replace x by $2x$ in the series for $x\cot x$:

$$(2x)\cot(2x) = 1 + \sum_{n\geq 1} b_n (2x)^n = 1 + \sum_{n\geq 1} 2^n b_n x^n.$$

So

$$1+\sum_{n\geq 1}(b_n-a_{n-1})x^n = 1+\sum_{n\geq 1}2^n b_n x^n.$$

The constant terms agree, and comparing coefficients of x^n for $n \geq 1$ gives $b_n - a_{n-1} = 2^n b_n$, that is,

$$a_{n-1} = (1-2^n)\,b_n \qquad (n \geq 1),$$

as required.

Ask yourself Why multiply by x before expanding, rather than working with $\cot x$ itself?

$\cot x$ blows up at 0, so it has no power series there; the question only gives us a series for $x\cot x$. Multiplying the whole identity by x makes every term of the form "function with a power series", and the cotangent term on the right becomes $(2x)\cot(2x)$, which is the same function evaluated at $2x$. That is exactly the shape we need.

The coefficients c_n

The companion identity is the sum instead of the difference:

$$\cot x + \tan x = \frac{\cos^2 x + \sin^2 x}{\sin x \cos x} = \frac{1}{\frac{1}{2}\sin 2x} = 2\operatorname{cosec} 2x.$$

Multiplying by x again,

$$x\cot x + x\tan x = (2x)\operatorname{cosec}(2x).$$

Using the same expansions as before, and the series for $x\operatorname{cosec} x$ with x replaced by $2x$,

$$1+\sum_{n\geq 1}(b_n+a_{n-1})x^n = 1+\sum_{n\geq 1}2^n c_n x^n,$$

so for $n \geq 1$

$$2^n c_n = b_n + a_{n-1}.$$

Now use the first result to eliminate b_n. Since $n \geq 1$, $1-2^n \neq 0$, so $b_n = \dfrac{a_{n-1}}{1-2^n}$, and

$$2^n c_n = a_{n-1}\left(\frac{1}{1-2^n}+1\right) = a_{n-1}\frac{2-2^n}{1-2^n} = a_{n-1}\frac{2\,(2^{n-1}-1)}{2^n-1}.$$

Dividing by 2^n, and writing $2/2^n = 1/2^{n-1}$,

$$c_n = \frac{2^{n-1}-1}{2^n-1}\cdot\frac{1}{2^{n-1}}\,a_{n-1} \qquad (n \geq 1),$$

which is the required result.

Checks

From $\tan x = x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \cdots$ we have $a_1 = 1$, $a_3 = \frac{1}{3}$. The first result predicts $b_2 = a_1/(1-4) = -\frac{1}{3}$ and $b_4 = a_3/(1-16) = -\frac{1}{45}$, matching $x\cot x = 1 - \frac{1}{3}x^2 - \frac{1}{45}x^4 - \cdots$. The second predicts $c_2 = \frac{1}{3}\cdot\frac{1}{2}\cdot 1 = \frac{1}{6}$ and $c_4 = \frac{7}{15}\cdot\frac{1}{8}\cdot\frac{1}{3} = \frac{7}{360}$, matching $x\operatorname{cosec} x = 1 + \frac{1}{6}x^2 + \frac{7}{360}x^4 + \cdots$. Note also $n = 1$: $a_0 = 0$ and $b_1 = c_1 = 0$, consistent with both formulae. We checked both relations exactly (with fractions) for all $n \le 23$ by dividing the series for sin and cos, and checked the series against $\tan 0.3 \approx 0.309336$ and $0.3\operatorname{cosec} 0.3 \approx 1.015159$.

Common slips

- Forgetting the shift of index: $x\tan x$ has coefficient a_{n-1}, not a_n, at x^n.
- Substituting $2x$ for x in $x\cot x$ but writing $2x\cot x$ or $x\cot 2x$; the term in the identity must be exactly $(2x)\cot(2x)$.
- Not noting that $1 - 2^n \neq 0$ before dividing by it.

Beyond STEP

The coefficients here are, up to simple factors, the Bernoulli numbers: $x\cot x = \sum_{k\ge 0}(-1)^k \frac{2^{2k}B_{2k}}{(2k)!}x^{2k}$, and the question's relations give the tangent and cosecant series in terms of them. The fact that we may equate coefficients of two power series that agree near 0 (the identity theorem for power series) is proved in a first analysis course. For more, see Abbott, *Understanding Analysis*, on power series, and Needham, *Visual Complex Analysis*, where these expansions arise naturally from complex functions.

WORKED SOLUTION

STEP II 1989, Question 3

Pure • Complex Hyperbolic Mapping

STEP II 1989, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q3

The key idea

The right-hand side is $2\sinh(x+\mathrm{i}y)$, so the whole question is about the map $u+\mathrm{i}v=\sinh(x+\mathrm{i}y)$. Expanding sinh of a sum splits it into a real and an imaginary part, giving u and v explicitly. Fixing x or y then leaves a parametric curve, and the identities $\cos^2+\sin^2=1$ and $\cosh^2-\sinh^2=1$ eliminate the parameter.

Solution

Finding u and v

Dividing by 2, $u+\mathrm{i}v=\sinh(x+\mathrm{i}y)$. Using $\sinh(\mathrm{i}y)=\mathrm{i}\sin y$ and $\cosh(\mathrm{i}y)=\cos y$,

$$\begin{aligned}\sinh(x+\mathrm{i}y) &= \sinh x\cosh(\mathrm{i}y)+\cosh x\sinh(\mathrm{i}y)\\ &= \sinh x\cos y+\mathrm{i}\cosh x\sin y.\end{aligned}$$

(If you prefer, expand $\frac{1}{2}\left(\mathrm{e}^{x}(\cos y+\mathrm{i}\sin y)-\mathrm{e}^{-x}(\cos y-\mathrm{i}\sin y)\right)$ directly; it's the same thing.) Since x, y, u, v are all real, equating real and imaginary parts gives

$$u=\sinh x\cos y,\qquad v=\cosh x\sin y.$$

The line $x=a$

With $x=a$ fixed and y free, $u=\sinh a\cos y$ and $v=\cosh a\sin y$. Since $a>0$, both $\sinh a$ and $\cosh a$ are positive, so we may divide:

$$\left(\frac{u}{\sinh a}\right)^2+\left(\frac{v}{\cosh a}\right)^2=\cos^2 y+\sin^2 y=1.$$

Conversely every point of this ellipse has the form $(\sinh a\cos y,\cosh a\sin y)$ for some y, since that is the standard parametrisation of the ellipse. So the line $x=a$ corresponds to the whole ellipse (traced once for each interval of length 2π in y), with semi-axes $\sinh a$ along the u-axis and $\cosh a$ along the v-axis.

The line $y = b$

Now $y = b$ is fixed and x is free: $u = \sinh x \cos b$, $v = \cosh x \sin b$. We are told $0 < \sin b < 1$, so $\sin b \neq 0$ and $\cos^2 b = 1 - \sin^2 b > 0$, so $\cos b \neq 0$ too. Dividing,

$$\left(\frac{v}{\sin b}\right)^2 - \left(\frac{u}{\cos b}\right)^2 = \cosh^2 x - \sinh^2 x = 1.$$

This is a hyperbola with its axis along the v-axis and vertices at $(0, \pm \sin b)$.

Ask yourself Why only *one* branch, and is all of that branch covered?

Since $\cosh x > 0$ and $\sin b > 0$, we always have $v > 0$: only the upper branch is reached. And all of it is reached: given any point (u, v) on the upper branch, choose $x = \sinh^{-1}(u / \cos b)$, so that $\sinh x \cos b = u$; then $\cosh x \sin b$ is the positive root of the hyperbola's equation for that u, which is v. So the line $y = b$ corresponds to exactly one branch, the one with $v > 0$.

The intersection and the right angle

The point $x = a$, $y = b$ lies on both lines, so its image lies on both curves:

$$(u, v) = (\sinh a \cos b,\ \cosh a \sin b).$$

To find the directions of the curves there, differentiate each parametrisation with respect to its own parameter. Along the ellipse ($x = a$, parameter y), at $y = b$:

$$\mathbf{t}_1 = \left(\frac{\mathrm{d}u}{\mathrm{d}y}, \frac{\mathrm{d}v}{\mathrm{d}y}\right) = (-\sinh a \sin b,\ \cosh a \cos b).$$

Along the hyperbola ($y = b$, parameter x), at $x = a$:

$$\mathbf{t}_2 = \left(\frac{\mathrm{d}u}{\mathrm{d}x}, \frac{\mathrm{d}v}{\mathrm{d}x}\right) = (\cosh a \cos b,\ \sinh a \sin b).$$

Neither vector is zero, because each has a component $\cosh a \cos b \neq 0$, so both are genuine tangent vectors. Their scalar product is

$$\mathbf{t}_1 \cdot \mathbf{t}_2 = -\sinh a \sin b \cosh a \cos b + \cosh a \cos b \sinh a \sin b = 0,$$

so the ellipse and the hyperbola branch cross at right angles at this point, as required.

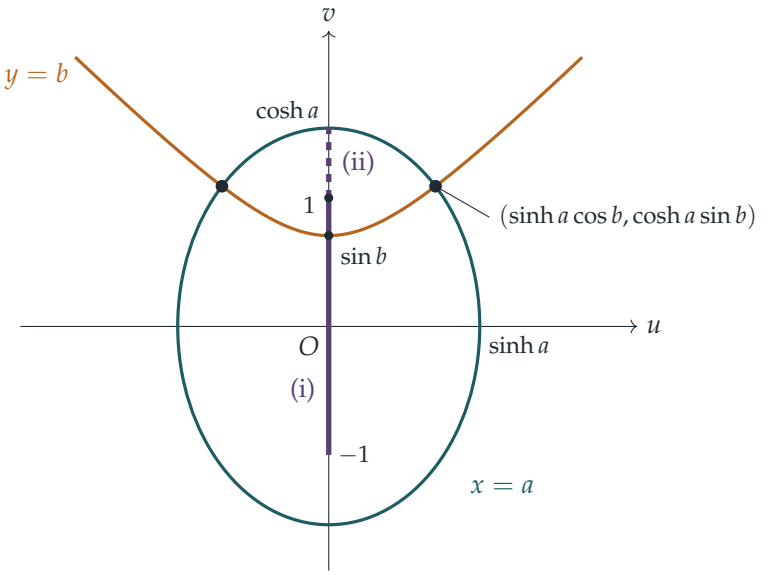
(By the symmetry $u \mapsto -u$ of both curves, they also meet at right angles at $(-\sinh a \cos b, \cosh a \sin b)$, the image of $x = -a$, $y = b$, or equally of $x = a$, $y = \pi - b$.)

The sketch

(i) $x = 0$ gives $u = 0$, $v = \sin y$: the segment of the v-axis from $(0, -1)$ to $(0, 1)$. Notice that it is the limit of the ellipses as $a \to 0$, since $\sinh a \to 0$ and $\cosh a \to 1$.

(ii) $y = \frac{1}{2}\pi$, $0 \leq x \leq a$ gives $u = 0$, $v = \cosh x$, which increases from 1 to $\cosh a$: the segment of the v-axis from $(0, 1)$ up to the top of the ellipse, $(0, \cosh a)$.

The hyperbola branch has its vertex at $(0, \sin b)$, which lies on segment (i). The sketch uses $a = 1$, $b = \frac{1}{4}\pi$.



Segment (i) is solid and segment (ii) dashed; together they run up the v-axis from -1 to the top of the ellipse. Taking $y = \frac{1}{2}\pi$ for all real x instead would give the ray $v \geq 1$ on the v-axis twice over, the degenerate case $\sin b = 1$ that the question excludes.

Checks

- Both conics have foci at $(0, \pm 1)$: for the ellipse $\cosh^2 a - \sinh^2 a = 1$, and for the hyperbola $\sin^2 b + \cos^2 b = 1$. Confocal ellipses and hyperbolas always cross at right angles, which agrees with our result.
- Numerically, with $a = 1$, $b = \frac{1}{4}\pi$: $\sinh(1 + \frac{1}{4}\pi\mathrm{i}) \approx 0.8310 + 1.0911\mathrm{i}$, matching $(\sinh 1 \cos \frac{1}{4}\pi,\ \cosh 1 \sin \frac{1}{4}\pi)$. We also checked the two curve equations and the zero scalar product at many random (a, b).

Common slips

- Writing $\sinh(\mathrm{i}y) = \sin y$ rather than $\mathrm{i} \sin y$, which swaps the roles of u and v.
- Saying "a hyperbola" without explaining why only the branch with $v > 0$ appears.
- Forgetting to check $\cos b \neq 0$ before dividing by it; that is exactly what the condition $\sin b < 1$ is for.

Beyond STEP

The map $z \mapsto \sinh z$ is *conformal*: wherever its derivative is non-zero it preserves angles, so the perpendicular grid of lines $x = \text{const}$ and $y = \text{const}$ must become a perpendicular grid of curves, here confocal ellipses and hyperbolas. This is why such "elliptic coordinates" appear when solving Laplace's equation around a flat plate or a slit. Needham's *Visual Complex Analysis* explains conformal maps beautifully, with many pictures of exactly this kind.

WORKED SOLUTION

STEP II 1989, Question 4

Pure • Rational Function Sketching

STEP II 1989, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q4

The key idea

A product or quotient of linear factors is easiest to differentiate logarithmically, because the log turns it into a sum. For the range, ask the question backwards: for which z does $\mathrm{f}(x) = z$ have a real solution? That equation is a quadratic in x, so the answer comes from a discriminant. The sketches then follow from the zeros, the poles, the behaviour for large $|x|$, and whether or not some values of z are missed.

One word on the printed text. The condition $a + d \neq c + d$ only says $a \neq c$, which we already know. The condition actually used is $a + b \neq c + d$, and we assume that throughout.

Solution

The derivative

Where $\mathrm{f}(x) \neq 0$, that is for $x \neq a, b$ (as well as $x \neq c, d$), we can take logarithms of moduli:

$$\ln|\mathrm{f}(x)| = \ln|x-a| + \ln|x-b| - \ln|x-c| - \ln|x-d|.$$

Differentiating, and using $\frac{\mathrm{d}}{\mathrm{d}x}\ln|x-k| = \frac{1}{x-k}$,

$$\frac{\mathrm{f}'(x)}{\mathrm{f}(x)} = \frac{1}{x-a} + \frac{1}{x-b} - \frac{1}{x-c} - \frac{1}{x-d}.$$

Now multiply by x. For $x \neq 0$ each term becomes

$$\frac{x}{x-k} = \frac{1}{1-k/x} = \left(1 - \frac{k}{x}\right)^{-1},$$

which gives the required identity

$$\frac{x\,\mathrm{f}'(x)}{\mathrm{f}(x)} = \left(1-\frac{a}{x}\right)^{-1} + \left(1-\frac{b}{x}\right)^{-1} - \left(1-\frac{c}{x}\right)^{-1} - \left(1-\frac{d}{x}\right)^{-1}.$$

Large $|x|$

When $|x|$ is much larger than $|k|$, the binomial expansion gives $(1-k/x)^{-1} = 1 + \frac{k}{x} + O\left(\frac{1}{x^2}\right)$. The four leading 1s cancel ($1+1-1-1=0$), leaving

$$\frac{x\,\mathrm{f}'(x)}{\mathrm{f}(x)} = \frac{a+b-c-d}{x} + O\left(\frac{1}{x^2}\right), \qquad \text{so} \qquad \mathrm{f}'(x) = \frac{\mathrm{f}(x)}{x^2}\left[(a+b-c-d) + O\left(\tfrac{1}{x}\right)\right].$$

Since $a+b-c-d \neq 0$, the bracket has the sign of $a+b-c-d$ once $|x|$ is large enough. Also $\mathrm{f}(x) \to 1$ as $|x| \to \infty$, so $\mathrm{f}(x) > 0$ there, and $x^2 > 0$. Hence $\mathrm{f}'(x)$ has the same sign as $a+b-c-d$ for $|x|$ large, at both ends of the real line.

It's also worth recording which side of the asymptote $y = 1$ the curve is on. Directly,

$$\mathrm{f}(x) - 1 = \frac{(c+d-a-b)x + ab - cd}{(x-c)(x-d)} \approx \frac{c+d-a-b}{x} \quad \text{for large } |x|,$$

and the numerator shows that $\mathrm{f}(x) = 1$ at exactly one point, $x_0 = \dfrac{cd-ab}{c+d-a-b}$.

The range condition: a method

Rearrange $\mathrm{f}(x) = z$ to $(x-a)(x-b) = z(x-c)(x-d)$, which is the quadratic

$$(1-z)x^2 - \big(a+b-(c+d)z\big)x + (ab - cdz) = 0.$$

- No root of this can be $x = c$ or $x = d$, because the left-hand side $(x-a)(x-b)$ is non-zero there. So real roots of the quadratic are exactly the real solutions of $\mathrm{f}(x) = z$.
- For $z \neq 1$ there is a real root if and only if the discriminant $\Delta(z) = \big(a+b-(c+d)z\big)^2 - 4(1-z)(ab-cdz)$ is ≥ 0. Multiply out, multiply by $(c-d)^2 > 0$, and collect powers of z. With $P = (a-c)(b-d)$ and $Q = (a-d)(b-c)$, the useful identity is $P - Q = (a-b)(c-d)$, so the constant term $(a-b)^2(c-d)^2$ equals $(P+Q)^2 - 4PQ$. The given inequality is then exactly $(c-d)^2\Delta(z) \geq 0$.
- For $z = 1$ the equation is linear with x-coefficient $c+d-a-b \neq 0$, so there is a root; and the inequality holds since both sides differ by $(c-d)^2(a+b-c-d)^2 > 0$.

The sketches

Write $G(z) = \big[(c-d)^2 z + P + Q\big]^2 - 4PQ$. As z varies, the square takes every value in $[0, \infty)$, so the smallest value of G is $-4PQ$. Therefore:

$$\text{the range of f is all of } \mathbb{R} \iff PQ \leq 0.$$

If $PQ > 0$, the values of z with $G(z) < 0$ form an open interval, and these are the missed values.

Ask yourself How do we know the sign of PQ without the actual numbers?

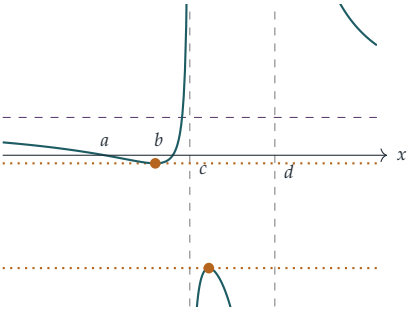
Regroup: $PQ = (a-c)(a-d)\cdot(b-c)(b-d)$. Since $a < c < d$, the first pair is a product of two negatives, so it's positive. So PQ has the sign of $(b-c)(b-d)$, which is negative exactly when b lies between the poles c and d. Everything depends on where b sits relative to $c < d$, and on the sign of $a+b-c-d$:

1. $a < b < c < d$: then $a+b < c+d$ automatically.
2. $a < c < b < d$: again $a+b < c+d$, since $a < c$ and $b < d$.
3. $a < c < d < b$ with $a+b < c+d$.
4. $a < c < d < b$ with $a+b > c+d$.

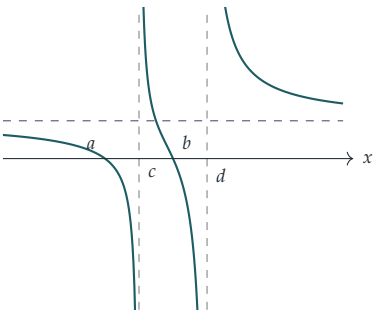
So the range is all of $\mathbb{R}$ only in case 2. In cases 1, 3 and 4 an interval of values is missed.

One more tool helps with the turning points. If f has a stationary point at x_1 with $f(x_1) = z_1$, then x_1 is a repeated root of $(x-a)(x-b) - z_1(x-c)(x-d) = 0$. (For $z_1 = 1$ this is linear, so it can't happen.) A repeated root means $G(z_1) = 0$. So the stationary values are exactly the ends of the missing interval. In case 2, $G > 0$ always, so there are no stationary points at all.

The sign of f just either side of each pole comes from counting negative factors. In each picture the vertical dashed lines are the poles $x = c$, $x = d$, the horizontal dashed line is the asymptote $y = 1$, and the dotted amber lines mark the ends of the missing interval, which are the values at the stationary points (amber dots). The curves are drawn for particular values (given under each picture), but the shapes are the same for every choice in that case.



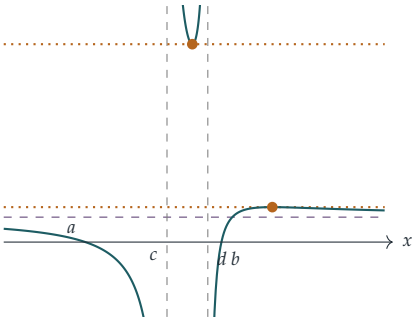
Case 1: $a, b, c, d = -2, 0, \frac{1}{2}, 3$



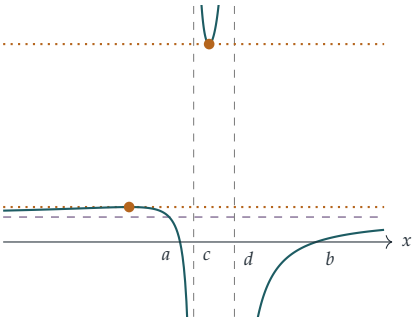
Case 2: $a, b, c, d = -1, 1, 0, 2$

Case 1 ($a < b < c < d$). Here f < 1 far to the left and f > 1 far to the right. On $x < c$ the curve comes in just below 1, falls to 0 at a, dips to a negative minimum between a and b, and rises through b to $+\infty$ at c. Between the poles both denominator factors have opposite signs and the numerator is positive, so f $\to -\infty$ at both ends, with a negative local maximum in between. To the right of d the curve falls from $+\infty$ to 1. The local maximum on (c, d) lies below the local minimum on (a, b), and the values strictly between them are missed. *Range not all of* $\mathbb{R}$.

Case 2 ($a < c < b < d$). There are no stationary points. The curve falls from just below 1 through a to $-\infty$ at c; then from $+\infty$ through b to $-\infty$ at d; then from $+\infty$ down to 1. The middle branch alone takes every real value. *Range is all of* $\mathbb{R}$.



Case 3: $a, b, c, d = -3, 2, 0, \frac{3}{2}$



Case 4: $a, b, c, d = -\frac{1}{2}, \frac{9}{2}, 0, \frac{3}{2}$

Case 3 ($a < c < d < b$, $a + b < c + d$). On $x < c$ the curve falls from just below 1 through a to $-\infty$. Between the poles the numerator is negative, so $f \to +\infty$ at both ends, with a local minimum m. To the right of d it rises from $-\infty$ through b, crosses $y = 1$ at x_0, reaches a local maximum M just above 1, and then falls back to 1 from above. The values in (M, m) are missed, where $1 < M < m$. *Range not all of* $\mathbb{R}$.

Case 4 ($a < c < d < b$, $a + b > c + d$). This is case 3 with the outer behaviour reversed. Far to the left f is just above 1 and increasing. It reaches a local maximum $M > 1$, crosses $y = 1$ at $x_0 < a$, falls through a to $-\infty$ at c. The middle branch is a U with minimum m, as before. To the right of d the curve rises from $-\infty$ through b towards 1 from below. Again (M, m) is missed. *Range not all of* $\mathbb{R}$.

Checks

- The identity: a finite-difference estimate of $xf'(x)/f(x)$ at random a, b, c, d, x agrees with the right-hand side to about 10^{-9}.
- Case 1 with $a, b, c, d = -2, 0, \frac{1}{2}, 3$: here $P = \frac{15}{2}$, $Q = \frac{5}{2}$, $(c-d)^2 = \frac{25}{4}$, and $G(z) = 0$ gives $z = -\frac{8}{5} \pm \frac{4}{5}\sqrt{3}$, about -0.2144 and -2.9856. These are exactly the stationary values found by solving $f'(x) = 0$ (at $x \approx -0.515$ and $x \approx 1.060$). Sampling f at 4×10^5 points never lands strictly between them.
- Cases 3 and 4 (the drawn values): the missing interval is $(1.4007, 7.9327)$, again matching the stationary values. In case 2, $f'(x) = 0$ has no real solutions.

Common slips

- Forgetting that $f(x) = z$ is only a quadratic when $z \neq 1$: the value $z = 1$ needs its own sentence.

- Assuming the curve approaches $y = 1$ from the same side at both ends. It doesn't: $\mathrm{f} - 1 \approx (c + d - a - b)/x$ changes sign with x.

Beyond STEP

Turning "what is the range of f?" into "for which z does $\mathrm{f}(x) = z$ have a solution?" is the idea of a preimage, and the observation that stationary values are where two solutions merge is the start of the theory of critical values. A rational function of degree 2 like this one is a two-to-one map, and its behaviour is far cleaner over the complex numbers, where every value is taken (counting multiplicity) exactly twice. Needham's *Visual Complex Analysis* develops this geometric viewpoint beautifully.

WORKED SOLUTION

STEP II 1989, Question 5

Pure • Polar Curves and Reflection

STEP II 1989, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q5

The key idea

Part (i) is just $\mathrm{d}y/\mathrm{d}x = (\mathrm{d}y/\mathrm{d}\theta)/(\mathrm{d}x/\mathrm{d}\theta)$ with $x = r\cos\theta$, $y = r\sin\theta$. For the mirror, polar coordinates centred on O are the natural choice, because the reflected ray always runs along the radius back to the pole. The law of reflection then tells us the direction of the normal at every point in terms of θ alone, and part (i) turns that into a first-order differential equation for $r(\theta)$ that separates.

Solution

Part (i)

A point with polar coordinates (r, θ) has Cartesian coordinates $x = r\cos\theta$, $y = r\sin\theta$. Along a curve $r = r(\theta)$, both are functions of θ, and by the product rule

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta, \qquad \frac{\mathrm{d}y}{\mathrm{d}\theta} = \frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta.$$

By the chain rule (for a parametrised curve), wherever $\mathrm{d}x/\mathrm{d}\theta \neq 0$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}\theta}{\mathrm{d}x/\mathrm{d}\theta} = \frac{\dfrac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta}{\dfrac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta} = \frac{\dfrac{\mathrm{d}r}{\mathrm{d}\theta}\tan\theta + r}{\dfrac{\mathrm{d}r}{\mathrm{d}\theta} - r\tan\theta},$$

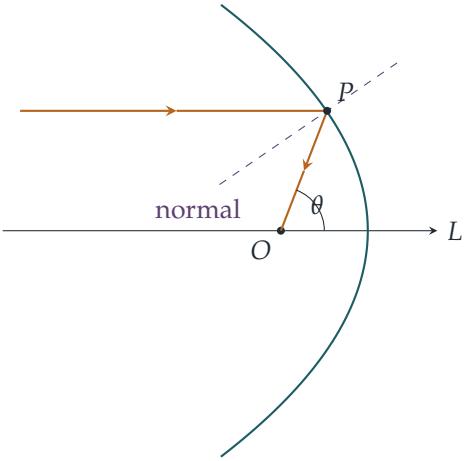
dividing top and bottom by $\cos\theta$, which is allowed provided $\cos\theta \neq 0$. This is the required formula.

Part (ii)

Reducing to a plane curve. Take any plane Π containing L, and a point P of the mirror in Π. The ray through P parallel to L lies in Π, and so does O; hence the reflected ray PO lies in Π too. The normal at P bisects the angle between two lines in Π, so it also lies in Π. The curve C in which the mirror meets Π has its tangent at P in both Π and

the mirror's tangent plane, so it is perpendicular to the normal. So, within the plane Π, the normal to C at P bisects the angle between the incident and reflected rays: we have a two-dimensional problem.

Coordinates. In Π use polar coordinates with pole O and initial line along L, oriented so that the incoming rays travel in the direction $\theta = 0$ (as in the figure, they come from the side of L beyond the mirror's opening). Let $P = (r, \theta)$ be on C.



The direction of the normal. From P, the incident ray goes back in the direction $(-1, 0)$ and the reflected ray goes towards O in the direction $(-\cos\theta, -\sin\theta)$. Both are unit vectors, so their sum points along the bisector of the angle between them, which is the normal:

$$(-1 - \cos\theta,\ -\sin\theta) = -2\cos\tfrac{\theta}{2}\left(\cos\tfrac{\theta}{2},\ \sin\tfrac{\theta}{2}\right),$$

using $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$ and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$. So the normal at P makes angle $\frac{\theta}{2}$ with L, the tangent is perpendicular to it, and the gradient of C at P is

$$m = -\cot\tfrac{\theta}{2}.$$

The differential equation. Write $r' = \mathrm{d}r/\mathrm{d}\theta$, $t = \tan\theta$ and $\tau = \tan\frac{\theta}{2}$, so that $t = \dfrac{2\tau}{1-\tau^2}$ and $m = -\dfrac{1}{\tau}$. By part (i),

$$\frac{r't + r}{r' - rt} = m \quad\Longrightarrow\quad r'(t-m) = -r(1+mt) \quad\Longrightarrow\quad \frac{r'}{r} = -\frac{1+mt}{t-m}.$$

Now

$$1 + mt = 1 - \frac{2}{1-\tau^2} = -\frac{1+\tau^2}{1-\tau^2}, \qquad t - m = \frac{2\tau}{1-\tau^2} + \frac{1}{\tau} = \frac{1+\tau^2}{\tau(1-\tau^2)},$$

and so

$$\frac{1}{r}\frac{\mathrm{d}r}{\mathrm{d}\theta} = \tan\tfrac{\theta}{2}.$$

Ask yourself Part (i) breaks down where $\cos\theta = 0$. Does that leave a gap?

No. The equation $r'/r = \tan\frac{\theta}{2}$ involves only continuous functions of θ on $-\pi < \theta < \pi$. It holds at every point except possibly the isolated values $\theta = \pm\frac{\pi}{2}$ (and any isolated points where the tangent is vertical), so by continuity of r' it holds there too. (Alternatively, the angle ψ between the radius and the tangent satisfies $\tan\psi = r/r'$, and the geometry gives $\psi = \frac{\pi}{2} - \frac{\theta}{2}$ directly, with no exceptional points.)

Solving. Integrating, with $\cos\frac{\theta}{2} > 0$ for $-\pi < \theta < \pi$,

$$\ln r = \int \tan\tfrac{\theta}{2}\,\mathrm{d}\theta = -2\ln\cos\tfrac{\theta}{2} + \text{const}, \qquad \text{so} \qquad r = \frac{k}{\cos^2\frac{\theta}{2}} = \frac{2k}{1+\cos\theta}$$

for a constant $k > 0$ (as $r > 0$). Put $\ell = 2k$. Then $r + r\cos\theta = \ell$, that is $r = \ell - x$. Squaring, using $r^2 = x^2 + y^2$,

$$x^2 + y^2 = (\ell - x)^2 \quad\Longleftrightarrow\quad y^2 = \ell^2 - 2\ell x.$$

This is a parabola with axis L and focus O (it is $Y^2 = 4aX$ with $X = \frac{\ell}{2} - x$, $a = \frac{\ell}{2}$, so the focus is at $x = \frac{\ell}{2} - a = 0$). Since Π was any plane containing L, the mirror meets every such plane in a parabola, as required. (In fact the constant ℓ is the same in each plane only if the mirror is a single paraboloid of revolution; the question does not need this.)

Checks

- At $\theta = 0$ we get $r = \frac{\ell}{2}$: the vertex is halfway between O and the directrix $x = \ell$, as it should be for a parabola with focus O.
- For $\ell = 2$ and $\theta = \frac{\pi}{2}$ the point is $P = (0, 2)$, the gradient of $y^2 = 4 - 4x$ there is $-1 = -\cot\frac{\pi}{4}$, and reflecting the direction $(1, 0)$ in this tangent gives $(0, -1)$, straight down to O.
- Numerically: the gradient formula in (i) agreed with finite differences on 2000 random curves (to 10^{-8}); a Runge–Kutta solution of $r' = r\tan\frac{\theta}{2}$ from $r(0) = 1.35$ gave $r(2.5) = 13.5776380$, matching $2.7/(1+\cos 2.5)$; and reflecting a ray parallel to L off $y^2 = \ell^2 - 2\ell x$ sent it through O at 1000 random points.

Common slips

- Taking the wrong bisector: the sum of the two unit vectors pointing *away* from P along the rays gives the normal; their difference gives the tangent.
- Forgetting to explain why the problem is planar: the plane through L contains the incident ray, O, hence the reflected ray and the normal.
- Writing $\ln r = -2\ln\cos\frac{\theta}{2}$ without a constant, which loses the whole family of parabolas.

Beyond STEP

The equation $r(1+\cos\theta) = \ell$ is the polar form of a conic with focus at the pole and eccentricity 1; replacing 1 by e gives ellipses and hyperbolas, which is exactly how planetary orbits appear when one solves Newton's equations in polar coordinates. The reflection argument is also the reason satellite dishes and car headlamps are paraboloids. Kleppner and Kolenkow, *An Introduction to Mechanics*, derives the conic orbits this way.

WORKED SOLUTION

STEP II 1989, Question 6

Pure • Integrals of Inverse Functions

STEP II 1989, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q6

The key idea

The right-hand side contains an integral of g with respect to y, so the natural move is to rewrite the left-hand integral in terms of $y = \mathrm{f}(x)$, that is, to substitute $x = \mathrm{g}(y)$. The new integrand contains $y\,\mathrm{g}'(y)$, which is crying out for integration by parts, and the boundary terms produce the rectangles. The second formula works in exactly the same way, with x replaced by $x^2/2$ and areas replaced by volumes of revolution about the y-axis.

Solution

The first formula

Since $\mathrm{f}'(x) > 0$ on $[a, b]$, f is strictly increasing there, so it is a bijection from $[a, b]$ onto $[\alpha, \beta]$ and the inverse $\mathrm{g}\colon [\alpha, \beta] \to [a, b]$ exists. Moreover g is differentiable, with

$$\mathrm{g}'(y) = \frac{1}{\mathrm{f}'(\mathrm{g}(y))},$$

which makes sense because f' never vanishes.

Substitute $x = \mathrm{g}(y)$, so that $\mathrm{f}(x) = y$ and $\mathrm{d}x = \mathrm{g}'(y)\,\mathrm{d}y$. When $x = a$ we have $y = \mathrm{f}(a) = \alpha$, and when $x = b$ we have $y = \beta$. Hence

$$\int_a^b \mathrm{f}(x)\,\mathrm{d}x = \int_\alpha^\beta y\,\mathrm{g}'(y)\,\mathrm{d}y.$$

Now integrate by parts, with $u = y$ and $\mathrm{d}v = \mathrm{g}'(y)\,\mathrm{d}y$, so $v = \mathrm{g}(y)$:

$$\begin{aligned}\int_\alpha^\beta y\,\mathrm{g}'(y)\,\mathrm{d}y &= \Big[y\,\mathrm{g}(y)\Big]_\alpha^\beta - \int_\alpha^\beta \mathrm{g}(y)\,\mathrm{d}y \\ &= \beta\,\mathrm{g}(\beta) - \alpha\,\mathrm{g}(\alpha) - \int_\alpha^\beta \mathrm{g}(y)\,\mathrm{d}y \\ &= b\beta - a\alpha - \int_\alpha^\beta \mathrm{g}(y)\,\mathrm{d}y,\end{aligned}$$

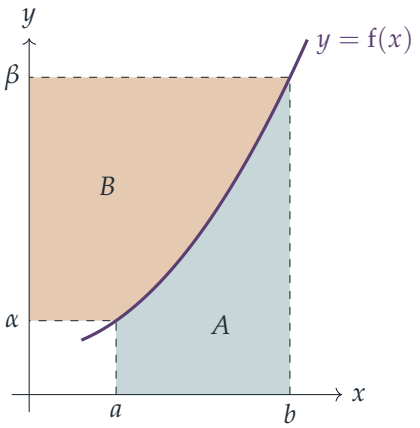
since $\mathrm{g}(\beta) = b$ and $\mathrm{g}(\alpha) = a$. This is the required formula.

Geometric meaning

Take $a > 0$ and $\alpha > 0$. As f is increasing, $0 < a < b$ and $0 < \alpha < \beta$, so the arc of $y = \mathrm{f}(x)$ from (a, α) to (b, β) lies in the first quadrant. Write the formula as

$$\underbrace{\int_a^b \mathrm{f}(x)\,\mathrm{d}x}_{\text{area } A} + \underbrace{\int_\alpha^\beta \mathrm{g}(y)\,\mathrm{d}y}_{\text{area } B} = b\beta - a\alpha.$$

Here A is the area between the curve and the x-axis for $a \le x \le b$, and B is the area between the curve and the y-axis for $\alpha \le y \le \beta$ (the curve is $x = \mathrm{g}(y)$ seen sideways). Together they fill the rectangle $[0, b] \times [0, \beta]$ with the smaller rectangle $[0, a] \times [0, \alpha]$ removed, and that L-shaped region has area $b\beta - a\alpha$.



The second formula

Use the same substitution $x = \mathrm{g}(y)$, $\mathrm{d}x = \mathrm{g}'(y)\,\mathrm{d}y$:

$$2\pi \int_a^b x\,\mathrm{f}(x)\,\mathrm{d}x = 2\pi \int_\alpha^\beta y\,\mathrm{g}(y)\,\mathrm{g}'(y)\,\mathrm{d}y.$$

? **Ask yourself** Which factor should we integrate this time?

Notice that $\mathrm{g}(y)\,\mathrm{g}'(y)$ is the derivative of $\frac{1}{2}[\mathrm{g}(y)]^2$ by the chain rule. So integrate by parts with $u = y$ and $v = \frac{1}{2}[\mathrm{g}(y)]^2$:

$$\begin{aligned} 2\pi \int_\alpha^\beta y\,\mathrm{g}(y)\,\mathrm{g}'(y)\,\mathrm{d}y &= 2\pi \left(\left[\tfrac{1}{2} y\,[\mathrm{g}(y)]^2 \right]_\alpha^\beta - \int_\alpha^\beta \tfrac{1}{2}[\mathrm{g}(y)]^2\,\mathrm{d}y \right) \\ &= \pi(\beta b^2 - \alpha a^2) - \pi \int_\alpha^\beta [\mathrm{g}(y)]^2\,\mathrm{d}y, \end{aligned}$$

again using $\mathrm{g}(\beta) = b$ and $\mathrm{g}(\alpha) = a$. This is the second formula.

Geometric meaning of the second formula

Again take $a > 0$, $\alpha > 0$, and rotate the picture above through a full turn about the y-axis.

- Region A sweeps out a solid whose volume, by the method of cylindrical shells, is $2\pi \int_a^b x\,\mathrm{f}(x)\,\mathrm{d}x$: the thin strip at distance x from the axis, of height $\mathrm{f}(x)$ and width $\mathrm{d}x$, becomes a shell of volume about $2\pi x\,\mathrm{f}(x)\,\mathrm{d}x$.
- Region B sweeps out a solid whose volume, by discs, is $\pi \int_\alpha^\beta [\mathrm{g}(y)]^2\,\mathrm{d}y$: the strip at height y becomes a disc of radius $\mathrm{g}(y)$.
- Together the two solids fill a cylinder of radius b and height β with a coaxial cylinder of radius a and height α removed from its base, which has volume $\pi b^2\beta - \pi a^2\alpha$.

So the formula says shell volume plus disc volume equals the difference of the two cylinders. The conditions $a > 0$ and $\alpha > 0$ ensure that both regions lie on one side of the axis and above the x-axis, so that the integrals really are volumes and the two solids do not overlap.

Checks

Try $\mathrm{f}(x) = x^2$ on $[1, 2]$, so $\alpha = 1$, $\beta = 4$ and $\mathrm{g}(y) = \sqrt{y}$. The first formula gives $\frac{7}{3}$ on the left and $8 - 1 - \frac{2}{3}(8 - 1) = \frac{7}{3}$ on the right. The second gives $2\pi \cdot \frac{15}{4} = \frac{15\pi}{2}$ on the left and $\pi(16 - 1) - \pi \cdot \frac{15}{2} = \frac{15\pi}{2}$ on the right. We also checked both formulae numerically by Simpson's rule, with g found by bisection, for several functions including $x^3 + x$ on $[\frac{1}{2}, 2]$, where both sides of the first formula equal 5.859375.

Common slips

- Forgetting to change the limits: after substituting $x = \mathrm{g}(y)$ they become α and β, not a and b.
- Not saying why g exists and is differentiable. The condition $\mathrm{f}' > 0$ is there exactly for this.
- Describing region B as lying under a curve. It lies between the curve and the y-axis.

Beyond STEP

The first formula is the integral version of the fact that the graphs of f and f^{-1} are reflections in $y = x$. It is the idea behind Young's inequality: for increasing f with $\mathrm{f}(0) = 0$ and $a, b \geq 0$, $ab \leq \int_0^a \mathrm{f} + \int_0^b \mathrm{f}^{-1}$, and behind the Legendre transform used in convex analysis and in Hamiltonian mechanics. It is also how integrals of inverse functions such as arcsin and ln can be found without any clever substitution. For a careful treatment of substitution and inverse functions, see Abbott, *Understanding*

Analysis.

WORKED SOLUTION

STEP II 1989, Question 7

Pure • Second-Order Differential Equations

STEP II 1989, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q7

The key idea

The terms $x\,y''$ and $-b\,y'$ look awkward together, but a change of independent variable $t = x^{\alpha}$ produces a first-derivative term from y'' that can be made to cancel $-b\,y'$ exactly. Choosing α to do this leaves simple harmonic motion in t. After that, the behaviour as $x \to 0$ comes down to how the factor x^b in $\mathrm{d}y/\mathrm{d}x$ behaves, and that depends on the sign of b.

Solution

Changing the variable

Put $t = x^{\alpha}$, with α to be chosen, and regard y as a function of t. By the chain rule, with $\dot{y} = \mathrm{d}y/\mathrm{d}t$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \alpha x^{\alpha-1}\dot{y},$$
$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \alpha(\alpha-1)x^{\alpha-2}\dot{y} + \alpha^2 x^{2\alpha-2}\ddot{y},$$

where the second line uses the product rule and $\frac{\mathrm{d}}{\mathrm{d}x}\dot{y} = \alpha x^{\alpha-1}\ddot{y}$. Substituting,

$$x\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - b\frac{\mathrm{d}y}{\mathrm{d}x} = \alpha^2 x^{2\alpha-1}\ddot{y} + \alpha(\alpha-1-b)\,x^{\alpha-1}\dot{y}.$$

The $\dot{y}$ term disappears if we choose $\alpha = b+1$, which is non-zero because $b > -1$. Then $2\alpha - 1 = 2b+1$, so the equation becomes

$$(b+1)^2 x^{2b+1}\ddot{y} + x^{2b+1}y = 0 \quad\Longrightarrow\quad \ddot{y} + \frac{y}{(b+1)^2} = 0,$$

dividing by $x^{2b+1} \neq 0$ (we have $x > 0$). This is simple harmonic motion in t with angular frequency $1/(b+1)$, so

$$\boxed{y = A\cos\left(\frac{x^{b+1}}{b+1}\right) + B\sin\left(\frac{x^{b+1}}{b+1}\right)}$$

with A, B arbitrary constants. Two constants for a second-order equation: this is the general solution.

Behaviour as $x \to 0$

Write $u = x^{b+1}/(b+1)$. Since $b+1 > 0$, $u \to 0^+$ as $x \to 0^+$, so

$$y \to A\cos 0 + B\sin 0 = A.$$

Hence $y \to 1$ forces $A = 1$. Differentiating, using $\mathrm{d}u/\mathrm{d}x = x^b$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = x^b\big(-\sin u + B\cos u\big) \qquad (A = 1).$$

Case $b > 0$. Now $x^b \to 0$, while the bracket tends to B, which is finite. So $\mathrm{d}y/\mathrm{d}x \to 0$ *whatever the value of B*. Every member of the family

$$y = \cos\left(\frac{x^{b+1}}{b+1}\right) + B\sin\left(\frac{x^{b+1}}{b+1}\right), \qquad B \in \mathbb{R},$$

satisfies both conditions. Solutions exist, but they are not unique, as required.

Ask yourself For an ordinary second-order equation, a value of y and a value of y' fix the solution. Why does that fail here?

The uniqueness theorem needs the equation in the form $y'' = f(x, y, y')$ with f well behaved. Dividing by x gives $y'' = (b/x)\,y' - x^{2b}y$, and the coefficient b/x blows up at $x = 0$. The point $x = 0$ is *singular*, so the usual guarantee doesn't apply there.

Case $b = 0$. Now $x^b = 1$ and $\mathrm{d}y/\mathrm{d}x \to B$. The condition forces $B = 0$, giving the unique solution $y = \cos x$. (This is just $y'' + y = 0$.)

Case $-1 < b < 0$. Now $x^b \to \infty$. If $B \neq 0$ then $Bx^b\cos u \to \pm\infty$, while the other term is small: since $\sin u \leq u$,

$$\left|x^b \sin u\right| \leq x^b \cdot \frac{x^{b+1}}{b+1} = \frac{x^{2b+1}}{b+1}.$$

More precisely $\sin u/u \to 1$, so $x^b \sin u \sim x^{2b+1}/(b+1)$. So we need $B = 0$, and then

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -x^b \sin u \sim -\frac{x^{2b+1}}{b+1}.$$

This tends to 0 if and only if $2b+1 > 0$, that is $b > -\frac{1}{2}$. At $b = -\frac{1}{2}$ it tends to -2, and for $b < -\frac{1}{2}$ it is unbounded. So for $-\frac{1}{2} < b < 0$ there is exactly one solution, $y = \cos\big(x^{b+1}/(b+1)\big)$, and for $-1 < b \leq -\frac{1}{2}$ there is none.

Conclusion

The conditions determine a unique solution precisely when

$$-\tfrac{1}{2} < b \leq 0,$$

and that solution is $y = \cos\left(x^{b+1}/(b+1)\right)$.



Checks

- $b = 0$ gives $y'' + y = 0$, whose solutions $A\cos x + B\sin x$ match the formula.
- Numerically, the general solution satisfies the equation to within rounding (finite differences at random x, b, A, B). With $b = -\frac{1}{4}$, $B = 0$ we found $y' \approx -4.2 \times 10^{-5}$ at $x = 10^{-9}$, tending to 0. With $b = -\frac{1}{2}$ the value stays at -2, and with $b = 1$, $B = 5$ it is 5×10^{-9} at $x = 10^{-9}$.

Common slips

- Stopping at "$b \leq 0$ gives uniqueness": for $-1 < b \leq -\frac{1}{2}$ even the cosine solution has $y' \not\to 0$, so there is no solution at all.
- Forgetting the product rule when differentiating $\alpha x^{\alpha-1}\dot{y}$ with respect to x. The extra term is exactly what cancels $-b\,y'$.

Beyond STEP

The point $x = 0$ is a *regular singular point* of this equation. Near such points, the Frobenius method looks for solutions $x^r \sum c_n x^n$, and the indicial equation here has roots $r = 0$ and $r = b + 1$. Which conditions at the singular point pin down a solution depends on these exponents, exactly as we saw. Boyce and DiPrima, *Elementary Differential Equations*, covers series solutions and regular singular points.

WORKED SOLUTION

STEP II 1989, Question 8

Pure • Groups of Transformations

STEP II 1989, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q8

The key idea

Everything rests on one identity: $\Omega^3 = -1$, so Ω satisfies the quadratic in the first line, and that lets us reduce any power of Ω to something linear in Ω. Once we know how R and T shuffle the four points, the counting is about the four points, not the formulae: the group must carry z_0 to every one of the four points, and it already contains three different maps that leave z_0 alone. For the last part, ask what kind of permutation R and T are.

Solution

The identity

Since $\Omega = e^{i\pi/3}$, we have $\Omega^3 = e^{i\pi} = -1$, so

$$0 = \Omega^3 + 1 = (\Omega + 1)(\Omega^2 - \Omega + 1).$$

As $\Omega \neq -1$, the second factor vanishes: $\Omega^2 - \Omega + 1 = 0$, as required.

We will use this in the forms

$$\Omega^2 = \Omega - 1, \qquad \Omega^3 = -1, \qquad \Omega^4 = -\Omega.$$

Note also $2\Omega - 1 = 2\cos\frac{\pi}{3} - 1 + 2i\sin\frac{\pi}{3} = i\sqrt{3} \neq 0$.

R and T permute the four points

For R, multiplying by $\Omega^2 = e^{2i\pi/3}$ is a rotation by 120° about 0:

$$R(0) = 0, \qquad R(1) = \Omega^2, \qquad R(\Omega^2) = \Omega^4 = -\Omega, \qquad R(-\Omega) = -\Omega^3 = 1.$$

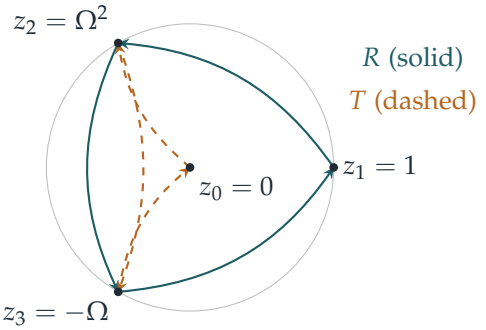
So R fixes z_0 and sends $z_1 \to z_2 \to z_3 \to z_1$.

For T, we substitute each point and simplify with $\Omega^2 = \Omega - 1$:

$$\begin{aligned} T(0) &= \frac{\Omega^2}{1} = z_2, \\ T(1) &= \frac{\Omega + \Omega^2}{2\Omega^2 + 1} = \frac{2\Omega - 1}{2\Omega - 1} = 1 = z_1, \\ T(\Omega^2) &= \frac{\Omega^3 + \Omega^2}{2\Omega^4 + 1} = \frac{\Omega - 2}{1 - 2\Omega} = -\Omega = z_3, \\ T(-\Omega) &= \frac{-\Omega^2 + \Omega^2}{-2\Omega^3 + 1} = \frac{0}{3} = 0 = z_0. \end{aligned}$$

In the third line, $\Omega^3 + \Omega^2 = -1 + (\Omega - 1) = \Omega - 2$, and the last step holds because $-\Omega(1 - 2\Omega) = -\Omega + 2\Omega^2 = -\Omega + 2\Omega - 2 = \Omega - 2$. All the denominators are non-zero (they are 1, $\mathrm{i}\sqrt{3}$, $-\mathrm{i}\sqrt{3}$ and 3).

So T fixes z_1 and sends $z_0 \to z_2 \to z_3 \to z_0$. Both R and T permute the four points; each is a 3-cycle.



The group has order at least 12

Let G be the smallest group of transformations containing R and T: everything we can build from R, T and their inverses by composition. Every element of G maps the set $\{z_0, z_1, z_2, z_3\}$ onto itself, because R and T do (and so do their inverses, being inverses of bijections of this finite set).

Three maps fixing z_0. The set $H = \{I, R, R^2\}$ lies in G, every element fixes z_0, and the three are different transformations (rotations by 0°, 120°, 240°).

Maps sending z_0 everywhere. Using the cycles found above,

$$g_0 = I, \qquad g_1 = R^2T, \qquad g_2 = T, \qquad g_3 = RT$$

satisfy $g_k(z_0) = z_k$: for example $RT(z_0) = R(z_2) = z_3$ and $R^2T(z_0) = R^2(z_2) = R(z_3) = z_1$.

Now consider the twelve products $g_k h$ with $k \in \{0, 1, 2, 3\}$ and $h \in H$. Since $h(z_0) = z_0$, we get $g_k h(z_0) = z_k$.

Ask yourself Could two of these twelve products be the same transformation?

No. If $g_k h = g_l h'$, apply both sides to z_0: we get $z_k = z_l$, so $k = l$. Then composing on the left with g_k^{-1} gives $h = h'$. So the twelve products are distinct elements of G, and $|G| \geq 12$. (In the language of group theory: the orbit of z_0 has size 4 and its stabiliser has size at least 3, so by the orbit–stabiliser theorem $|G| \geq 4 \times 3$.)

The final part

Yes: exactly half of the 24 permutations of the four points can never be produced.

Ask yourself What do R and T have in common as permutations?

Each is a 3-cycle, and a 3-cycle is an *even* permutation: for instance $(z_1\, z_2\, z_3) = (z_1\, z_3)(z_1\, z_2)$, a product of two transpositions. Their inverses are also 3-cycles. A composition of even permutations is even, so every permutation of the points produced by R and T is even. There are only 12 even permutations of four objects, so the other 12, the odd ones, are never produced. For example, swapping z_0 and z_1 while fixing z_2 and z_3 is a single transposition, hence odd, and cannot be obtained.

In fact, combined with the previous part this pins G down exactly. R and T are Möbius maps (T has $ad - bc = \Omega - 2\Omega^4 = 3\Omega \neq 0$), and a Möbius map fixing three points is the identity, so each element of G is determined by how it permutes the four points. Hence $|G| \leq 12$, and with the previous part, $|G| = 12$: G acts on the points as the full group of even permutations, A_4.

Checks

- Numerically, $|\Omega^2 - \Omega + 1| \approx 2 \times 10^{-16}$, and evaluating R and T at the four points gives exactly the two 3-cycles above.
- $R^3 = I$ obviously; and $T^3 = I$ too, as a 3-cycle should (checked at a random point, to 10^{-16}).
- Generating all compositions of R and T by computer gives exactly 12 distinct maps, all inducing even permutations, with z_0 moved to all four points and 3 maps fixing it.

Common slips

- Arguing "G contains elements of order 3, so $|G| \geq 12$". Elements of order 3 alone only give 3 dividing $|G|$; it is the transitivity on four points that supplies the factor 4.
- Leaving powers of Ω unsimplified, so that $T(\Omega^2) = -\Omega$ looks false. Reduce everything to the form $a + b\Omega$ first.

Beyond STEP

The count $|G| = (\text{orbit size}) \times (\text{stabiliser size})$ is the orbit–stabiliser theorem, and the even permutations form the alternating group A_4, also the rotation group of a regular tetrahedron. There is a lovely reason this happens here: Möbius maps preserve the cross-ratio of four points, and for these four points the cross-ratio is $e^{-i\pi/3}$, the special "equianharmonic" value for which exactly the even permutations preserve it. For groups and group actions, see Allenby, *Rings, Fields and Groups*; for Möbius maps and the Riemann sphere, Needham, *Visual Complex Analysis*.

WORKED SOLUTION

STEP II 1989, Question 9

Pure • Matrix Exponential Systems

STEP II 1989, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q9

The key idea

Everything hinges on one small calculation: $\mathbf{A}^2$ turns out to be the identity. Then every even power of $\mathbf{A}$ is $\mathbf{I}$ and every odd power is $\mathbf{A}$, so the series splits into the even and odd parts of the exponential series, which are $\cosh t$ and $\sinh t$. For the differential equation we copy the scalar integrating-factor method, with the matrix $\mathbf{F}$ playing the role of $\mathrm{e}^{\int a\,\mathrm{d}t}$.

Solution

The matrix F

Multiplying out,

$$\mathbf{A}^2 = \begin{pmatrix} -3 & -1 \\ 8 & 3 \end{pmatrix}\begin{pmatrix} -3 & -1 \\ 8 & 3 \end{pmatrix} = \begin{pmatrix} 9-8 & 3-3 \\ -24+24 & -8+9 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I}.$$

Hence $\mathbf{A}^{2k} = (\mathbf{A}^2)^k = \mathbf{I}$ and $\mathbf{A}^{2k+1} = \mathbf{A}^{2k}\mathbf{A} = \mathbf{A}$ for every $k \geq 0$. Writing $\mathbf{I} = \frac{t^0}{0!}\mathbf{A}^0$ and splitting the series into even and odd n (each entry of the matrix series is a rearrangement of absolutely convergent scalar series, so this is allowed),

$$\begin{aligned} \mathbf{F} &= \sum_{k=0}^{\infty} \frac{t^{2k}}{(2k)!}\mathbf{A}^{2k} + \sum_{k=0}^{\infty} \frac{t^{2k+1}}{(2k+1)!}\mathbf{A}^{2k+1} \\ &= \mathbf{I}\sum_{k=0}^{\infty} \frac{t^{2k}}{(2k)!} + \mathbf{A}\sum_{k=0}^{\infty} \frac{t^{2k+1}}{(2k+1)!} = \mathbf{I}\cosh t + \mathbf{A}\sinh t, \end{aligned}$$

using the Maclaurin series $\cosh t = \sum t^{2k}/(2k)!$ and $\sinh t = \sum t^{2k+1}/(2k+1)!$. This is the required form.

The inverse

Let $\mathbf{G} = \mathbf{I}\cosh t - \mathbf{A}\sinh t$. Since $\mathbf{I}$ and $\mathbf{A}$ commute, we can multiply out as with ordinary numbers:

$$\mathbf{FG} = \mathbf{I}\cosh^2 t - \mathbf{A}^2\sinh^2 t + (\mathbf{A} - \mathbf{A})\cosh t \sinh t = \mathbf{I}(\cosh^2 t - \sinh^2 t) = \mathbf{I},$$

using $\mathbf{A}^2 = \mathbf{I}$ and $\cosh^2 t - \sinh^2 t = 1$. The same computation gives $\mathbf{GF} = \mathbf{I}$, so $\mathbf{F}^{-1} = \mathbf{I}\cosh t - \mathbf{A}\sinh t$.

The derivative

Differentiating entry by entry (the matrices $\mathbf{I}$ and $\mathbf{A}$ are constant),

$$\frac{\mathrm{d}\mathbf{F}}{\mathrm{d}t} = \mathbf{I}\sinh t + \mathbf{A}\cosh t, \qquad \text{while} \qquad \mathbf{FA} = \mathbf{A}\cosh t + \mathbf{A}^2\sinh t = \mathbf{A}\cosh t + \mathbf{I}\sinh t.$$

These agree, so $\dfrac{\mathrm{d}\mathbf{F}}{\mathrm{d}t} = \mathbf{FA}$. Note also that $\mathbf{FA} = \mathbf{AF}$, since $\mathbf{F}$ is built from $\mathbf{I}$ and $\mathbf{A}$.

The differential equation

In the scalar equation $r' + ar = 0$ we multiply by e^{at}, whose derivative is $\mathrm{e}^{at}a$, so that the left-hand side becomes an exact derivative. The matrix $\mathbf{F}$ has exactly this property, $\mathbf{F}' = \mathbf{FA}$, so we multiply the equation on the *left* by $\mathbf{F}$:

$$\mathbf{F}\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} + \mathbf{FA}\mathbf{r} = \mathbf{0}.$$

By the product rule (which holds for matrix products, as long as we keep the order of the factors),

$$\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{Fr}) = \frac{\mathrm{d}\mathbf{F}}{\mathrm{d}t}\mathbf{r} + \mathbf{F}\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \mathbf{FA}\mathbf{r} + \mathbf{F}\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t},$$

which is precisely the left-hand side. So the equation says

$$\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{Fr}) = \mathbf{0},$$

and $\mathbf{Fr}$ is a constant vector $\mathbf{c}$.

Ask yourself Why multiply on the left rather than the right, and how do we find $\mathbf{c}$?

The product $\mathbf{rF}$ doesn't even make sense ($\mathbf{r}$ is a 2×1 column), and the product rule produced $\mathbf{FAr}$, which matches the term $\mathbf{F}(\mathbf{Ar})$ only when $\mathbf{F}$ is on the left. For the constant, put $t = 0$: there $\cosh 0 = 1$ and $\sinh 0 = 0$, so $\mathbf{F} = \mathbf{I}$ and

$$\mathbf{c} = \mathbf{I}\,\mathbf{r}(0) = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}.$$

Now multiply $\mathbf{Fr} = \mathbf{c}$ on the left by $\mathbf{F}^{-1}$, which we found above:

$$\mathbf{r} = \mathbf{F}^{-1}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \cosh t\begin{pmatrix}\alpha\\ \beta\end{pmatrix} - \sinh t\begin{pmatrix}-3 & -1\\ 8 & 3\end{pmatrix}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \cosh t\begin{pmatrix}\alpha\\ \beta\end{pmatrix} - \sinh t\begin{pmatrix}-3\alpha - \beta\\ 8\alpha + 3\beta\end{pmatrix}.$$

Reading off the components,

$$x(t) = \alpha\cosh t + (3\alpha+\beta)\sinh t,$$
$$y(t) = \beta\cosh t - (8\alpha+3\beta)\sinh t,$$

as required.

Checks

- At $t = 0$ the formulae give $x = \alpha$, $y = \beta$, as they should.
- Direct substitution: $x' = \alpha\sinh t + (3\alpha+\beta)\cosh t$, and $-(-3x-y) = 3x+y = (3\alpha+\beta)\cosh t + \alpha\sinh t$ (using $9\alpha+3\beta-8\alpha-3\beta = \alpha$). So $x' = -(\text{first row of } \mathbf{Ar})$, and similarly for y.
- Numerically, summing the series to 60 terms agrees with $\mathbf{I}\cosh t + \mathbf{A}\sinh t$, and a Runge–Kutta solution of the system with $\alpha = 1$, $\beta = 0$ gives $x(2) \approx 14.6428$, matching $\cosh 2 + 3\sinh 2$.

Common slips

- Multiplying by $\mathbf{F}$ on the right, or writing $\mathbf{AF}$ and $\mathbf{FA}$ interchangeably without noting why that's allowed here (they commute because $\mathbf{F}$ is a combination of $\mathbf{I}$ and $\mathbf{A}$).
- Forgetting the $\mathbf{I}$ term (the $n = 0$ term) when splitting the series, which loses the $\cosh t$ constant term 1.
- Sign errors when multiplying $-\mathbf{A}$ into $(\alpha, \beta)^{\mathrm{T}}$.

Beyond STEP

The matrix $\mathbf{F}$ is the matrix exponential $\mathrm{e}^{t\mathbf{A}}$, and the method here shows that $\mathbf{r}' = \mathbf{Mr}$ always has solution $\mathbf{r} = \mathrm{e}^{t\mathbf{M}}\mathbf{r}(0)$; here $\mathbf{M} = -\mathbf{A}$, which is why the answer involves $\mathbf{F}^{-1} = \mathrm{e}^{-t\mathbf{A}}$. In general one computes $\mathrm{e}^{t\mathbf{M}}$ by diagonalising $\mathbf{M}$: our $\mathbf{A}$ has eigenvalues ± 1 (that is what $\mathbf{A}^2 = \mathbf{I}$ says), which is where $\mathrm{e}^{\pm t}$, and so cosh and sinh, come from. Strang's *Introduction to Linear Algebra* has a good chapter on matrix exponentials and systems of differential equations, and Boyce and DiPrima's *Elementary Differential Equations* treats linear systems in detail.

WORKED SOLUTION

STEP II 1989, Question 10

Pure • Lines and Planes

STEP II 1989, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q10

The key idea

Two lines in space can fail to meet in a single point in three ways: they can be the same line, they can be parallel, or they can be *skew*, missing each other in different planes. So "exactly one point" needs two conditions: the directions are not parallel, and the lines lie in a common plane. The cross product handles both at once. For the planes, the line of intersection runs along the cross product of the normals, and the final part is the first part again in disguise.

Solution

First part: when two lines meet in exactly one point

The lines meet where $\mathbf{a} + \lambda\mathbf{u} = \mathbf{b} + \mu\mathbf{v}$, that is, where

$$\lambda\mathbf{u} - \mu\mathbf{v} = \mathbf{b} - \mathbf{a}.$$

For these to be genuine lines we need $\mathbf{u} \neq \mathbf{0}$ and $\mathbf{v} \neq \mathbf{0}$. Then:

- If $\mathbf{u}$ and $\mathbf{v}$ are parallel ($\mathbf{u} \times \mathbf{v} = \mathbf{0}$), the lines are either parallel and distinct (no common point) or identical (infinitely many). Either way, not exactly one.
- If $\mathbf{u} \times \mathbf{v} \neq \mathbf{0}$, then $\mathbf{u}$ and $\mathbf{v}$ span a plane, and the equation has a solution exactly when $\mathbf{b} - \mathbf{a}$ lies in that plane, i.e. is perpendicular to $\mathbf{u} \times \mathbf{v}$. When a solution exists it is unique, because $\mathbf{u}$ and $\mathbf{v}$ are independent.

So the conditions are

$$\mathbf{u} \neq \mathbf{0}, \quad \mathbf{v} \neq \mathbf{0}, \quad \mathbf{u} \times \mathbf{v} \neq \mathbf{0}, \quad (\mathbf{b} - \mathbf{a}) \cdot (\mathbf{u} \times \mathbf{v}) = 0.$$

(The first two are implied by the third, but it's worth saying them.) If the last condition fails, the lines are skew.

Second part: the two planes

The normals are $\mathbf{n}_1 = (1, 1, b)$ and $\mathbf{n}_2 = (b, b, 1)$, and

$$\mathbf{n}_1 \times \mathbf{n}_2 = \left(1 - b^2,\ b^2 - 1,\ 0\right) = (1 - b^2)(1, -1, 0).$$

Two planes fail to meet in a line exactly when their normals are parallel, which happens here when $1 - b^2 = 0$, so $b = 1$ or $b = -1$.

- $b = 1$: both equations become $x + y + z = 3$.
- $b = -1$: the equations are $x + y - z = 1$ and $-x - y + z = -1$, the same plane again.

So in both cases the two planes coincide, rather than being parallel and separate.

Now take $b \neq \pm 1$. The line runs along $\mathbf{n}_1 \times \mathbf{n}_2$, so we can use $\mathbf{u} = (1, -1, 0)$. To find points on it, notice that x and y appear only through $s = x + y$:

$$s + bz = b + 2, \qquad bs + z = 2b + 1.$$

Subtracting b times the first from the second gives $(1 - b^2)z = 1 - b^2$, so $z = 1$ (we may divide since $b^2 \neq 1$), and then $s = 2$. The line of intersection is therefore $x + y = 2$, $z = 1$, independent of b.

We need a point $\mathbf{a}$ on this line with $\mathbf{a} \cdot \mathbf{u} = 0$, i.e. $x - y = 0$. With $x + y = 2$ this gives $x = y = 1$, so

$$\mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \qquad b \neq \pm 1.$$

The point $\mathbf{a}$ is the foot of the perpendicular from the origin to the line.

Final part

The points whose coordinates satisfy both equations $(*)$ form the line above when $b \neq \pm 1$, and a whole plane when $b = \pm 1$. We treat the cases separately.

? **Ask yourself** What set of points satisfies both equations when $b = \pm 1$?

It is a whole plane, not the empty set, because we found that the planes coincide. That changes the answer completely in those cases.

Case $b \neq \pm 1$. We need the given line to meet the line $\mathbf{r} = (1, 1, 1) + \lambda(1, -1, 0)$ in exactly one point. Apply the first part with $\mathbf{a} = (1, 1, 1)$, $\mathbf{u} = (1, -1, 0)$, $\mathbf{b} = (0, 0, c)$, $\mathbf{v} = (1, d, 0)$. Then

$$\mathbf{u} \times \mathbf{v} = (0,\ 0,\ d + 1), \qquad (\mathbf{b} - \mathbf{a}) \cdot (\mathbf{u} \times \mathbf{v}) = (-1, -1, c - 1) \cdot (0, 0, d + 1) = (c - 1)(d + 1).$$

We need $\mathbf{u} \times \mathbf{v} \neq \mathbf{0}$, so $d \neq -1$, and then $(c - 1)(d + 1) = 0$ forces $c = 1$. So the conditions are $c = 1$ and $d \neq -1$. (Directly: the given line has $z = c$, so it meets $z = 1$ only if $c = 1$, and then $x + y = \mu(1 + d) = 2$ has the single solution $\mu = 2/(1 + d)$ when $d \neq -1$.)

Case $b = \pm 1$. Now we need the given line to meet a plane in exactly one point. A point $(\mu, d\mu, c)$ lies on $x + y + z = 3$ (for $b = 1$) when $\mu(1 + d) = 3 - c$, and on $x + y - z = 1$ (for $b = -1$) when $\mu(1 + d) = 1 + c$. Each equation has exactly one solution for μ if and only if $1 + d \neq 0$; if $d = -1$ the line is parallel to the plane and there are no solutions or infinitely many. Here c can be anything.

Answer. There is exactly one such point if and only if

$$d \neq -1 \quad \text{and} \quad \big(c = 1 \text{ or } b = \pm 1\big),$$

which can be written compactly as $d \neq -1$ and $(b^2 - 1)(c - 1) = 0$.

Checks

- Substitute $(1 + \lambda, 1 - \lambda, 1)$ into $(*)$: $2 + b = b + 2$ and $2b + 1 = 2b + 1$, for every b and λ.
- With $b = 2$, $c = 1$, $d = 3$ the point is $\mu = \frac{1}{2}$, i.e. $(\frac{1}{2}, \frac{3}{2}, 1)$: $\frac{1}{2} + \frac{3}{2} + 2 = 4 = b + 2$ and $1 + 3 + 1 = 5 = 2b + 1$.
- A computer tested every b, c, d in $\{-3, \ldots, 3, \frac{1}{2}, -\frac{5}{3}\}$ in exact arithmetic (729 cases), solving for μ directly, and the count was one exactly when the stated conditions hold.

Common slips

- Saying that two lines meet in one point just because they aren't parallel. In three dimensions they can be skew.
- Assuming that when $b = \pm 1$ the planes are parallel and separate, and concluding there are no points at all. Check the right-hand sides: the planes coincide.
- Dividing by $1 - b^2$ without saying why it is non-zero.

Beyond STEP

This is a small system of linear equations in disguise: two equations in three unknowns have a line of solutions when the coefficient matrix has rank 2, and a plane (or nothing) when it has rank 1. Adding the line's parametrisation turns it into an overdetermined system in μ, and "exactly one solution" is a statement about rank and consistency. Strang's *Introduction to Linear Algebra* builds all of this from the row picture (planes meeting) and the column picture (combinations of vectors) that appear here.

WORKED SOLUTION

STEP II 1989, Question 11

Mechanics • Connected Particles and Impulse

STEP II 1989, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q11

The key idea

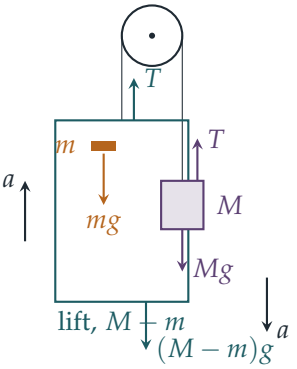
The lift and counterweight balance at the start, so the lift's mass M must include the tile. Once the tile lets go, the lift is lighter than the counterweight and accelerates *upwards* while the tile falls freely, so the floor rises to meet the tile. For the collision, treat lift, cable and counterweight as one system moving along the cable. Its "momentum along the cable" changes only through gravity, and the impulse from the collision is internal to it.

Solution

Setting up the model

We model the lift, counterweight and tile as particles, and the cable as light and inextensible over a smooth, light pulley. The guides are smooth, so they exert no vertical force. Before the tile falls, the system is at rest with masses M and M on the two sides. So the lift's mass M includes the tile. (If it did not, the lift side would weigh $(M+m)g$ against Mg and could not be at rest.)

After detachment, the lift has mass $M-m$, the counterweight has mass M, and the tile is a free particle.



The time to reach the floor

Let the lift accelerate upwards at a, so the counterweight accelerates downwards at a. Let T be the cable tension. Newton's second law gives

$$\begin{aligned} \text{lift (up):} \quad & T - (M-m)g = (M-m)a, \\ \text{counterweight (down):} \quad & Mg - T = Ma. \end{aligned}$$

Adding the two equations eliminates T:

$$a = \frac{mg}{2M-m}, \qquad T = M(g-a) = \frac{2M(M-m)g}{2M-m} > 0,$$

so the cable stays taut.

Ask yourself Why can't we just say the tile falls a distance h under gravity?

Because the floor is moving too. The tile accelerates down at g and the floor accelerates up at a, both from rest. So the gap closes with constant relative acceleration

$$g + a = g + \frac{mg}{2M-m} = \frac{2Mg}{2M-m}.$$

The gap starts at h, so the tile lands when $\frac{1}{2}(g+a)t^2 = h$, which gives

$$t^2 = \frac{2h}{g+a} = \frac{(2M-m)h}{Mg}, \qquad t = \sqrt{\frac{(2M-m)h}{Mg}}.$$

This is *not* the expression printed in the question, which has $2(M-m)$ in place of $2M-m$. The two agree only when $m = 0$, and our value is confirmed below by stepping the motion numerically. The printed form would need the lift to accelerate at $mg/(M-m)$. That is what you get if you forget that the counterweight must be accelerated too, and treat the tension as staying at Mg. We believe the printed formula is a misprint or slip in the original question. None of the rest of the question depends on it.

The collision

Just before impact, the lift moves up and the counterweight moves down, both with speed $u = at$. The tile moves down with speed $v = gt$. Then

$$(2M - m)u = (2M - m) \cdot \frac{mg}{2M - m}\, t = mgt = mv.$$

Now look at the collision. Take "positive" as lift-up / counterweight-down, the direction in which the cable carries them together. The impact is very short, so the finite forces (the weights) give negligible impulse. The only impulses are the tile–floor impulse, which is internal to the lift-and-tile, and an impulsive tension J in the cable. Let w be the common speed afterwards (lift-and-tile, mass M, up; counterweight down). The collision is perfectly inelastic, so the tile ends up moving with the floor. Then

$$\begin{aligned} \text{lift and tile:} \quad & Mw - \big[(M - m)u - mv\big] = J, \\ \text{counterweight:} \quad & Mw - Mu = -J. \end{aligned}$$

Adding these,

$$2Mw = (M - m)u + Mu - mv = (2M - m)u - mv = 0,$$

so $w = 0$: the lift is brought to rest. (Also $J = Mu > 0$, so the cable really is jerked taut rather than going slack, which is consistent with our assumption.) After the impact the masses are M and M again, in balance, so everything stays at rest.

Ask yourself Why is the total "momentum along the cable" zero just before impact?

Before impact, the only external forces on the three bodies that act along the motion are gravity, with net effect $Mg - (M - m)g - mg = 0$ in the positive sense defined above. (The pulley's reaction does no work and the cable is internal.) So that momentum stays at its initial value, 0, the whole time. The collision cannot change it either.

Energy lost

The system starts at rest and ends at rest, so the mechanical energy lost equals the decrease in gravitational potential energy between the start and the end. During the fall, let the lift rise s_1 (so the counterweight falls s_1) and let the tile fall s_2. The tile closes the gap of h, so $s_1 + s_2 = h$. The change in potential energy is

$$(M - m)g\, s_1 - Mg\, s_1 - mg\, s_2 = -mg(s_1 + s_2) = -mgh.$$

Nothing moves after the collision, so the energy lost is mgh, as required. All of the kinetic energy present just before impact is destroyed in the collision. Note that this argument does not use the value of t at all.

Checks

- If $m \to 0$, then $a \to 0$ and $t \to \sqrt{2h/g}$, which is free fall onto a stationary floor. Both formulas pass this test, so it doesn't tell them apart.
- With $M = 5$, $m = 1$, $h = 1$ and $g = 9.8$, stepping the motion numerically gives $t \approx 0.4286\,\text{s}$, matching $\sqrt{9/49} \approx 0.4286$. The printed formula gives $\sqrt{8/49} \approx 0.404\,\text{s}$.
- In the same simulation, the kinetic energy just before impact equals mgh to five significant figures, and the post-collision speed is zero to rounding error.

Common slips

- Taking the lift to have mass M *plus* the tile: then the system could not start at rest.
- Forgetting the counterweight's inertia, so that the lift's acceleration comes out as $mg/(M - m)$.
- Using conservation of linear momentum for the lift and tile alone in the collision, which ignores the impulsive tension in the cable.

Beyond STEP

The "momentum along the cable" used here is a *generalised momentum*: for a system with one degree of freedom, the coordinate is the length of cable that has passed over the pulley. Lagrangian mechanics makes this systematic. A conserved generalised momentum goes with every coordinate on which the forces have no net effect, and impulsive constraints are handled the same way. Kleppner and Kolenkow, *An Introduction to Mechanics*, gives a careful treatment of systems joined by strings and pulleys, and of impulse and energy in collisions.

WORKED SOLUTION

STEP II 1989, Question 12

Mechanics • Rigid Body Equilibrium

STEP II 1989, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q12

The key idea

Only three forces act on the lamina: its weight and the two peg reactions. Smooth pegs push at right angles to the sides they touch, and in a rectangle that means each reaction runs *parallel to the other side*. A body held in equilibrium by three non-parallel forces needs their lines of action to meet at one point. So the point where the two reactions cross must lie on the vertical through the centre. Both results follow from that one fact together with the condition that the pegs are level.

Solution

Setting up

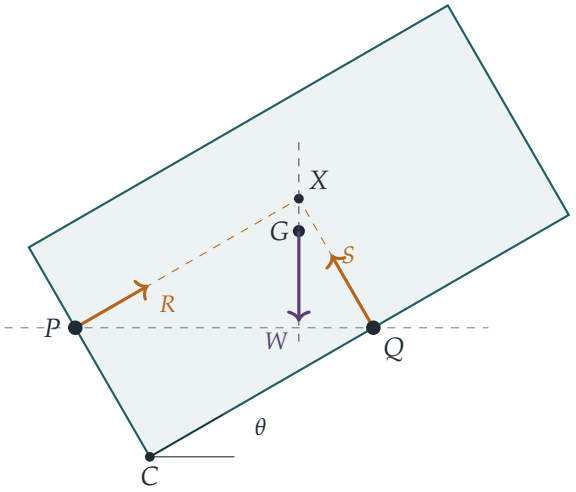
We model the lamina as a rigid uniform rectangle of weight W, with centre G, and the pegs as smooth points P and Q at the same height. Let C be the lowest corner. The side of length $2a$ through C rests on P, with $CP = p$, and the side of length $2b$ through C rests on Q, with $CQ = q$. The $2b$ side makes the acute angle θ with the horizontal, so the $2a$ side (perpendicular to it) makes the angle θ with the vertical.

It is easiest to use coordinates along the sides: from C, measure u along the $2b$ side and v along the $2a$ side. Then

$$C = (0,0), \qquad P = (0,p), \qquad Q = (q,0), \qquad G = (b,a).$$

The point (u,v) has horizontal and vertical displacement from C

$$x = u\cos\theta - v\sin\theta, \qquad y = u\sin\theta + v\cos\theta.$$



The reactions meet on the vertical through G

The reaction R at P is perpendicular to the $2a$ side, so it is parallel to the $2b$ side: its line of action is $v = p$. Similarly the reaction S at Q acts along $u = q$. These lines meet at $X = (q, p)$, the fourth corner of the rectangle $CPXQ$.

Taking moments about X, the reactions have zero moment, so the weight must too: the vertical through G passes through X. In other words, X and G have the same horizontal coordinate:

$$q\cos\theta - p\sin\theta = b\cos\theta - a\sin\theta, \qquad \text{that is,} \qquad (q-b)\cos\theta = (p-a)\sin\theta. \qquad (1)$$

(Resolving gives $R = W\sin\theta$ and $S = W\cos\theta$, both positive, so the pegs really can supply these forces.)

The first result

Since $0 < \theta < \frac{\pi}{2}$, both $\cos\theta$ and $\sin\theta$ are positive, so by (1) $q - b$ and $p - a$ have the same sign. If $p < a$, the right-hand side of (1) is negative, so $q - b < 0$, that is, $q < b$. That is the required result: if the corner is less than a from the peg on the $2a$ side, it is less than b from the other peg.

Ask yourself Why isn't that the end of the problem? Where does d come in?

Equation (1) uses only the moment condition. It is not yet about d, and p and q are not independent: the pegs are level, and they are d apart. We now bring in those two facts.

The relation with d

The pegs are at the same height, so the y-coordinates of P and Q agree:

$$p\cos\theta = q\sin\theta, \qquad \text{so} \qquad p = q\tan\theta. \qquad (2)$$

Their horizontal separation is d:

$$d = x_Q - x_P = q\cos\theta + p\sin\theta = q\cos\theta + \frac{q\sin^2\theta}{\cos\theta} = \frac{q}{\cos\theta}. \tag{3}$$

(Equivalently, CPQ is right-angled at C with hypotenuse $PQ = d$ horizontal, so $q = d\cos\theta$ and $p = d\sin\theta$.)

Now substitute $q = d\cos\theta$ and $p = d\sin\theta$ into the left-hand side of the first form of (1):

$$\begin{aligned} b\cos\theta - a\sin\theta &= q\cos\theta - p\sin\theta \\ &= d\cos^2\theta - d\sin^2\theta \\ &= d\cos 2\theta, \end{aligned}$$

which is the required result.

Checks

- Square case $a = b$, $\theta = \frac{\pi}{4}$: both sides are zero. By symmetry the diagonal CG is vertical, so this is consistent, and it holds for every d, as it should when the square is placed symmetrically.
- With $a = 1$, $b = 2$, $\theta = \frac{\pi}{6}$, equilibrium needs $q = d\cos\theta$ and $d = (b\cos\theta - a\sin\theta)/\cos 2\theta \approx 2.4641$. This gives $p \approx 1.2321 > a$ and $q \approx 2.1340 > b$, consistent with (1). Taking moments about G with $R = W\sin\theta$ and $S = W\cos\theta$ gives zero.
- Numerically, more than 10^5 random equilibria (random a, b, θ with the pegs on the sides) all satisfy the identity to within 10^{-14} and never violate "$p < a \Rightarrow q < b$".

Common slips

- Taking the reaction at a peg to be vertical. A smooth contact pushes perpendicular to the *surface* (here, the side of the lamina), not perpendicular to the ground.
- Mixing up which side makes angle θ with the horizontal: the $2a$ side makes θ with the *vertical*.

Beyond STEP

The three-force principle is a special case of the general statics of rigid bodies: a force system is in equilibrium exactly when its resultant force and its resultant moment about any one point both vanish. At university this becomes the language of wrenches and of virtual work. The virtual-work approach would find this equilibrium by minimising the height of G subject to the lamina keeping contact with the pegs. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and torque carefully.

WORKED SOLUTION

STEP II 1989, Question 13

Mechanics • Moments of Inertia

STEP II 1989, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q13

This question has a full guided treatment in *Beyond STEP*, Section 11.7, *Simpson's Rule in a Rod*: https://beyondstep.keithmcnulty.org/ch11/simpsons-rod.html

The key idea

Everything rests on two facts about moments of inertia. The parallel axis theorem says the moment of inertia about any axis is fixed by the mass, the centre of mass and the moment of inertia about the parallel axis through the centre of mass. And moments of inertia are *additive*, so a body can be cut into pieces and each piece swapped for something equivalent. A lamina is a stack of thin rods, so once we know how to replace one rod, we can replace the lamina.

Solution

Equivalence about every axis

Let ℓ be any axis, and let ℓ_O be the parallel axis through O, a distance d from ℓ. Both the body and the particle system have mass m and centre of mass O, so we can apply the parallel axis theorem to each of them:

$$I_\ell^{\text{body}} = I_{\ell_O}^{\text{body}} + md^2, \qquad I_\ell^{\text{particles}} = I_{\ell_O}^{\text{particles}} + md^2.$$

By hypothesis $I_{\ell_O}^{\text{body}} = I_{\ell_O}^{\text{particles}}$, because ℓ_O passes through O. So the two right-hand sides are equal, and $I_\ell^{\text{body}} = I_\ell^{\text{particles}}$ for every axis ℓ, as required.

The uniform rod

Put the rod along the x-axis from $x = -a$ to $x = a$, so its centre of mass O is the origin. Place particles of masses p, q, r at $x = -a$, 0, a. For total mass m and centre of mass at

O we need

$$p+q+r=m, \qquad -pa+ra=0,$$

so $r=p$ and $q=m-2p$.

Now take any axis through O making an angle θ with the rod. A point of the rod at position x is a distance $|x|\sin\theta$ from this axis. So every moment of inertia about an axis through O is $\sin^2\theta$ times $\sum(\text{mass})\,x^2$, and we only need that one sum to agree. For the rod, with mass $\frac{m}{2a}\,\mathrm{d}x$ in each element,

$$\int_{-a}^{a}\frac{m}{2a}\,x^2\,\mathrm{d}x=\frac{m}{2a}\cdot\frac{2a^3}{3}=\tfrac{1}{3}ma^2,$$

and for the particles it is $pa^2+q\cdot 0+pa^2=2pa^2$. These agree for every θ when $p=\frac{1}{6}m$. So the rod is dynamically equivalent to

$$\tfrac{1}{6}m \text{ at each end}, \qquad \tfrac{2}{3}m \text{ at the midpoint}.$$

By the first part, the two then have the same moment of inertia about *any* axis, not only axes through O.

The rectangular lamina

Let the rectangle have centre O and sides $2a$ (parallel to the x-axis) and $2b$, so its mass per unit area is $M/(4ab)$. Cut it into thin strips parallel to the x-axis. The strip at height y with width δy is a uniform rod of length $2a$ and mass $\delta M=\frac{M}{2b}\,\delta y$.

> **Ask yourself** The centre of a strip isn't at O. Why may we still replace each strip by its three particles?

This is exactly what the first part is for. Each strip has the same moment of inertia as its three particles about *every* axis, not just axes through the strip's own centre. Moments of inertia add, so the lamina has the same moment of inertia, about any axis, as the collection of all the replacement particles. That collection also has total mass M and centre of mass O, because each strip's replacement has the same mass and centre of mass as the strip.

The strip at height y is replaced by $\frac{1}{6}\delta M=\frac{M}{12b}\,\delta y$ at each of $(\pm a,y)$ and $\frac{2}{3}\delta M=\frac{M}{3b}\,\delta y$ at $(0,y)$. As y runs from $-b$ to b, these particles fill out three uniform rods of length $2b$:

- one along each side $x=\pm a$, each of mass $\frac{M}{12b}\cdot 2b=\frac{1}{6}M$;
- one along the centre line $x=0$, of mass $\frac{M}{3b}\cdot 2b=\frac{2}{3}M$.

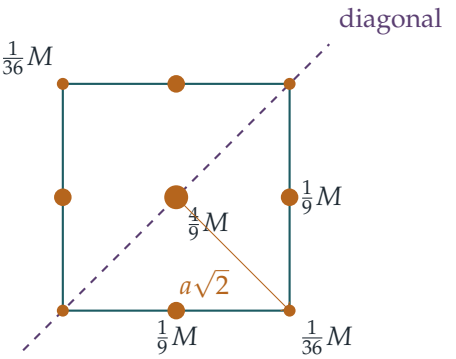
Now apply the rod result to each of these rods, by the same additivity argument:

- each side rod ($\frac{1}{6}M$) becomes $\frac{1}{6}\cdot\frac{1}{6}M=\frac{1}{36}M$ at each of its ends, which are corners, and $\frac{2}{3}\cdot\frac{1}{6}M=\frac{1}{9}M$ at its midpoint, the midpoint of a side of length $2b$;
- the central rod ($\frac{2}{3}M$) becomes $\frac{1}{6}\cdot\frac{2}{3}M=\frac{1}{9}M$ at each of its ends, the midpoints of the sides of length $2a$, and $\frac{2}{3}\cdot\frac{2}{3}M=\frac{4}{9}M$ at the centre.

So there is $\frac{1}{36}M$ at each corner, $\frac{1}{9}M$ at the midpoint of each side and $\frac{4}{9}M$ at the centre. The total is $\frac{4}{36}M + \frac{4}{9}M + \frac{4}{9}M = M$, and by symmetry the centre of mass is O. This is the required equivalent system.

The square about a diagonal

For the square of side $2a$, put the corners at $(\pm a, \pm a)$ and take the diagonal $y = x$. The distance of (x_0, y_0) from this line is $|x_0 - y_0|/\sqrt{2}$.



- The corners (a, a), $(-a, -a)$ and the centre lie on the diagonal and contribute nothing.
- The corners $(a, -a)$, $(-a, a)$ are at distance $2a/\sqrt{2} = a\sqrt{2}$.
- Each midpoint $(\pm a, 0)$, $(0, \pm a)$ is at distance $a/\sqrt{2}$.

Hence

$$I_{\text{diagonal}} = 2 \cdot \tfrac{1}{36}M \cdot 2a^2 + 4 \cdot \tfrac{1}{9}M \cdot \tfrac{1}{2}a^2 = \tfrac{1}{9}Ma^2 + \tfrac{2}{9}Ma^2 = \tfrac{1}{3}Ma^2.$$

The non-uniform rod

Let the rod have length $2a$, lying along the x-axis from $x = 0$ to $x = 2a$, with line density kx. Then

$$m = \int_0^{2a} kx \, \mathrm{d}x = 2ka^2, \qquad \text{so} \qquad k = \frac{m}{2a^2}.$$

Its centre of mass is at

$$\bar{x} = \frac{1}{m}\int_0^{2a} kx^2 \, \mathrm{d}x = \frac{k}{m} \cdot \frac{8a^3}{3} = \frac{4a}{3},$$

and

$$\int_0^{2a} kx^3 \, \mathrm{d}x = \frac{k\,(2a)^4}{4} = 4ka^4 = 2ma^2.$$

As with the uniform rod, every point is on one line, so about an axis through the centre of mass at angle θ to the rod the moment of inertia is $\sin^2\theta \sum(\text{mass})(x - \bar{x})^2$. If the masses and centres of mass agree, $\sum(\text{mass})(x - \bar{x})^2 = \sum(\text{mass})x^2 - m\bar{x}^2$, so it is enough to match $\sum(\text{mass})\,x^2$, which is the moment of inertia about the perpendicular axis through the end $x = 0$.

Let the particles be p_0 at $x = 0$, p_1 at $x = a$ and p_2 at $x = 2a$. Matching the mass, the first moment and the second moment gives the equations

$$\begin{aligned} p_0 + p_1 + p_2 &= m, \\ p_1 a + 2p_2 a &= \tfrac{4}{3}ma, \\ p_1 a^2 + 4p_2 a^2 &= 2ma^2. \end{aligned}$$

Subtracting the second (divided by a) from the third (divided by a^2) gives $2p_2 = \frac{2}{3}m$, so $p_2 = \frac{1}{3}m$. Then $p_1 = \frac{4}{3}m - \frac{2}{3}m = \frac{2}{3}m$, and finally $p_0 = m - \frac{2}{3}m - \frac{1}{3}m = 0$.

The particle at the light end has zero mass, so only two particles are needed: $\frac{2}{3}m$ at the midpoint and $\frac{1}{3}m$ at the heavy end.

Checks

- The standard result for a square lamina of side $2a$ about an axis through its centre in its plane, parallel to a side, is $\frac{1}{3}Ma^2$. By the perpendicular axis theorem and symmetry, every in-plane axis through the centre gives the same value, so the diagonal must too. This agrees.
- For the non-uniform rod, the two particles give mass m, first moment $\frac{2}{3}m \cdot a + \frac{1}{3}m \cdot 2a = \frac{4}{3}ma$ (so the centre of mass is at $\frac{4}{3}a$), and second moment $\frac{2}{3}ma^2 + \frac{4}{3}ma^2 = 2ma^2$, all as required.
- Numerically, with a fine grid of point masses, we compared each body with its particle system about 100–200 random axes in space (not through the centre). The rod and the variable rod agreed to better than 10^{-7}, the rectangle to 4×10^{-5} (grid error), and the square about its diagonal gave 0.33332 for $M = a = 1$.

Common slips

- Replacing each strip by its three particles without saying why that's allowed. The strips aren't centred at O, so you need the first part's "any axis" result.
- In the last part, imposing only the mass and centre of mass conditions. Two equations can't fix three masses: the moment of inertia condition is what forces $p_0 = 0$.

Beyond STEP

For a rigid body moving in space, what matters dynamically is its mass, its centre of mass and its *inertia tensor*, a symmetric 3×3 matrix that gives the moment of inertia about every axis through the centre of mass. Two bodies sharing these move identically under the same forces, and that is why "dynamically equivalent" is a good name. Matching the first few moments of a distribution with a few point masses is also the idea behind numerical integration. The weights $\frac{1}{6}, \frac{2}{3}, \frac{1}{6}$ for the uniform rod are exactly Simpson's rule, and choosing the positions as well as the weights leads to Gaussian

quadrature. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops the inertia tensor and rigid-body motion.

WORKED SOLUTION

STEP II 1989, Question 14

Mechanics • Motion in a Circle

STEP II 1989, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

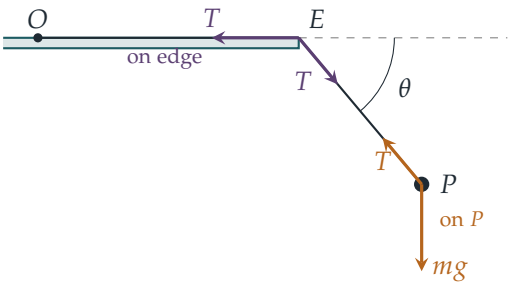
step.maths.org/questions/89-s2-q14

The key idea

Once the particle is released, the part of the string lying on the table stays put: it is held taut between the fixed end and the edge. So the edge acts as a fixed pivot, and the particle moves on a circle of radius a about it, exactly like a simple pendulum released from the horizontal. Energy gives the speed and the radial equation of motion gives the tension. The edge is smooth, so the tension is the same on both sides of it, and the force on the edge is the vector sum of two equal tensions pulling along the two pieces of string.

Solution

Model. The string is light and inextensible, the table-top is smooth and thin, and the particle is a point mass. Let O be the fixed end of the string and E the point of the edge where the string passes over it, so $OE = l - a$ and the hanging part EP has length a. Let θ be the angle the hanging part has turned through below the horizontal (the direction it started in, pointing away from the table).



The part OE stays on the table and stays still, since O is fixed and the string remains taut (we check below that $T > 0$). So P moves on a circle of centre E and radius a.

Speed. The table is smooth and the tension is always perpendicular to the motion of P, so neither does any work, and mechanical energy is conserved. When the string has turned through θ, P has fallen a height $a\sin\theta$, so starting from rest,

$$\tfrac{1}{2}mv^2 = mga\sin\theta \qquad \Longrightarrow \qquad v^2 = 2ga\sin\theta.$$

Tension. Resolving towards the centre E along PE: the tension acts along PE, and the component of the weight along PE is $mg\sin\theta$ *away* from E (for $0 < \theta < \pi$, P is below the level of E). So

$$T - mg\sin\theta = \frac{mv^2}{a} = 2mg\sin\theta, \qquad \text{giving} \qquad \boxed{T = 3mg\sin\theta.}$$

This is positive for $0 < \theta < \pi$, so the string stays taut throughout. At $\theta = \pi$ we have $v = 0$: the particle just reaches the underside of the table, level with E, and swings back. So the motion takes place for $0 \le \theta \le \pi$.

Force on the edge. The edge is smooth, so the tension is T on both sides of it. The string presses on the edge with two pulls of size T: one along EO, horizontal towards the fixed end, and one along EP. Take x horizontal away from the table and y vertically up. The unit vectors along EO and EP are $(-1, 0)$ and $(\cos\theta, -\sin\theta)$, so the force on the edge is

$$\mathbf{F} = T\,(\cos\theta - 1,\ -\sin\theta).$$

Its magnitude is

$$|\mathbf{F}| = T\sqrt{(1-\cos\theta)^2 + \sin^2\theta} = T\sqrt{2 - 2\cos\theta} = 2T\sin\tfrac{\theta}{2}$$

using $1 - \cos\theta = 2\sin^2\frac{\theta}{2}$, and $\sin\frac{\theta}{2} \ge 0$ for $0 \le \theta \le \pi$. (Geometrically, two equal forces T separated by angle $\pi - \theta$ add to $2T\cos\frac{\pi-\theta}{2}$, the same thing.) Substituting T and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$,

$$|\mathbf{F}| = 6mg\sin\theta\sin\tfrac{\theta}{2} = 12mg\,s^2\sqrt{1-s^2}, \qquad \text{where } s = \sin\tfrac{\theta}{2} \in [0,1].$$

Ask yourself How do we maximise $s^2\sqrt{1-s^2}$ cleanly?

Since it is non-negative, it is enough to maximise its square, $h(u) = u^2(1-u)$ with $u = s^2 \in [0,1]$. Then $h'(u) = 2u - 3u^2 = u(2-3u)$, which is positive for $0 < u < \frac{2}{3}$ and negative for $\frac{2}{3} < u \le 1$. So h has its maximum on $[0,1]$ at $u = \frac{2}{3}$, where $h = \frac{4}{9}\cdot\frac{1}{3} = \frac{4}{27}$. Hence the largest value of $s^2\sqrt{1-s^2}$ is $\sqrt{4/27} = \dfrac{2}{3\sqrt{3}}$, and

$$|\mathbf{F}|_{\max} = 12mg\cdot\frac{2}{3\sqrt{3}} = \frac{8mg}{\sqrt{3}},$$

as required. It occurs when $\sin^2\frac{\theta}{2} = \frac{2}{3}$, that is $\cos\theta = 1 - 2\sin^2\frac{\theta}{2} = -\frac{1}{3}$, so $\theta \approx 109.5°$: the particle has already passed below the edge and is swinging back in under the table.

Checks

- At $\theta = \frac{\pi}{2}$ (string vertical) the speed is $\sqrt{2ga}$ and $T = 3mg$, the familiar result for a pendulum released from the horizontal. The force on the edge is then $T\sqrt{2} = 3\sqrt{2}\,mg \approx 4.24mg$, a little less than $8mg/\sqrt{3} \approx 4.62mg$, consistent with the maximum lying beyond the vertical.
- At $\theta = 0$ and $\theta = \pi$ the particle is momentarily at rest level with the edge, so $T = 0$ and the edge feels nothing, as the formula says.
- We integrated the equation of motion $\ddot{\theta} = (g/a)\cos\theta$ numerically: T from the radial equation matched $3mg\sin\theta$ to 10^{-12}, and the largest edge force came out at $\theta \approx 1.9106$ (that is, $\arccos(-\frac{1}{3})$) with value $4.6188\,mg$, matching $8/\sqrt{3}$.

Common slips

- Taking the force on the edge to be just T, or just the vertical part of the pull from the hanging string. The string on the table pulls the edge horizontally too, and both pulls must be added as vectors.
- Stopping the maximisation at $\theta \leq \frac{\pi}{2}$. The particle carries on under the table, and the maximum is in that later stage.

Beyond STEP

The tension here is a *constraint force*: it does no work and is whatever it needs to be to keep the particle on its circle. In Lagrangian mechanics such forces drop out of the equations of motion entirely, and are recovered afterwards (as we did with the radial equation) or by using Lagrange multipliers. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats circular motion, constraints and energy methods carefully at this level.

WORKED SOLUTION

STEP II 1989, Question 15

Probability and Statistics • Geometric Probability

STEP II 1989, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q15

The key idea

A point chosen "at random on the perimeter" is uniform with respect to length, so everything starts from the total length $\pi + 2$. The triangle always has a vertex at the centre O, so its area is $\frac{1}{2}ab\sin C$, where a and b are the distances from O and C is the angle at O. Each conditional case then reduces to a simple expectation, and the law of total expectation stitches them together.

Solution

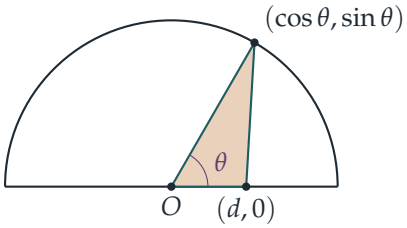
Put O at the origin, with the diameter along the x-axis from $(-1, 0)$ to $(1, 0)$ and the arc in $y \geq 0$. The perimeter has length $2 + \pi$: the diameter contributes 2 and the arc π. Since each point is uniform along the perimeter,

$$\mathrm{P}(\text{a point lies on the diameter}) = \frac{2}{\pi + 2}, \qquad \mathrm{P}(\text{a point lies on the arc}) = \frac{\pi}{\pi + 2}.$$

Given that it lies on the arc, the point is $(\cos\theta, \sin\theta)$ with θ uniform on $[0, \pi]$. Given that it lies on the diameter, it is $(d, 0)$ with d uniform on $[-1, 1]$.

The two points are independent, and "exactly one on the diameter" can happen with either the first or the second point on the diameter, so

$$\mathrm{P}(\text{exactly one on the diameter}) = 2 \cdot \frac{2}{\pi + 2} \cdot \frac{\pi}{\pi + 2} = \frac{4\pi}{(\pi + 2)^2}.$$



Part (i)

Let one point be $(d, 0)$ on the diameter and the other $(\cos\theta, \sin\theta)$ on the arc. The sides from O have lengths $|d|$ and 1, and the angle between them is θ (if $d < 0$ the angle is $\pi - \theta$, which has the same sine). So

$$A = \tfrac{1}{2}\,|d| \sin\theta.$$

Given this case, d and θ are independent, so the expectation of the product is the product of the expectations. Since d is uniform on $[-1, 1]$, $|d|$ is uniform on $[0, 1]$ and $\mathrm{E}|d| = \frac{1}{2}$. Also

$$\mathrm{E}(\sin\theta) = \frac{1}{\pi}\int_0^\pi \sin\theta \,\mathrm{d}\theta = \frac{2}{\pi}.$$

Therefore

$$\mathrm{E}(A \mid \text{exactly one on the diameter}) = \frac{1}{2}\cdot\frac{1}{2}\cdot\frac{2}{\pi} = \frac{1}{2\pi},$$

as required.

Part (ii)

Now both points are on the arc, at angles θ_1 and θ_2, independent and uniform on $[0, \pi]$. Both are at distance 1 from O, and the angle between them at O is $X = |\theta_1 - \theta_2|$, which lies in $[0, \pi]$. So

$$A = \tfrac{1}{2}\sin X.$$

? **Ask yourself** Why can we use the given density for X?

The angle θ measures arc length along the unit semicircle, so choosing two points uniformly on the arc is the same as choosing two points uniformly on a line segment of length π. The angle X is exactly the distance between them, so X has density $2(\pi - x)/\pi^2$ on $[0, \pi]$. Hence

$$\mathrm{E}(\sin X) = \frac{2}{\pi^2}\int_0^\pi (\pi - x)\sin x \,\mathrm{d}x.$$

Splitting the integral, $\int_0^\pi \pi \sin x \,\mathrm{d}x = 2\pi$, and integrating by parts with $u = x$,

$$\int_0^\pi x\sin x \,\mathrm{d}x = \Big[-x\cos x\Big]_0^\pi + \int_0^\pi \cos x \,\mathrm{d}x = \pi + 0 = \pi.$$

So $\int_0^\pi (\pi - x)\sin x \,\mathrm{d}x = 2\pi - \pi = \pi$, giving $\mathrm{E}(\sin X) = \dfrac{2}{\pi}$ and

$$\mathrm{E}(A \mid \text{neither on the diameter}) = \frac{1}{2}\cdot\frac{2}{\pi} = \frac{1}{\pi},$$

as required.

The expected value of A

There is a third case: if both points lie on the diameter, the "triangle" is degenerate (all three points are on the x-axis), so $A = 0$. The probability that neither point is on the diameter is $\left(\frac{\pi}{\pi+2}\right)^2$. By the law of total expectation,

$$\begin{aligned}\mathrm{E}(A) &= \frac{4\pi}{(\pi+2)^2}\cdot\frac{1}{2\pi} + \frac{\pi^2}{(\pi+2)^2}\cdot\frac{1}{\pi} + \frac{4}{(\pi+2)^2}\cdot 0 \\ &= \frac{2+\pi}{(\pi+2)^2} \\ &= \frac{1}{2+\pi},\end{aligned}$$

which is the result we wanted.

Checks

- The three case probabilities $\frac{4}{(\pi+2)^2}$, $\frac{4\pi}{(\pi+2)^2}$ and $\frac{\pi^2}{(\pi+2)^2}$ add up to $\frac{(2+\pi)^2}{(\pi+2)^2} = 1$.
- $\mathrm{E}(A) \le \frac{1}{2}$, the largest possible area (two radii at right angles), as it must be; in fact $\frac{1}{2+\pi} \approx 0.1945$.
- A Monte Carlo simulation of 2×10^6 pairs of points gave P(exactly one) ≈ 0.4757 (exact 0.4754), conditional means 0.1591 and 0.3184 (exact $\frac{1}{2\pi} \approx 0.1592$ and $\frac{1}{\pi} \approx 0.3183$), and $\mathrm{E}(A) \approx 0.1944$.

Common slips

- Forgetting the factor 2 in P(exactly one): either point can be the one on the diameter.
- Treating "on the diameter" as having probability $\frac{1}{2}$: the choice is uniform by length, not by which piece of the boundary.
- Using $\sin\theta_1 \sin\theta_2$ or similar in part (ii) instead of the sine of the angle *between* the radii.

Beyond STEP

Splitting an expectation over a partition of cases, $\mathrm{E}(A) = \sum_i \mathrm{E}(A \mid B_i)\,\mathrm{P}(B_i)$, is the law of total expectation, one of the workhorses of first-year probability; its general form, $\mathrm{E}(A) = \mathrm{E}\big(\mathrm{E}(A \mid Y)\big)$, turns up everywhere from branching processes to Markov chains. The density of X quoted in part (ii) is itself a nice exercise: it comes from the area of a strip in the square $[0,\pi]^2$. Questions like this belong to geometric probability; Grimmett and Welsh, *Probability: An Introduction*, covers conditional expectation and joint densities at the right level.

WORKED SOLUTION

STEP II 1989, Question 16

Probability and Statistics • Binomial Distribution

STEP II 1989, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q16

The key idea

Split the batch into the two parts that behave differently: the n widgets in the sample and the N that are never sampled. The number of faults in each part is binomial, and the two counts are independent. Everything depends on the sample: if it has two or more faults the batch ends up perfect, and otherwise nothing changes. So a batch with $k \geq 1$ faults at the end must have had 0 or 1 faults in the sample, with the rest among the other N.

Solution

Write $q = 1 - p$. Let X be the number of faulty widgets in the sample and Y the number of faulty widgets among the other N in the batch. Each widget is faulty independently with probability p, so

$$X \sim \mathrm{B}(n, p), \qquad Y \sim \mathrm{B}(N, p),$$

and X and Y are independent, because they count faults in disjoint sets of widgets.

Let K be the number of faulty widgets in the batch after quality control. If $X \geq 2$, every fault is corrected and $K = 0$. If $X \leq 1$, nothing is corrected and $K = X + Y$. So

$$K = \begin{cases} X + Y, & X \leq 1, \\ 0, & X \geq 2. \end{cases}$$

The distribution of K

Take $1 \leq k \leq N$. Since $k \geq 1$, the event $K = k$ cannot come from the case $X \geq 2$, so it happens exactly when $X = 0,\ Y = k$ or $X = 1,\ Y = k - 1$. These are disjoint, and by

independence

$$\begin{aligned}\mathrm{P}(K=k) &= \mathrm{P}(X=0)\,\mathrm{P}(Y=k) + \mathrm{P}(X=1)\,\mathrm{P}(Y=k-1)\\ &= q^n\binom{N}{k}p^k q^{N-k} + npq^{n-1}\binom{N}{k-1}p^{k-1}q^{N-k+1}\\ &= \left[\binom{N}{k} + n\binom{N}{k-1}\right]p^k q^{N+n-k}.\end{aligned}$$

Both terms have the same power of p and of q, which is what makes them combine. Now put the binomial coefficients over the common denominator $k!\,(N-k+1)!$:

$$\binom{N}{k} = \frac{N!\,(N-k+1)}{k!\,(N-k+1)!}, \qquad n\binom{N}{k-1} = \frac{n\,N!\,k}{k!\,(N-k+1)!}.$$

Adding, the numerator is $N!\,(N-k+1+nk) = N!\,[N+1+k(n-1)]$, so

$$\mathrm{P}(K=k) = \frac{[N+1+k(n-1)]\,N!}{(N-k+1)!\,k!}\,p^k(1-p)^{N+n-k},$$

as required.

? **Ask yourself** What happens when $k=0$?

The formula only holds for $k \geq 1$. At $k=0$ it gives $\frac{(N+1)N!}{(N+1)!}q^{N+n} = q^{N+n}$, which is the probability that there were no faults at all. But $K=0$ also happens whenever $X \geq 2$, so in fact

$$\mathrm{P}(K=0) = q^{N+n} + \mathrm{P}(X \geq 2) = q^{N+n} + 1 - q^n - npq^{n-1}.$$

The "$k \leq N$" in the question should be read as $1 \leq k \leq N$.

The case $k = N+1$

For $K = N+1$ we need $X \leq 1$ and $X+Y = N+1$. Since $Y \leq N$, this forces $X=1$ and $Y=N$, so

$$\mathrm{P}(K=N+1) = npq^{n-1}\cdot p^N = np^{N+1}q^{n-1}.$$

Putting $k=N+1$ into the formula gives the coefficient

$$\frac{[N+1+(N+1)(n-1)]\,N!}{0!\,(N+1)!} = \frac{(N+1)\,n\,N!}{(N+1)!} = n,$$

and the power of q is $q^{N+n-(N+1)} = q^{n-1}$. So the formula gives $np^{N+1}q^{n-1}$, which is correct. (For $k \geq N+2$ the probability is zero, since $K \leq N+1$.)

The expected number of faults

We could sum $k\,\mathrm{P}(K=k)$, but it is much quicker to use the description of K in terms of X and Y. Writing $\mathbf{1}_{\{X\leq 1\}}$ for the indicator of the event $X \leq 1$,

$$K = (X+Y)\,\mathbf{1}_{\{X\leq 1\}} = X\,\mathbf{1}_{\{X\leq 1\}} + Y\,\mathbf{1}_{\{X\leq 1\}}.$$

For the first term, $X\,\mathbf{1}_{\{X\le 1\}}$ equals 1 when $X = 1$ and 0 otherwise, so its expectation is $\mathrm{P}(X = 1) = npq^{n-1}$. For the second term, Y is independent of X, so

$$\mathrm{E}\left[Y\,\mathbf{1}_{\{X\le 1\}}\right] = \mathrm{E}[Y]\,\mathrm{P}(X \le 1) = Np\left(q^n + npq^{n-1}\right).$$

Adding, and taking out the factor pq^{n-1},

$$\begin{aligned}\mathrm{E}[K] &= npq^{n-1} + Np\,q^{n-1}(q + np)\\ &= pq^{n-1}\left[n + N(1-p) + Nnp\right]\\ &= \left[N + n + pN(n-1)\right]p(1-p)^{n-1},\end{aligned}$$

as required.

Checks

- The probabilities should sum to 1. With $n = 2$, $N = 1$: $\mathrm{P}(K = 1) = (2+1)pq^2 = 3pq^2$, $\mathrm{P}(K = 2) = 2p^2q$ and $\mathrm{P}(K = 0) = q^3 + p^2$, and these add to $q^3 + 3pq^2 + 3p^2q + p^3 = 1$ (using $p^2 = p^2(p+q)$).
- If n is huge, almost every batch with a fault in the sample is fixed, and the expectation tends to 0, which matches the factor $(1-p)^{n-1}$.
- We checked every formula with exact fractions for $2 \le n \le 6$, $0 \le N \le 6$ and several values of p, and simulated the procedure: with $n = 4$, $N = 6$, $p = 0.3$ the formula gives $\mathrm{E}[K] \approx 1.585$, and 200 000 simulated batches gave 1.581.

Common slips

- Applying the "two or more" rule to the total number of faults in the batch. Only the sample count X decides whether faults are corrected.
- Forgetting that $K = 0$ has an extra contribution from corrected batches, so the formula fails at $k = 0$.
- Computing $\mathrm{E}[K]$ by brute-force summation. It works, but splitting K with indicators and using independence is far shorter.

Beyond STEP

Writing a random variable as a sum of pieces multiplied by indicators, and using linearity of expectation, is one of the most useful tricks in first-year probability. It avoids finding a distribution when only the mean is needed. The question is also a small model of acceptance sampling in quality control, where we choose n to balance testing cost against the faults that slip through. For more, see Grimmett and Welsh, *Probability: An Introduction*, or Blitzstein and Hwang, *Introduction to Probability*, which uses indicators throughout.

WORKED SOLUTION

STEP III 1989, Question 1

Pure • Surfaces of Revolution

STEP III 1989, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q1

The key idea

A sphere is the surface of revolution of a semicircle, so the zone's area is a standard surface-of-revolution integral. The surprise is that the integrand turns out to be constant, which is exactly why the answer depends only on the distance between the planes. In the second part the surface splits into a cone (from OP) and a spherical zone (from the arc PR). The first result then does the hard work, and the tangent condition supplies the right-angled triangle we need.

A word on the printed equation: with c^2 on the right the circle passes through O and the constant a never appears. The intended curve is clearly

$$(x-c)^2+y^2=a^2,$$

the circle with centre $C=(c,0)$ and radius a. Since $c>a$, the origin lies outside it, so tangents from O exist. We solve the question for this curve.

Solution

The area of a zone

Take a sphere of radius r with centre at the origin, and choose the x-axis perpendicular to the two planes, which are $x=h_1$ and $x=h_2$ with $-r\le h_1<h_2\le r$. The sphere is formed by rotating the upper semicircle $y=\sqrt{r^2-x^2}$ through 2π about the x-axis, so the zone has area

$$S=\int_{h_1}^{h_2}2\pi y\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x.$$

Here $\dfrac{\mathrm{d}y}{\mathrm{d}x}=-\dfrac{x}{\sqrt{r^2-x^2}}$, so for $|x|<r$

$$y\sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}=\sqrt{r^2-x^2}\sqrt{\frac{r^2-x^2+x^2}{r^2-x^2}}=\sqrt{r^2-x^2}\cdot\frac{r}{\sqrt{r^2-x^2}}=r.$$

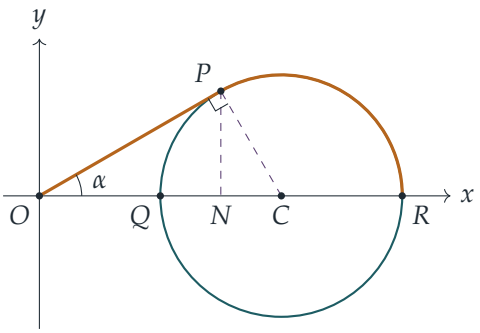
Ask yourself What happens at $x = \pm r$, where $\mathrm{d}y/\mathrm{d}x$ is infinite?

The derivative blows up at the poles, but the product $y\sqrt{1+(y')^2}$ equals r on all of $(-r, r)$, so it is bounded. If a plane is tangent to the sphere ($h_1 = -r$ or $h_2 = r$), the integral is improper only in a harmless way: integrate up to $h_2 - \varepsilon$ and let $\varepsilon \to 0$. (Alternatively, parametrise by $x = r\cos\theta$, $y = r\sin\theta$, which avoids the problem altogether.) Therefore

$$S = \int_{h_1}^{h_2} 2\pi r \,\mathrm{d}x = 2\pi r\,(h_2 - h_1),$$

which is $2\pi \times$ (radius) $\times$ (distance between the planes), as required.

The surface



Let $C = (c, 0)$ be the centre and N the foot of the perpendicular from P to the x-axis. Then $Q = (c-a, 0)$ and $R = (c+a, 0)$. The arc PR is the one that does not pass through Q (shown in amber): the other arc from P to R runs through Q and round the bottom of the circle, and rotating it would sweep out the whole sphere.

A tangent is perpendicular to the radius at the point of contact, so the angle OPC is a right angle. In triangle OPC we have $OC = c$ and $CP = a$, so by Pythagoras

$$OP = \sqrt{c^2 - a^2}.$$

Let α be the angle POC. Then $\sin\alpha = \frac{a}{c}$ and $\cos\alpha = \frac{\sqrt{c^2-a^2}}{c}$, which gives

$$PN = OP\sin\alpha = \frac{a\sqrt{c^2-a^2}}{c}, \qquad ON = OP\cos\alpha = \frac{c^2-a^2}{c}.$$

The cone. Rotating OP about the x-axis gives a cone with slant height $l = OP$ and base radius PN. Its curved surface area is $\pi \times$ (base radius) $\times$ (slant height), which is

$$\pi \cdot \frac{a\sqrt{c^2-a^2}}{c} \cdot \sqrt{c^2-a^2} = \frac{\pi a(c^2-a^2)}{c}.$$

The zone. Rotating the arc PR gives the part of the sphere (centre C, radius a) lying between the plane through N perpendicular to the axis and the tangent plane at R.

These planes are a distance

$$NR = (c+a) - \frac{c^2 - a^2}{c} = \frac{ac + a^2}{c} = \frac{a(c+a)}{c}$$

apart, so by the first result the zone has area

$$2\pi a \cdot \frac{a(c+a)}{c} = \frac{2\pi a^2 (c+a)}{c}.$$

Total. Adding the two pieces,

$$\begin{aligned} \text{Area} &= \frac{\pi a}{c}\left[(c^2 - a^2) + 2a(c+a)\right] \\ &= \frac{\pi a}{c}\left(c^2 + 2ac + a^2\right) \\ &= \frac{\pi a (c+a)^2}{c}. \end{aligned}$$

Checks

- Take $a = 1$, $c = 2$. Then $P = \left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right)$, the cone has area $\frac{3\pi}{2}$ and the zone 3π, so the total is $\frac{9\pi}{2} \approx 14.137$. Approximating the curve by 20 000 short segments and adding the areas of the resulting frustums gives 14.13717.
- As $a \to c$, P approaches O and the cone disappears. The surface becomes the whole sphere of radius c, and indeed $\pi c (2c)^2 / c = 4\pi c^2$.
- As $c \to \infty$ with a fixed, the area grows like $\pi a c$, which is the area of a long thin cone of slant height about c and base radius about a. That is what we should expect.

Common slips

- Using OC or ON instead of OP as the slant height of the cone.
- Measuring the zone's thickness from C or from Q rather than from N, the foot of the perpendicular from P.
- Using the formula $2\pi r h$ with $r = c$ rather than $r = a$: the sphere being cut is the one of radius a centred at C, not anything centred at O.

Beyond STEP

The zone result is Archimedes' theorem. Projecting a sphere horizontally onto the circumscribing cylinder preserves area, which is the idea behind the Lambert equal-area map projection. It also explains why a point chosen uniformly on a sphere has a uniformly distributed height. Areas of surfaces of revolution are the first case of the surface integral $\iint |\mathbf{r}_u \times \mathbf{r}_v| \, \mathrm{d}u \, \mathrm{d}v$ from a first-year vector calculus course. For a beautifully geometric view of curvature and surfaces, see Needham, *Visual Differential Geometry and Forms*.

WORKED SOLUTION

STEP III 1989, Question 2

Pure • Planes in Three Dimensions

STEP III 1989, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q2

The key idea

Put in coordinates: east, north and up. Then each plane is a graph $z = px + qy + r$, and three points fix each one. How steep a plane is depends only on how fast z changes horizontally, so the angle comes from p and q. For the second part there is a neat shortcut: the *depth* of the seam below the ground is also a linear function of position, and the seam reaches the surface exactly where that depth is zero.

Solution

Setting up

Take A as the origin, with the x-axis pointing east, the y-axis north and the z-axis vertically up (all lengths in metres). Then

$$A = (0,0,0), \qquad B = (0,100,-5), \qquad C = (60,100,-10),$$

since B is 5 m below A and C is a further 5 m below B. The bore-holes are vertical, so the points where they strike the seam are directly below these:

$$A' = (0,0,-95), \qquad B' = (0,100,-50), \qquad C' = (60,100,-86).$$

The angle of the seam

The seam is not vertical, since A', B', C' lie above three points that are not collinear in plan. So we can write it as $z = px + qy + r$.

- At A': $r = -95$.
- From A' to B', y increases by 100 and z by 45, so $100q = 45$ and $q = \frac{9}{20}$.
- From B' to C', x increases by 60 and z changes by -36, so $60p = -36$ and $p = -\frac{3}{5}$.

So the seam is

$$z = -\tfrac{3}{5}x + \tfrac{9}{20}y - 95,$$

and a normal vector to it is $\mathbf{n} = \left(\frac{3}{5}, -\frac{9}{20}, 1\right)$, or, scaling by 20, $(12, -9, 20)$.

The angle between two planes equals the angle between their normals. The horizontal plane has normal $\mathbf{k} = (0,0,1)$, so the angle θ of the seam to the horizontal satisfies

$$\cos\theta = \frac{|(12,-9,20)\cdot(0,0,1)|}{\sqrt{12^2+9^2+20^2}} = \frac{20}{\sqrt{144+81+400}} = \frac{20}{\sqrt{625}} = \frac{20}{25} = \frac{4}{5}.$$

Hence the seam is inclined at $\cos^{-1}\left(\frac{4}{5}\right)$ to the horizontal, as required.

(Another way to see it: along the line of greatest slope, z changes by $\sqrt{p^2+q^2} = \sqrt{\frac{9}{25}+\frac{81}{400}} = \frac{3}{4}$ per metre horizontally. So $\tan\theta = \frac{3}{4}$, a 3–4–5 triangle, and $\cos\theta = \frac{4}{5}$.)

Where the seam reaches the surface

The ground is a plane through the origin. In the same way as before, going north from A to B drops it by 5 in 100, and going east from B to C drops it by 5 in 60, so

$$z = -\tfrac{1}{12}x - \tfrac{1}{20}y.$$

Ask yourself Do we really need to solve two plane equations at the same time?

Not quite. Let $d(x,y)$ be the depth of the seam below the ground at the point with horizontal position (x,y). It is the difference of two linear functions, so it is linear itself:

$$d(x,y) = \left(-\tfrac{1}{12}x - \tfrac{1}{20}y\right) - \left(-\tfrac{3}{5}x + \tfrac{9}{20}y - 95\right) = \tfrac{31}{60}x - \tfrac{1}{2}y + 95.$$

As a check, $d(0,0) = 95$, $d(0,100) = 45$ and $d(60,100) = 45+31 = 76$, which are the three bore-hole depths. (Indeed we could have written d down straight from these: the depth falls by 50 in 100 m going north and rises by 31 in 60 m going east.)

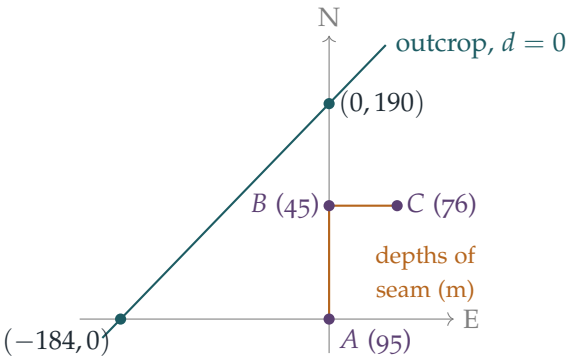
The seam comes to the surface where $d = 0$, which in plan is the line

$$\tfrac{31}{60}x - \tfrac{1}{2}y + 95 = 0, \qquad \text{that is,} \qquad y = \tfrac{31}{30}x + 190.$$

Along this line, moving 30 m east goes with moving 31 m north. So the line makes an angle φ with due north, where

$$\tan\varphi = \frac{\text{east}}{\text{north}} = \frac{30}{31}, \qquad \varphi = \tan^{-1}\left(\tfrac{30}{31}\right) \approx 44.06^\circ.$$

The bearing of the line is therefore $\tan^{-1}\left(\frac{30}{31}\right) \approx 044^\circ$. Read in the opposite direction it is 224°; a line has both bearings, and either describes it. The line runs just west of north-east, and in plan it passes 190 m due north of A.



Checks

- The seam's normal $(12, -9, 20)$ has length exactly 25, so $\cos\theta = \frac{20}{25}$, a clean 3–4–5 answer, as the question promised.
- The depth formula d reproduces all three bore-hole depths, so both planes are right.
- Numerically, $\tan^{-1}(30/31) = 44.0608°$. Random points on the line $y = \frac{31}{30}x + 190$ give the same height on both planes, to rounding error.

Common slips

- Using depths below *each* hole's top as if the tops were level. B and C are 5 m and 10 m lower than A, so the seam is 50 m and 86 m below A's level there, not 45 m and 76 m.
- Giving the bearing of a horizontal line in the seam (where $-\frac{3}{5}x + \frac{9}{20}y$ is constant, bearing $\tan^{-1}\frac{3}{4} \approx 037°$) instead of the outcrop. Because the ground slopes too, these differ.
- Mixing up $\tan\varphi = \text{east}/\text{north}$ with north/east. Bearings are measured clockwise from north.

Beyond STEP

This is the geologist's problem of "strike and dip", and the tools are those of first-year vector geometry: planes as $\mathbf{n} \cdot \mathbf{r} = c$, angles between planes through their normals, and lines of intersection. The depth trick is an instance of a general principle: the difference of two affine functions is affine, so the intersection of two non-parallel planes is the zero set of one linear function. Strang's *Introduction to Linear Algebra* builds this picture into the geometry of linear equations.

WORKED SOLUTION

STEP III 1989, Question 3

Pure • Matrix Groups

STEP III 1989, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q3

The key idea

$\mathbf{M}$ is a rotation through $2\pi/m$, so $\mathbf{M}^m = \mathbf{I}$, and the sum of its powers is a geometric series: multiply it by $\mathbf{I} - \mathbf{M}$ and everything telescopes. For the recurrence, write out $\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3$ and the pattern is plain. The group then comes from one identity: substituting one of these expressions into another just adds the indices, so the elements behave like the integers modulo m.

Solution

The sum of powers of $\mathbf{M}$

$\mathbf{M}$ is the matrix of an anticlockwise rotation through $\theta = 2\pi/m$ about the origin. Applying it k times rotates through $k\theta$, so

$$\mathbf{M}^k = \begin{pmatrix} \cos k\theta & -\sin k\theta \\ \sin k\theta & \cos k\theta \end{pmatrix}, \qquad \text{and in particular} \qquad \mathbf{M}^m = \begin{pmatrix} \cos 2\pi & -\sin 2\pi \\ \sin 2\pi & \cos 2\pi \end{pmatrix} = \mathbf{I}.$$

(The formula for $\mathbf{M}^k$ follows by induction from the compound-angle formulae, since $\mathbf{M}^{k+1} = \mathbf{M}\mathbf{M}^k$.)

Let $\mathbf{S} = \mathbf{I} + \mathbf{M} + \cdots + \mathbf{M}^{m-1}$. Since all these matrices are powers of $\mathbf{M}$, they commute, and the product telescopes:

$$(\mathbf{I} - \mathbf{M})\mathbf{S} = \mathbf{I} - \mathbf{M}^m = \mathbf{I} - \mathbf{I} = \mathbf{O}.$$

? **Ask yourself** Can we just "cancel" the factor $\mathbf{I} - \mathbf{M}$?

Only if $\mathbf{I} - \mathbf{M}$ is invertible: products of non-zero matrices can be $\mathbf{O}$. Here

$$\det(\mathbf{I} - \mathbf{M}) = (1 - \cos\theta)^2 + \sin^2\theta = 2 - 2\cos\theta,$$

and since $m > 1$ we have $0 < \theta \leq \pi$, so $\cos\theta \neq 1$ and the determinant is non-zero.

Multiplying $(\mathbf{I}-\mathbf{M})\mathbf{S}=\mathbf{O}$ on the left by $(\mathbf{I}-\mathbf{M})^{-1}$ gives $\mathbf{S}=\mathbf{O}$, which is the required result.

A formula for $\mathbf{X}_k$

Working out the first few terms,

$$\mathbf{X}_1=\mathbf{P}\mathbf{X}_0+\mathbf{Q},\quad \mathbf{X}_2=\mathbf{P}^2\mathbf{X}_0+\mathbf{P}\mathbf{Q}+\mathbf{Q},\quad \mathbf{X}_3=\mathbf{P}^3\mathbf{X}_0+(\mathbf{P}^2+\mathbf{P}+\mathbf{I})\mathbf{Q},$$

suggests that, for $k\geq 0$,

$$\mathbf{X}_k=\mathbf{P}^k\mathbf{X}_0+\mathbf{S}_k\mathbf{Q},\qquad \text{where}\quad \mathbf{S}_k=\mathbf{I}+\mathbf{P}+\cdots+\mathbf{P}^{k-1}\quad(\mathbf{S}_0=\mathbf{O}).$$

For $k=0$ this reads $\mathbf{X}_0=\mathbf{X}_0$, which is true. Suppose it holds for some $k\geq 0$. Then

$$\begin{aligned}\mathbf{X}_{k+1}=\mathbf{P}\mathbf{X}_k+\mathbf{Q}&=\mathbf{P}^{k+1}\mathbf{X}_0+(\mathbf{P}+\mathbf{P}^2+\cdots+\mathbf{P}^k)\mathbf{Q}+\mathbf{Q}\\&=\mathbf{P}^{k+1}\mathbf{X}_0+\mathbf{S}_{k+1}\mathbf{Q},\end{aligned}$$

so it holds for $k+1$, and by induction for all $k\geq 0$. Keep the order of the factors: $\mathbf{P}$ multiplies on the left throughout, and $\mathbf{P}$ and $\mathbf{Q}$ need not commute.

The group

Think of $\mathbf{X}_i$ as the expression $\mathbf{P}^i\mathbf{X}_0+\mathbf{S}_i\mathbf{Q}$, a rule that turns any matrix $\mathbf{X}_0$ into another. Substituting $\mathbf{X}_j$ for $\mathbf{X}_0$ gives

$$\begin{aligned}\mathbf{X}_i*\mathbf{X}_j&=\mathbf{P}^i\left(\mathbf{P}^j\mathbf{X}_0+\mathbf{S}_j\mathbf{Q}\right)+\mathbf{S}_i\mathbf{Q}\\&=\mathbf{P}^{i+j}\mathbf{X}_0+\left(\mathbf{P}^i\mathbf{S}_j+\mathbf{S}_i\right)\mathbf{Q}.\end{aligned}$$

Now $\mathbf{P}^i\mathbf{S}_j=\mathbf{P}^i+\cdots+\mathbf{P}^{i+j-1}$, and adding $\mathbf{S}_i=\mathbf{I}+\cdots+\mathbf{P}^{i-1}$ gives exactly $\mathbf{S}_{i+j}$. So, for any $\mathbf{P}$,

$$\mathbf{X}_i*\mathbf{X}_j=\mathbf{X}_{i+j}.$$

(This is no surprise: doing j steps of the recurrence and then i more is doing $i+j$ steps.)

Now take $\mathbf{P}=\mathbf{M}$. By the first part, $\mathbf{M}^m=\mathbf{I}$ and $\mathbf{S}_m=\mathbf{O}$, so

$$\mathbf{X}_m=\mathbf{X}_0,$$

that is, $\mathbf{X}_m$ is the expression that leaves every matrix unchanged. Then for any k, $\mathbf{X}_{k+m}=\mathbf{X}_k*\mathbf{X}_m=\mathbf{X}_k$, so the sequence of expressions repeats with period m and

$$G=\{\mathbf{X}_1,\mathbf{X}_2,\mathbf{X}_3,\ldots\}=\{\mathbf{X}_1,\mathbf{X}_2,\ldots,\mathbf{X}_m\}$$

is a finite set. We check the group axioms.

- *Closure.* $\mathbf{X}_i*\mathbf{X}_j=\mathbf{X}_{i+j}$, and $i+j\geq 1$, so this lies in G.
- *Associativity.* Both $(\mathbf{X}_i*\mathbf{X}_j)*\mathbf{X}_k$ and $\mathbf{X}_i*(\mathbf{X}_j*\mathbf{X}_k)$ equal $\mathbf{X}_{i+j+k}$. (More generally, substitution is composition of functions, which is always associative.)

- *Identity.* $\mathbf{X}_m * \mathbf{X}_j = \mathbf{X}_{m+j} = \mathbf{X}_j$ and $\mathbf{X}_j * \mathbf{X}_m = \mathbf{X}_j$, so $\mathbf{X}_m$ is the identity.
- *Inverses.* For $1 \le i < m$, $\mathbf{X}_i * \mathbf{X}_{m-i} = \mathbf{X}_{m-i} * \mathbf{X}_i = \mathbf{X}_m$, and $\mathbf{X}_m$ is its own inverse. Each inverse lies in G.

So G is a finite group under $*$.

In fact it has exactly m elements: $\mathbf{X}_1, \ldots, \mathbf{X}_m$ are different expressions because their coefficients $\mathbf{M}^1, \ldots, \mathbf{M}^m$ are rotations through different angles in $(0, 2\pi]$. Since every element is a "power" of $\mathbf{X}_1$, G is the cyclic group of order m.

Checks

- With $m = 2$, $\mathbf{M} = -\mathbf{I}$ and the sum is $\mathbf{I} + (-\mathbf{I}) = \mathbf{O}$. With $m = 4$, $\mathbf{M}$ is a quarter-turn and $\mathbf{I} + \mathbf{M} + \mathbf{M}^2 + \mathbf{M}^3 = \mathbf{I} + \mathbf{M} - \mathbf{I} - \mathbf{M} = \mathbf{O}$.
- Numerically, the sum of powers is $\mathbf{O}$ (to rounding) for every m from 2 to 39; the formula for $\mathbf{X}_k$ matches the recurrence for random $\mathbf{P}$, $\mathbf{Q}$, $\mathbf{X}_0$; and with $\mathbf{P} = \mathbf{M}$, random $\mathbf{Q}$ and $\mathbf{X}_0$, running the recurrence m steps returns exactly to $\mathbf{X}_0$.

Common slips

- Cancelling $\mathbf{I} - \mathbf{M}$ without checking it is invertible.
- Writing $\mathbf{Q}\mathbf{S}_k$ or $\mathbf{X}_0\mathbf{P}^k$: matrices do not commute, and the recurrence puts $\mathbf{P}$ on the left.
- Taking $\mathbf{X}_0$ as the identity: $\mathbf{X}_0$ is not in G as listed; the identity element is $\mathbf{X}_m$, which equals $\mathbf{X}_0$ only because $\mathbf{S}_m = \mathbf{O}$.

Beyond STEP

Each $\mathbf{X}_i$ is an *affine map* $\mathbf{X} \mapsto \mathbf{A}\mathbf{X} + \mathbf{B}$, and the affine maps with invertible $\mathbf{A}$ form a group under composition; here we found a cyclic subgroup of order m generated by a single map whose mth power is the identity. The fact that $\mathbf{I} + \mathbf{M} + \cdots + \mathbf{M}^{m-1} = \mathbf{O}$ is the matrix version of the mth roots of unity summing to zero, and the general idea, a group acting on a set by transformations, is central to a first course in group theory. Allenby's *Rings, Fields and Groups* develops this.

WORKED SOLUTION

STEP III 1989, Question 4

Pure • Curve Sketching and Roots

STEP III 1989, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q4

The key idea

Splitting the fraction as a polynomial plus a proper fraction gives the oblique asymptote at once, and it also makes the derivative easy. The derivative turns out to be greater than 2 everywhere, so each of the three branches climbs from $-\infty$ to $+\infty$. Each equation in the second half is then "curve = some simpler curve" in disguise. Strict monotonicity of the difference on each branch then counts the crossings exactly.

Solution

The curve

Since $2x(x^2-5) = 2x(x^2-4) - 2x$,

$$y = 2x - \frac{2x}{x^2-4}.$$

The function is odd, so the curve has rotational symmetry of order 2 about the origin. It meets the axes only where $x(x^2-5)=0$, that is at $x=0$ and $x=\pm\sqrt{5}$.

Asymptotes. The denominator vanishes at $x=\pm 2$ (where the numerator is $\mp 4 \neq 0$), so $x=2$ and $x=-2$ are vertical asymptotes. As $|x|\to\infty$, $\frac{2x}{x^2-4}\to 0$, so $y=2x$ is an oblique asymptote. Also $y-2x = -\frac{2x}{x^2-4}$ is negative for large positive x and positive for large negative x. So the curve approaches $y=2x$ from below on the right and from above on the left.

Gradient. Differentiating the split form,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2 - \frac{2(x^2-4) - 2x\cdot 2x}{(x^2-4)^2} = 2 + \frac{2(x^2+4)}{(x^2-4)^2} > 2.$$

So there are no stationary points, and y is strictly increasing on each of the intervals $x<-2$, $-2<x<2$ and $x>2$. Differentiating again gives $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\dfrac{4x(x^2+12)}{(x^2-4)^3}$, so the only point of inflection is the origin.

Tangent at the origin. At $x = 0$ the gradient is $2 + \frac{2\cdot 4}{16} = \frac{5}{2}$, so the tangent there is

$$y = \tfrac{5}{2}x.$$

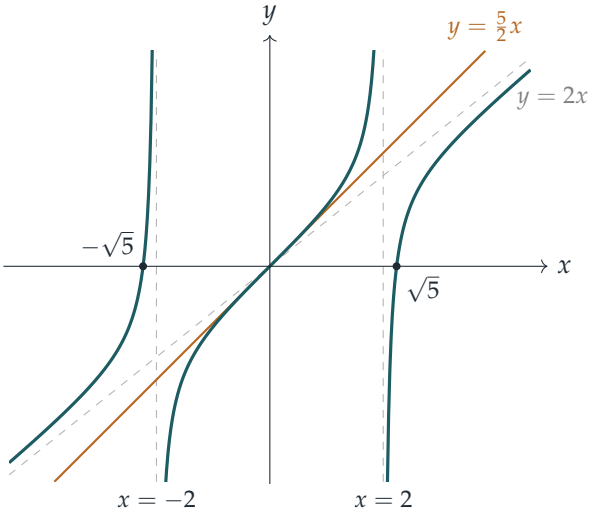
Behaviour near $x = \pm 2$. Near $x = 2$ the term $-\frac{2x}{x^2-4}$ dominates. It tends to $-\infty$ as $x \to 2^+$ and to $+\infty$ as $x \to 2^-$, and by symmetry it tends to $+\infty$ as $x \to -2^-$ and to $-\infty$ as $x \to -2^+$. So:

- for $x < -2$, y rises from $-\infty$ to $+\infty$, crossing the axis at $-\sqrt{5}$;
- for $-2 < x < 2$, y rises from $-\infty$ to $+\infty$ through the origin;
- for $x > 2$, y rises from $-\infty$ to $+\infty$, crossing the axis at $\sqrt{5}$.

One more fact will be useful. For $0 < x < 2$,

$$y - \tfrac{5}{2}x = \frac{x\left[4(x^2-5) - 5(x^2-4)\right]}{2(x^2-4)} = \frac{-x^3}{2(x^2-4)} > 0,$$

so on this stretch the curve lies above its tangent at the origin (and, by symmetry, below it for $-2 < x < 0$).



Part (i)

Neither $x = 0$ (LHS 0, RHS 8) nor $x = \pm 2$ (LHS -16, RHS 0) is a root. So we may divide by $2x(x^2-4)$, and the equation becomes

$$\frac{2x(x^2-5)}{x^2-4} = \frac{5x-2}{2x} = \frac{5}{2} - \frac{1}{x}.$$

The roots are therefore the x-coordinates of the intersections of the curve with the hyperbola $H\colon y = \frac{5}{2} - \frac{1}{x}$. We go branch by branch.

- $x < -2$: here H has gradient $\frac{1}{x^2} < \frac{1}{4}$, while the curve has gradient > 2. So the difference (curve $-$ H) is strictly increasing. It tends to $-\infty$ as $x \to -\infty$ and to $+\infty$ as $x \to -2^-$, because H stays between $\frac{5}{2}$ and 3. Exactly one root.
- $-2 < x < 0$: the curve has $y < 0$ (all three factors x, $x^2 - 5$, $x^2 - 4$ are negative), but H has $y > \frac{5}{2}$. No roots.
- $0 < x < 2$: by the fact above, $y > \frac{5}{2}x$. Also

$$\tfrac{5}{2}x - \left(\tfrac{5}{2} - \tfrac{1}{x}\right) = \frac{5x^2 - 5x + 2}{2x} > 0,$$

  since $5x^2 - 5x + 2$ has discriminant $25 - 40 < 0$. So the curve lies strictly above H. No roots.
- $x > 2$: as in the first case, the difference is strictly increasing (gradient $> 2 - \frac{1}{4}$). It goes from $-\infty$ to $+\infty$, because H stays between 2 and $\frac{5}{2}$. Exactly one root.

So equation (i) has exactly **two** real roots, one less than -2 and one greater than 2.

Ask yourself How can the equation in (ii) be read off the curve when it has squares on both sides?

Divide by $(x^2 - 4)^2$. The left side becomes y^2, so the equation says $y^2 = x^2 + 1$.

Part (ii)

Again $x = \pm 2$ is not a root (LHS 16, RHS 0). So the equation is $y^2 = x^2 + 1$, where y is the curve's ordinate. The roots are the intersections of the curve with the two branches $y = \pm\sqrt{x^2 + 1}$ of the rectangular hyperbola $y^2 - x^2 = 1$.

Upper branch. Let $f(x) = y - \sqrt{x^2 + 1}$. Then

$$f'(x) = \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{x}{\sqrt{x^2 + 1}} > 2 - 1 > 0,$$

so f is strictly increasing on each of the three intervals. On each one, y runs from $-\infty$ to $+\infty$ while $\sqrt{x^2 + 1}$ stays finite (it grows only like $|x|$, against $2x$ as $x \to +\infty$; and as $x \to -\infty$ both terms of f are negative). So f goes from $-\infty$ to $+\infty$, and there is exactly one root on each interval: three in all.

Lower branch. The map $(x, y) \mapsto (-x, -y)$ sends the curve to itself and the lower branch to the upper one. So there are also exactly three intersections with the lower branch.

The two sets of roots don't overlap: at a common x the curve has a single y, which can't be both positive and negative. So equation (ii) has exactly **six** real roots. (As a check, it is a polynomial of degree 6, namely $3x^6 - 33x^4 + 92x^2 - 16 = 0$, so all its roots are real.)

Ask yourself The variable in (iii) is z, not x, and z might be negative. How do we bring the curve in?

For $z > 0$ we put $z = x^2$ with $x > 0$. We deal with $z \le 0$ separately and directly.

Part (iii)

The left side is never negative, so any root must have $z + 1 \ge 0$, that is $z \ge -1$. Also $z = 0$ is not a root (LHS 0, RHS 16), and neither is $z = 4$ (LHS 64, RHS 0).

Roots with $z > 0$. Put $z = x^2$ with $x > 0$, so that each positive root z corresponds to exactly one $x > 0$. The equation becomes $4x^4(x^2 - 5)^2 = (x^2 - 4)^2(x^2 + 1)$. Dividing by $x^2(x^2 - 4)^2$ gives

$$y^2 = 1 + \frac{1}{x^2}, \qquad \text{that is} \qquad y = \pm s(x), \quad s(x) = \sqrt{1 + x^{-2}}.$$

Here s decreases from $+\infty$ (as $x \to 0^+$) to 1 (as $x \to \infty$), and $s'(x) = -\dfrac{1}{x^3\sqrt{1 + x^{-2}}}$, so $|s'(x)| < \frac{1}{x^3}$.

- $y = s(x)$: the function $y - s$ has derivative $y' - s' > 2$. On $(0, 2)$ it runs from $-\infty$ (as $y \to 0$ and $s \to \infty$) to $+\infty$. On $(2, \infty)$ it also runs from $-\infty$ to $+\infty$. One root on each, so two roots.
- $y = -s(x)$ on $(0, 2)$: the curve has $y > 0 > -s$ there. No roots.
- $y = -s(x)$ on $(2, \infty)$: the function $y + s$ has derivative $y' + s' > 2 - \frac{1}{8} > 0$ and runs from $-\infty$ to $+\infty$. One root.

So there are exactly three positive roots z.

Roots with $-1 \le z < 0$. Write $z = -t$ with $0 < t \le 1$. The equation becomes

$$4t^2(t + 5)^2 = (t + 4)^2(1 - t).$$

On $(0, 1]$ the left side is strictly increasing, from 0 (as $t \to 0$) to 144. The right side is strictly decreasing, from 16 to 0. Their difference is strictly increasing and changes sign, so there is exactly one root.

Hence equation (iii) has exactly **four** real roots: three positive and one in $(-1, 0)$. Expanded, it is the quartic $4z^4 - 41z^3 + 107z^2 - 8z - 16 = 0$, so all four of its roots are real.

Checks

- At $x = 1000$, $y - 2x \approx -0.002$, which confirms the approach to $y = 2x$ from below.
- Numerically, (i) has roots $x \approx -2.143$ and 2.401, one on each outer branch as predicted.

- (ii) has roots $x \approx \pm 0.431,\ \pm 2.155,\ \pm 2.484$, one per branch of the curve for each branch of the hyperbola.
- (iii) has roots $z \approx -0.332,\ 0.473,\ 4.799,\ 5.310$. The last three give $x = \sqrt{z} \approx 0.688,\ 2.191,\ 2.304$, with curve values $y \approx 1.77,\ -1.10,\ 1.09$, matching the case analysis.

Common slips

- Dividing by x or $x^2 - 4$ without first checking that $x = 0$ and $x = \pm 2$ aren't roots.
- In (ii), counting only the intersections with $y = +\sqrt{x^2 + 1}$ and forgetting the lower branch, which gives 3 instead of 6.
- In (iii), substituting $z = x^2$ and forgetting that z may be negative, which misses the root in $(-1, 0)$.

Beyond STEP

The counting arguments here rest on the intermediate value theorem (a continuous function that changes sign has a zero) together with strict monotonicity (it can have at most one). Making these two facts rigorous is one of the first jobs of a university analysis course. Abbott's *Understanding Analysis* treats both carefully and readably. Descartes' rule of signs and Sturm sequences are algebraic ways to count real roots of polynomials, and they make an interesting comparison with the geometric method used here.

WORKED SOLUTION

STEP III 1989, Question 5

Pure • Chebyshev Polynomials

STEP III 1989, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q5

The key idea

Put $x = \cosh\theta$. Then $e^{\pm\theta}$ are exactly $x \pm \sqrt{x^2-1}$, and $y = \cosh n\theta$ is the average of their nth powers. Expanding those powers with the binomial theorem, the odd powers of $\sqrt{x^2-1}$ cancel, so what is left is a polynomial in x. Once we have that polynomial in a tidy form, each coefficient is just a short sum of binomial coefficients.

Solution

The closed form

Let $\theta = \cosh^{-1} x$, so $\theta \geq 0$ and $x = \cosh\theta$. Then $\sinh\theta \geq 0$ and $\sinh^2\theta = x^2 - 1$, so $\sinh\theta = \sqrt{x^2-1}$. Hence

$$e^{\theta} = \cosh\theta + \sinh\theta = x + \sqrt{x^2-1}, \qquad e^{-\theta} = \cosh\theta - \sinh\theta = x - \sqrt{x^2-1}.$$

Therefore

$$y = \cosh n\theta = \frac{e^{n\theta} + e^{-n\theta}}{2} = \frac{(x+\sqrt{x^2-1})^n + (x-\sqrt{x^2-1})^n}{2},$$

as required.

Why y is an odd polynomial

Write $s = \sqrt{x^2-1}$. By the binomial theorem,

$$(x+s)^n + (x-s)^n = \sum_{j=0}^{n} \binom{n}{j} x^{n-j} s^j \left(1 + (-1)^j\right) = 2 \sum_{j \text{ even}} \binom{n}{j} x^{n-j} s^j.$$

The odd powers of s cancel, and every even power $s^{2m} = (x^2-1)^m$ is a polynomial in x. So

$$y = \sum_{m=0}^{\lfloor n/2 \rfloor} \binom{n}{2m} x^{n-2m} (x^2-1)^m.$$

Now take $n = 2k+1$, so m runs from 0 to k. Each term is an odd power $x^{2k+1-2m}$ times a polynomial in x^2, so it contains only odd powers of x, the highest being x^{2k+1}. Hence $y = a_0x + a_1x^3 + \cdots + a_kx^{2k+1}$.

To read off coefficients, put $j = k - m$ (so $j = 0, \ldots, k$) and use $\binom{2k+1}{2k-2j} = \binom{2k+1}{2j+1}$:

$$y = \sum_{j=0}^{k} \binom{2k+1}{2j+1} x^{2j+1}(x^2-1)^{k-j}.$$

The term with index j starts at x^{2j+1}. So x^{2r+1} comes only from the terms with $j \le r$, and from the jth of these we need the $(x^2)^{r-j}$ term of $(x^2-1)^{k-j}$, which is $\binom{k-j}{r-j}(-1)^{k-r}$. Hence

$$a_r = (-1)^{k-r} \sum_{j=0}^{r} \binom{2k+1}{2j+1}\binom{k-j}{r-j}. \qquad (*)$$

a_0

With $r = 0$, only $j = 0$ contributes:

$$a_0 = (-1)^k \binom{2k+1}{1} = (-1)^k(2k+1).$$

Part (i)

With $r = 1$, $j = 0, 1$ contribute:

$$\begin{aligned} a_1 &= (-1)^{k-1}\left[(2k+1)\binom{k}{1} + \binom{2k+1}{3}\right] = (-1)^{k-1}(2k+1)\left[k + \frac{2k(2k-1)}{6}\right] \\ &= (-1)^{k-1}(2k+1)\,k\,\frac{3+2k-1}{3} = (-1)^{k-1}\,\frac{2k(k+1)(2k+1)}{3}, \end{aligned}$$

as required.

Part (ii)

Here we need $k \ge 2$ for x^5 to appear. With $r = 2$, $j = 0, 1, 2$ contribute:

$$a_2 = (-1)^k\left[(2k+1)\binom{k}{2} + \binom{2k+1}{3}\binom{k-1}{1} + \binom{2k+1}{5}\right].$$

Each term has the factor $(2k+1)k(k-1)$; taking it out,

$$\begin{aligned} a_2 &= (-1)^k(2k+1)k(k-1)\left[\frac{1}{2} + \frac{2k-1}{3} + \frac{(2k-1)(2k-3)}{30}\right] \\ &= (-1)^k(2k+1)k(k-1)\,\frac{15 + (20k-10) + (4k^2-8k+3)}{30} \\ &= (-1)^k(2k+1)k(k-1)\,\frac{4(k+1)(k+2)}{30} = (-1)^k\,\frac{2(k-1)k(k+1)(k+2)(2k+1)}{15}. \end{aligned}$$

For the third term we used $\binom{2k+1}{5} = \frac{(2k+1)(2k)(2k-1)(2k-2)(2k-3)}{120} = \frac{(2k+1)k(k-1)(2k-1)(2k-3)}{30}$.

Ask yourself Our answer has a factor $(k+1)$ that the printed formula in (ii) lacks. Which one is right?

Test $k = 2$. Then $y = \cosh(5\theta)$ with $x = \cosh\theta$, and $\cosh 5\theta = 16x^5 - 20x^3 + 5x$, so $a_2 = 16$. Our formula gives $2 \cdot 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5/15 = 16$. The printed one gives $2 \cdot 1 \cdot 2 \cdot 4 \cdot 5/15 = \frac{16}{3}$, which can't be right because the a_r are integers. So the paper has a misprint: the factor $(k+1)$ is missing. The correct result is the one shown above.

The sum of the coefficients

Set $x = 1$ in the polynomial. Since the identity holds for $x \geq 1$ and $x = 1$ is allowed,

$$\sum_{r=0}^{k} a_r = y(1) = \cosh\bigl(n \cosh^{-1} 1\bigr) = \cosh 0 = 1.$$

Checks

- $k = 1$: $\cosh 3\theta = 4x^3 - 3x$. The formulas give $a_0 = -3$ and $a_1 = 2 \cdot 1 \cdot 2 \cdot 3/3 = 4$, and $-3 + 4 = 1$.
- $k = 3$: $\cosh 7\theta = 64x^7 - 112x^5 + 56x^3 - 7x$. The formulas give $a_0 = -7$, $a_1 = 56$ and $a_2 = -2 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 7/15 = -112$, and the coefficients sum to 1.
- We checked $(*)$ by computer against the recurrence $T_{m+1} = 2xT_m - T_{m-1}$ for every $k \leq 14$, and compared the polynomial with $\cosh(n \cosh^{-1} x)$ at random $x \geq 1$.

Common slips

- Forgetting that x^{2r+1} gets contributions from several terms of the expansion, not just one.
- Sign errors: every contribution to a_r carries the same sign $(-1)^{k-r}$, so factor it out first.
- Trying to force the algebra to reach the printed form of (ii). Testing a small case such as $k = 2$ shows it is misprinted.

Beyond STEP

The polynomial y is the Chebyshev polynomial $T_n(x)$, usually defined by $T_n(\cos\phi) = \cos n\phi$. Our hyperbolic version is the same identity for $x \geq 1$, and the two agree because a polynomial is fixed by its values on any interval. Chebyshev polynomials are central in approximation theory, since among monic polynomials of degree n, $2^{1-n}T_n$ has the smallest maximum size on $[-1, 1]$. They reappear in any first course in numerical analysis. For more STEP-style practice with this kind of algebra, try Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 1989, Question 6

Pure • Polar Equations of Conics

STEP III 1989, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q6

The key idea

In polar coordinates a line not through the origin has $1/r$ equal to a combination of $\cos\theta$ and $\sin\theta$, and writing it as $A\cos\theta + B\cos(\theta-\gamma)$ lets us match the conic's own $\cos\theta$ term exactly. Once the $e\cos\theta$ parts agree, the chord equation only has to fix up a single cosine, and symmetry about $\theta = \gamma$ does the rest. Letting the chord shrink to a point gives the tangent, and the tangent equations make the bisector property almost automatic.

Solution

Throughout, (r,θ) are polar coordinates with $x = r\cos\theta$, $y = r\sin\theta$, and we allow $r < 0$ (the point (r,θ) is the same as $(-r,\theta+\pi)$). The conic has its focus S at the origin.

Lines in polar form

Since $lx + my = 1$ cannot pass through the origin, $r \neq 0$ on it, and dividing by r gives

$$\frac{1}{r} = l\cos\theta + m\sin\theta.$$

Expanding the proposed form,

$$A\cos\theta + B\cos(\theta-\gamma) = (A + B\cos\gamma)\cos\theta + B\sin\gamma\,\sin\theta.$$

These agree for all θ if $B\sin\gamma = m$ and $A + B\cos\gamma = l$. Since $\sin\gamma \neq 0$ we may take

$$B = \frac{m}{\sin\gamma}, \qquad A = l - m\cot\gamma,$$

which is the required polar equation. Conversely, any equation $1/r = A\cos\theta + B\cos(\theta-\gamma)$ with A, B not making both coefficients vanish is a line $lx + my = 1$ with $l = A + B\cos\gamma$, $m = B\sin\gamma$. Note that the form is unchanged by $(r,\theta) \to (-r,\theta+\pi)$, so it describes the whole line whichever sign of r we use.

The chord PQ

Look for the chord in the form $a/r = A\cos\theta + B\cos(\theta - \gamma)$. At P and Q we have $\theta = \gamma \mp \delta$ and $a/r = 1 + e\cos\theta$, so we need

$$A\cos(\gamma \mp \delta) + B\cos(\mp\delta) = 1 + e\cos(\gamma \mp \delta).$$

Because $\cos(\mp\delta) = \cos\delta$ for both points, the choice $A = e$, $B\cos\delta = 1$ satisfies both equations at once, and $\cos\delta \neq 0$ lets us solve for B. Two distinct points determine a line, so the chord PQ is

$$\boxed{\frac{a}{r} = e\cos\theta + \sec\delta\,\cos(\theta - \gamma).}$$

(If $\cos\delta = 0$ then P and Q are at $\theta = \gamma \mp \frac{\pi}{2}$, and PQ is a focal chord; a line through the origin has no equation of the form $1/r = \ldots$, which is why that case is excluded.)

Ask yourself How do we know the chord meets the conic only at P and Q?

At each θ the conic has exactly one point, with $a/r = 1 + e\cos\theta$. The chord passes through it exactly when $1 + e\cos\theta = e\cos\theta + \sec\delta\cos(\theta - \gamma)$, that is, when $\cos(\theta - \gamma) = \cos\delta$, so $\theta = \gamma \pm \delta$. The $e\cos\theta$ terms cancel, which is the whole point of choosing $A = e$.

The tangent at $\theta = \gamma$

The tangent at the point G where $\theta = \gamma$ is the limit of the chord PQ as P and Q close in on G, that is, as $\delta \to 0$. The chord equation depends continuously on δ and $\sec\delta \to 1$, so the tangent is

$$\boxed{\frac{a}{r} = e\cos\theta + \cos(\theta - \gamma).}$$

As a confirmation, this meets the conic where $\cos(\theta - \gamma) = 1$, i.e. only at G, as a tangent should. More decisively, in cartesian form the tangent is $(e + \cos\gamma)x + (\sin\gamma)y = a$, with normal vector $(e + \cos\gamma,\ \sin\gamma)$. The conic is $r = a - ex$; differentiating along the curve gives $(x\,\mathrm{d}x + y\,\mathrm{d}y)/r = -e\,\mathrm{d}x$, i.e. $(e + x/r)\,\mathrm{d}x + (y/r)\,\mathrm{d}y = 0$. At G, $x/r = \cos\gamma$ and $y/r = \sin\gamma$, so the curve's normal there is the same vector.

The final part

Put the focus S at the origin, so the conic is $a/r = 1 + e\cos\theta$, and let L and M be the points with $\theta = \alpha$ and $\theta = \beta$. Write

$$\gamma = \tfrac{1}{2}(\alpha + \beta), \qquad \delta = \tfrac{1}{2}(\beta - \alpha),$$

so that L and M are at $\theta = \gamma - \delta$ and $\gamma + \delta$, exactly as for P and Q above. By the previous result the tangents at L and M are

$$\frac{a}{r} = e\cos\theta + \cos(\theta - \alpha), \qquad \frac{a}{r} = e\cos\theta + \cos(\theta - \beta).$$

At T both hold, so $\cos(\theta-\alpha)=\cos(\theta-\beta)$. Hence either $\theta-\alpha=\theta-\beta+2k\pi$, which would make $L=M$, or

$$\theta-\alpha=-(\theta-\beta)+2k\pi \quad\Longrightarrow\quad \theta=\tfrac{1}{2}(\alpha+\beta)+k\pi=\gamma+k\pi.$$

So T lies on the line through S in the direction $\theta=\gamma$, with

$$r_T=\frac{a}{e\cos\gamma+\cos\delta}$$

(the tangents are parallel, and T does not exist, exactly when $e\cos\gamma+\cos\delta=0$).

The rays SL and SM point in the directions $\alpha=\gamma-\delta$ and $\beta=\gamma+\delta$ (taking the usual case $r>0$ at L and M, as for an ellipse or parabola). The line $\theta=\gamma$ makes the angle δ with each of them, so the line ST bisects the angle LSM. (If $r_T<0$, T lies on the opposite ray, and ST bisects the reflex angle LSM; the angles LST and MST are still equal.)

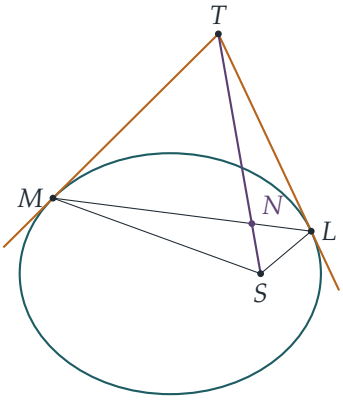
To find where ST meets LM, call the point N. The chord LM is $a/r=e\cos\theta+\sec\delta\cos(\theta-\gamma)$, and on ST we have $\theta=\gamma$, so

$$\frac{a}{SN}=e\cos\gamma+\sec\delta \quad\Longrightarrow\quad SN=\frac{a\cos\delta}{1+e\cos\gamma\cos\delta}.$$

Using $2\cos\gamma\cos\delta=\cos(\gamma-\delta)+\cos(\gamma+\delta)=\cos\alpha+\cos\beta$, this is

$$\boxed{N \text{ is at } \theta=\tfrac{1}{2}(\alpha+\beta),\quad r=\frac{2a\cos\frac{1}{2}(\beta-\alpha)}{2+e\cos\alpha+e\cos\beta}.}$$

When LM is a focal chord ($\cos\delta=0$) this gives $N=S$, as it must.



Checks

- Try a circle ($e=0$, radius a): the tangent becomes $a/r=\cos(\theta-\gamma)$, i.e. $x\cos\gamma+y\sin\gamma=a$, the familiar tangent to $x^2+y^2=a^2$.

- The angle bisector theorem in triangle LSM says $LN : NM = SL : SM$. With $a = 1.5$, $e = 0.6$, $\alpha = 40^\circ$, $\beta = 160^\circ$ (the figure), $SN \approx 0.791$, and the ratio checks numerically.
- A computer tested the chord and tangent equations, the direction of ST, the angle equality and the formula for SN on many thousands of random conics and points.

Common slips

- Solving $\cos(\theta - \alpha) = \cos(\theta - \beta)$ with only the $+$ sign. That branch gives nothing (it needs $\alpha = \beta$), and the useful solution is the $-$ sign.
- Not using the freedom in A: taking $A = e$ is what makes the $e\cos\theta$ terms cancel when intersecting with the conic.

Beyond STEP

The result that ST bisects $\angle LSM$ is a classical focal property of conics, and the line LM is the *polar* of T with respect to the conic. Pole–polar duality is the gateway to projective geometry, where all non-degenerate conics are equivalent and tangency and incidence become purely algebraic. Brannan, Esplen and Gray's *Geometry* is a good first-year account of conics and projective geometry.

WORKED SOLUTION

STEP III 1989, Question 7

Pure • Shear Matrices

STEP III 1989, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q7

The key idea

Everything is easiest in terms of two unit vectors: $\mathbf{u}$ along the line $L\colon y = x\tan\alpha$, and $\mathbf{n}$ perpendicular to it. The shear moves each point parallel to $\mathbf{u}$ by an amount proportional to its signed distance from L, so $\mathbf{M} = \mathbf{I} + k\,\mathbf{u}\mathbf{n}^{\mathsf{T}}$. For the second part, a matrix commutes with $\mathbf{I} + k\mathbf{A}$ exactly when it commutes with $\mathbf{A}$, which strips out the clutter.

Solution

Write $s = \sin\alpha$ and $c = \cos\alpha$. Since α is acute, $s > 0$ and $c > 0$. Let

$$\mathbf{u} = \begin{pmatrix} c \\ s \end{pmatrix}, \qquad \mathbf{n} = \begin{pmatrix} -s \\ c \end{pmatrix},$$

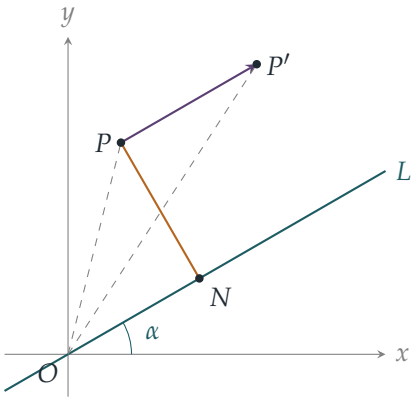
so $\mathbf{u}$ is the unit vector along L at angle α to the x-axis, and $\mathbf{n}$ is the unit normal pointing to the upper-left side of L.

Finding $\mathbf{M}$

For $P = (x, y)$ let

$$d = \mathbf{n}\cdot\overrightarrow{OP} = y\cos\alpha - x\sin\alpha,$$

the *signed* distance from P to L. Then $PN = |d|$, and $d > 0$ exactly when P lies above L.



Take P above L, so $d > 0$. By (i), $\overrightarrow{PP'}$ points along $\mathbf{u}$, so $\overrightarrow{PP'} = \lambda\mathbf{u}$ with $\lambda > 0$. Condition (ii) is consistent with this: the two-dimensional cross product is

$$\overrightarrow{OP} \times \overrightarrow{OP'} = \overrightarrow{OP} \times (\overrightarrow{OP} + \lambda\mathbf{u}) = \lambda\,(xs - yc) = -\lambda d < 0,$$

so the turn from OP to OP' is clockwise, as required. Condition (iii) gives $\lambda = k\,PN = kd$. (For these conditions to hold together we need $k > 0$.) Hence, for every P above L,

$$\overrightarrow{OP'} = \overrightarrow{OP} + kd\,\mathbf{u} = \overrightarrow{OP} + k\,\mathbf{u}\,(\mathbf{n} \cdot \overrightarrow{OP}).$$

Ask yourself We have only used points above L. Why does that fix $\mathbf{M}$ completely?

T is linear, and so is the map $\mathbf{v} \mapsto \mathbf{v} + k\,\mathbf{u}(\mathbf{n} \cdot \mathbf{v})$. They agree on the open half-plane above L, which contains two linearly independent vectors (for example $\mathbf{n}$ and $\mathbf{n} + \mathbf{u}$). Two linear maps that agree on a basis are equal. So for *all* vectors,

$$\mathbf{M} = \mathbf{I} + k\,\mathbf{u}\mathbf{n}^{\mathsf{T}} = \mathbf{I} + k\begin{pmatrix} c \\ s \end{pmatrix}\begin{pmatrix} -s & c \end{pmatrix} = \mathbf{I} + k\begin{pmatrix} -sc & c^2 \\ -s^2 & sc \end{pmatrix}.$$

That is,

$$\mathbf{M} = \begin{pmatrix} 1 - k\sin\alpha\cos\alpha & k\cos^2\alpha \\ -k\sin^2\alpha & 1 + k\sin\alpha\cos\alpha \end{pmatrix},$$

as required. As a sanity check, $\mathbf{n} \cdot \mathbf{u} = 0$ gives $\mathbf{M}\mathbf{u} = \mathbf{u}$, so L is fixed pointwise. Also $\det\mathbf{M} = 1 - k^2s^2c^2 + k^2s^2c^2 = 1$, as a shear should have.

The commuting condition

Let $\mathbf{A} = \mathbf{u}\mathbf{n}^{\mathsf{T}} = \begin{pmatrix} -sc & c^2 \\ -s^2 & sc \end{pmatrix}$ and $\mathbf{X} = \begin{pmatrix} p & q \\ r & t \end{pmatrix}$. Since $\mathbf{M} = \mathbf{I} + k\mathbf{A}$ and $\mathbf{I}$ commutes with everything,

$$\mathbf{X}\mathbf{M} - \mathbf{M}\mathbf{X} = k(\mathbf{X}\mathbf{A} - \mathbf{A}\mathbf{X}).$$

Because $k \neq 0$, $\mathbf{X}$ commutes with $\mathbf{M}$ if and only if it commutes with $\mathbf{A}$. Multiplying out,

$$\mathbf{XA} = \begin{pmatrix} -psc - qs^2 & pc^2 + qsc \\ -rsc - ts^2 & rc^2 + tsc \end{pmatrix}, \qquad \mathbf{AX} = \begin{pmatrix} -psc + rc^2 & -qsc + tc^2 \\ -ps^2 + rsc & -qs^2 + tsc \end{pmatrix},$$

so

$$\mathbf{XA} - \mathbf{AX} = \begin{pmatrix} -(qs^2 + rc^2) & -c^2(t-p) + 2qsc \\ -s^2(t-p) - 2rsc & qs^2 + rc^2 \end{pmatrix}.$$

This is zero if and only if

$$\begin{aligned} &\text{(a)} \quad qs^2 + rc^2 = 0, \\ &\text{(b)} \quad c^2(t-p) = 2qsc, \\ &\text{(c)} \quad s^2(t-p) = -2rsc. \end{aligned}$$

Since $c \neq 0$, (b) is $t - p = 2q\tan\alpha$. Since $s \neq 0$, (c) is $t - p = -2r\cot\alpha$.

Ask yourself There are three equations but the stated condition contains only two. What happened to (a)?

It follows from the other two. If $2q\tan\alpha = -2r\cot\alpha$, multiplying by $\frac{1}{2}sc$ gives $qs^2 = -rc^2$, which is (a). So

$$\mathbf{XM} = \mathbf{MX} \iff t - p = 2q\tan\alpha = -2r\cot\alpha.$$

Both directions hold: every step above is an equivalence. This is the required necessary and sufficient condition.

Checks

- With $\alpha = \frac{\pi}{4}$, $k = 2$ we get $\mathbf{M} = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}$. The point $(0,1)$, at distance $\frac{1}{\sqrt{2}}$ above $y = x$, goes to $(1,2)$. That is a move of $\sqrt{2} = 2 \times \frac{1}{\sqrt{2}}$ in direction $45°$, and the turn is clockwise.
- The condition is satisfied by $\mathbf{X} = \mathbf{M}$ itself: $t - p = 2ksc$, $2q\tan\alpha = 2kc^2 \cdot \frac{s}{c} = 2ksc$ and $-2r\cot\alpha = 2ks^2 \cdot \frac{c}{s} = 2ksc$. It is also satisfied by $\mathbf{I}$, with all three expressions zero.
- Numerically, for random α, k and P above L, $\mathbf{M}$ reproduced (i)–(iii). Random matrices satisfying the condition commuted with $\mathbf{M}$ to within 10^{-8}, and random matrices violating it did not.

Common slips

- Using the unsigned distance PN for every point. The displacement is k times the *signed* distance. Points below L move the other way, which is what makes T linear.
- Proving only "if" (or only "only if") for the commuting condition, or forgetting to show that the $(1,1)$ entry gives nothing new.

Beyond STEP

The matrices that commute with $\mathbf{M}$ here are exactly those of the form $a\mathbf{I} + b\mathbf{A}$: check that $t - p = 2q\tan\alpha = -2r\cot\alpha$ is a two-parameter family. This is an instance of a general fact. For a 2×2 matrix that is not a multiple of $\mathbf{I}$, the matrices commuting with it are exactly $a\mathbf{I} + b\mathbf{M}$. For $n \times n$ matrices, the analogue (commuting matrices are polynomials in $\mathbf{M}$) holds when the minimal polynomial equals the characteristic polynomial. Strang's *Introduction to Linear Algebra* develops rank-one matrices $\mathbf{u}\mathbf{n}^{\mathsf{T}}$ and the ideas behind this.

WORKED SOLUTION

STEP III 1989, Question 8

Pure • Coupled Differential Equations

STEP III 1989, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q8

The key idea

The first equation gives y in terms of x and $\mathrm{d}x/\mathrm{d}t$. So if we differentiate it once more, the $\mathrm{d}y/\mathrm{d}t$ that appears can be replaced using the second equation, and y drops out entirely. That leaves a second-order linear equation with constant coefficients. Its auxiliary equation has a repeated root, and one of the forcing terms matches that root. This resonance is the real test in the question.

Solution

Eliminating y

Write dots for derivatives with respect to t. Differentiating $\dot{x} = 4(x - y)$ gives

$$\ddot{x} = 4\dot{x} - 4\dot{y} = 4\dot{x} - 4x + 48\left(\mathrm{e}^{2t} + \mathrm{e}^{-2t}\right),$$

using the second equation for $\dot{y}$. Hence

$$\boxed{\ddot{x} - 4\dot{x} + 4x = 48\left(\mathrm{e}^{2t} + \mathrm{e}^{-2t}\right),}$$

which contains no y.

Solving for x

Complementary function. The auxiliary equation $m^2 - 4m + 4 = (m-2)^2 = 0$ has the repeated root $m = 2$, so the complementary function is

$$x_{\mathrm{c}} = (A + Bt)\,\mathrm{e}^{2t}.$$

Particular integral. Since the equation is linear, we can deal with the two forcing terms one at a time.

For $48\mathrm{e}^{-2t}$, try $x = D\mathrm{e}^{-2t}$. Substituting gives $D(4 + 8 + 4)\mathrm{e}^{-2t} = 48\mathrm{e}^{-2t}$, so $D = 3$.

Ask yourself Why won't Ce^{2t}, or even Cte^{2t}, work for the $48e^{2t}$ term?

Both e^{2t} and te^{2t} are already in the complementary function, so the left-hand side sends them to 0. Because 2 is a *double* root, we have to go up to t^2e^{2t}. Try $x = Ct^2e^{2t}$. A neat way to substitute is to write $x = ue^{2t}$. Then

$$\dot{x} = (\dot{u} + 2u)e^{2t}, \qquad \ddot{x} = (\ddot{u} + 4\dot{u} + 4u)e^{2t},$$

so

$$\ddot{x} - 4\dot{x} + 4x = \ddot{u}\,e^{2t}.$$

With $u = Ct^2$ this is $2Ce^{2t}$, so $2C = 48$ and $C = 24$. (The substitution also shows why the complementary function is $(A + Bt)e^{2t}$: we need $\ddot{u} = 0$.)

So the general solution is

$$x = (A + Bt)\,e^{2t} + 24t^2e^{2t} + 3e^{-2t}.$$

Initial conditions. We are given only $x(0) = 0$ and $y(0) = 0$. The second-order equation needs $\dot{x}(0)$, and the first original equation supplies it: $\dot{x}(0) = 4\big(x(0) - y(0)\big) = 0$.

From $x(0) = 0$ we get $A + 3 = 0$, so $A = -3$. Differentiating,

$$\dot{x} = \big(B + 2A + 2Bt + 48t + 48t^2\big)e^{2t} - 6e^{-2t},$$

so $\dot{x}(0) = B + 2A - 6 = 0$, giving $B = 12$. Therefore

$$\boxed{x = \Big(24t^2 + 12t - 3\Big)\,e^{2t} + 3e^{-2t}.}$$

Finding y

There is no need to solve another differential equation, because the first equation gives $y = x - \frac{1}{4}\dot{x}$. With $A = -3$ and $B = 12$,

$$\dot{x} = \Big(48t^2 + 72t + 6\Big)\,e^{2t} - 6e^{-2t},$$

so

$$\begin{aligned} y &= \Big(24t^2 + 12t - 3\Big)\,e^{2t} + 3e^{-2t} - \Big(12t^2 + 18t + \tfrac{3}{2}\Big)\,e^{2t} + \tfrac{3}{2}e^{-2t} \\ &= \Big(12t^2 - 6t - \tfrac{9}{2}\Big)\,e^{2t} + \tfrac{9}{2}e^{-2t}. \end{aligned}$$

That is,

$$\boxed{y = \tfrac{3}{2}\Big(8t^2 - 4t - 3\Big)\,e^{2t} + \tfrac{9}{2}e^{-2t}.}$$

Checks

- At $t = 0$: $y(0) = -\frac{9}{2} + \frac{9}{2} = 0$. This was never imposed directly, so it is a genuine check.
- The second equation, which we used only to eliminate y, should hold too. Differentiating gives
$$\dot{y} = \left(24t^2 + 12t - 15\right)\mathrm{e}^{2t} - 9\mathrm{e}^{-2t},$$
and
$$x - 12\left(\mathrm{e}^{2t} + \mathrm{e}^{-2t}\right) = \left(24t^2 + 12t - 15\right)\mathrm{e}^{2t} - 9\mathrm{e}^{-2t}.$$
These agree.
- Numerically, stepping the original pair forward from $(0, 0)$ with a Runge–Kutta method gives $x(1) \approx 244.2449$ and $y(1) \approx 11.6926$. The formulae give $33\mathrm{e}^2 + 3\mathrm{e}^{-2}$ and $\frac{3}{2}\mathrm{e}^2 + \frac{9}{2}\mathrm{e}^{-2}$, which have the same values.

Common slips

- Trying $Ct\,\mathrm{e}^{2t}$ for the resonant term. The root 2 is repeated, so we need $t^2\mathrm{e}^{2t}$.
- Not seeing where the second initial condition comes from. The value $\dot{x}(0) = 0$ is hidden in the first equation together with $x(0) = y(0)$.
- Solving a separate second-order equation for y. That brings in two new constants, which then have to be matched to x. Using $y = x - \frac{1}{4}\dot{x}$ avoids this.

Beyond STEP

Eliminating y is one way to solve a linear system. The undergraduate approach writes it as $\dot{\mathbf{v}} = M\mathbf{v} + \mathbf{f}(t)$ with $M = \begin{pmatrix} 4 & -4 \\ 1 & 0 \end{pmatrix}$. Here M has the repeated eigenvalue 2 but only one eigenvector, so it cannot be diagonalised, and that is exactly where the $t\,\mathrm{e}^{2t}$ terms come from. The Jordan normal form and the matrix exponential e^{Mt} make this precise. Boyce and DiPrima, *Elementary Differential Equations*, covers systems of linear equations, including the repeated-eigenvalue case.

WORKED SOLUTION

STEP III 1989, Question 9

Pure • Summing Power Series

STEP III 1989, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q9

The key idea

Each series is a familiar power series $\sum a_r x^r$ evaluated at a particular x, so the job is to spot which one. The geometric series $\frac{1}{1-x} = 1 + x + x^2 + \cdots$ is the parent of the first two. Differentiating it brings down a factor r, and integrating it divides by r. For the third, the products of odd numbers over factorials are what the binomial expansion with a negative fractional power produces.

Solution

Throughout we use, for $|x| < 1$,

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots = \sum_{r=1}^{\infty} x^{r-1}.$$

The question tells us not to worry about convergence, so we freely differentiate and integrate term by term. (All three values of x we use lie well inside the interval of convergence, so this is in fact legitimate.)

Part (i)

The general term is $r\left(\frac{1}{2}\right)^{r-1}$, so we want $\sum r x^{r-1}$ at $x = \frac{1}{2}$. Differentiating the geometric series term by term,

$$\sum_{r=1}^{\infty} r x^{r-1} = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{1-x}\right) = \frac{1}{(1-x)^2}.$$

Putting $x = \frac{1}{2}$:

$$1 + \frac{2}{2} + \frac{3}{2^2} + \frac{4}{2^3} + \cdots = \frac{1}{\left(\frac{1}{2}\right)^2} = 4.$$

Part (ii)

The general term is $\frac{1}{r}\left(\frac{1}{2}\right)^{r-1}$. Dividing by r is undone by differentiation, so this time we integrate. Integrating the geometric series from 0 to x,

$$\sum_{r=1}^{\infty} \frac{x^r}{r} = \int_0^x \frac{\mathrm{d}t}{1-t} = -\ln(1-x).$$

The powers here are one too high, so we divide by x (for $x \neq 0$):

$$\sum_{r=1}^{\infty} \frac{1}{r} x^{r-1} = -\frac{\ln(1-x)}{x}.$$

Putting $x = \frac{1}{2}$:

$$1 + \frac{1}{2}\times\frac{1}{2} + \frac{1}{3}\times\frac{1}{2^2} + \cdots = -\frac{\ln\frac{1}{2}}{\frac{1}{2}} = 2\ln 2.$$

Part (iii)

Write

$$c_k = \frac{1\times 3\times\cdots\times(2k-1)}{k!}, \qquad c_0 = 1.$$

Ask yourself Where have you seen products of odd numbers divided by $k!$ before?

In the binomial expansion of $(1+y)^{-1/2}$. The coefficient of y^k is

$$\binom{-\frac{1}{2}}{k} = \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\cdots\left(-\frac{2k-1}{2}\right)}{k!} = \frac{(-1)^k}{2^k}\cdot\frac{1\times 3\times\cdots\times(2k-1)}{k!} = \left(-\tfrac{1}{2}\right)^k c_k.$$

So, taking $y = -2x$, the signs and the powers of 2 cancel:

$$(1-2x)^{-1/2} = \sum_{k=0}^{\infty}\binom{-\frac{1}{2}}{k}(-2x)^k = \sum_{k=0}^{\infty} c_k x^k = 1 + x + \frac{1\times 3}{2!}x^2 + \frac{1\times 3\times 5}{3!}x^3 + \cdots,$$

valid for $|2x| < 1$.

Now look carefully at the given series. Its general term is $c_k \cdot \frac{1}{3^{k-1}} = 3c_k\left(\frac{1}{3}\right)^k$, and it starts at $k = 2$ (the first term is $\frac{1\times 3}{2!}\times\frac{1}{3}$). So it is 3 times the series above at $x = \frac{1}{3}$, with the $k = 0$ and $k = 1$ terms removed:

$$\begin{aligned}\sum_{k=2}^{\infty} c_k \frac{1}{3^{k-1}} &= 3\left(\sum_{k=0}^{\infty} c_k\left(\tfrac{1}{3}\right)^k - c_0 - c_1\cdot\tfrac{1}{3}\right)\\ &= 3\left(\left(1-\tfrac{2}{3}\right)^{-1/2} - 1 - \tfrac{1}{3}\right)\\ &= 3\left(\sqrt{3} - \tfrac{4}{3}\right)\\ &= 3\sqrt{3} - 4.\end{aligned}$$

Here $\left(\frac{1}{3}\right)^{-1/2} = \sqrt{3}$, and $x = \frac{1}{3}$ satisfies $|2x| < 1$.

Checks

- Part (i): the partial sums 1, 2, 2.75, 3.25, 3.5625, … creep up towards 4, and summing 200 terms gives 4 to machine accuracy.
- Part (ii): $2\ln 2 \approx 1.3863$, and the partial sums 1, 1.25, 1.3333, 1.3646, … agree.
- Part (iii): the terms are 0.5, 0.2778, 0.1620, …, and summing 400 terms gives 1.196152…, matching $3\sqrt{3} - 4 \approx 1.1962$. A quick sanity check: the answer must exceed the first term 0.5 and be positive, which rules out forgetting to remove the $k = 1$ term (that would give $3\sqrt{3} - 3 \approx 2.196$, too big since the terms shrink roughly geometrically).

Common slips

- In (ii), forgetting to divide by x: $-\ln(1 - x)$ at $x = \frac{1}{2}$ gives $\ln 2$, which is half the answer.
- In (iii), not noticing that the series starts at $k = 2$, or missing the extra factor of 3 from $\frac{1}{3^{k-1}} = 3 \cdot \frac{1}{3^k}$.

Beyond STEP

Differentiating and integrating a power series term by term is justified inside its radius of convergence, because a power series converges uniformly on every closed interval strictly inside that radius. This is a standard result in first-year analysis, and it is why "questions of convergence need not be considered" is safe here. The binomial series for non-integer powers is a power series of exactly this kind. Abbott's *Understanding Analysis* covers uniform convergence and power series clearly.

WORKED SOLUTION

STEP III 1989, Question 10

Pure • Riemann Sums

STEP III 1989, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q10

This question has a full guided treatment in *Beyond STEP*, Section 4.1, *What Is an Integral?*: https://beyondstep.keithmcnulty.org/ch04/riemann.html

The key idea

A product of consecutive integers is the "falling power" version of r^5, and it sums by telescoping, exactly as r^5 integrates: the difference of two consecutive six-term products is six times a five-term product. Parts (i) and (ii) give a product sum just above and just below $\sum r^5$. Part (iii) then squeezes the two Riemann sums of x^5 between these bounds, and both bounds tend to the same limit.

Solution

Part (i)

Write $P(r) = r(r+1)(r+2)(r+3)(r+4)(r+5)$. Then $P(r)$ and $P(r-1) = (r-1)r(r+1)(r+2)(r+3)(r+4)$ share the five factors $r(r+1)\cdots(r+4)$, so

$$\begin{aligned} P(r) - P(r-1) &= r(r+1)(r+2)(r+3)(r+4)\big[(r+5) - (r-1)\big] \\ &= 6\,r(r+1)(r+2)(r+3)(r+4). \end{aligned}$$

Summing from $r = 1$ to n, the right-hand side telescopes:

$$\begin{aligned} \sum_{r=1}^{n} r(r+1)(r+2)(r+3)(r+4) &= \tfrac{1}{6}\sum_{r=1}^{n}\big[P(r) - P(r-1)\big] = \tfrac{1}{6}\big[P(n) - P(0)\big] \\ &= \tfrac{1}{6}\,n(n+1)(n+2)(n+3)(n+4)(n+5), \end{aligned}$$

since $P(0) = 0$. (Induction on n works equally well; the inductive step is the same factorisation.)

For the deduction: if $r \geq 1$ then each factor satisfies $r + k > r > 0$ for $k = 1, 2, 3, 4$, so

$$r^5 < r(r+1)(r+2)(r+3)(r+4) \qquad (r \geq 1).$$

Adding these strict inequalities for $r = 1, \ldots, n$ gives $\sum_{r=1}^{n} r^5 < \frac{1}{6}n(n+1)(n+2)(n+3)(n+4)(n+5)$, as required.

Part (ii)

Now we want a product that sits *below* r^5, so we shift the factors down: use $(r-4)(r-3)(r-2)(r-1)r$. With $Q(r) = (r-4)(r-3)(r-2)(r-1)r(r+1)$, the same factorisation gives

$$\begin{aligned} Q(r) - Q(r-1) &= (r-4)(r-3)(r-2)(r-1)r\big[(r+1) - (r-5)\big] \\ &= 6\,(r-4)(r-3)(r-2)(r-1)r. \end{aligned}$$

Summing from $r = 0$ to $n-1$:

$$\begin{aligned} \sum_{r=0}^{n-1} (r-4)(r-3)(r-2)(r-1)r &= \tfrac{1}{6}\big[Q(n-1) - Q(-1)\big] \\ &= \tfrac{1}{6}\,(n-5)(n-4)(n-3)(n-2)(n-1)n, \end{aligned}$$

because $Q(-1) = (-5)(-4)(-3)(-2)(-1)\cdot 0 = 0$.

Ask yourself Is $(r-4)(r-3)(r-2)(r-1)r < r^5$ true for every r in the sum?

Not quite, and this is where $n > 1$ comes in. We compare term by term:

- $r = 0$: both sides are 0, so we only have equality.
- $r = 1, 2, 3, 4$: the product contains a zero factor, so it is $0 < r^5$.
- $r \geq 5$: every factor satisfies $0 < r - k < r$ for $k = 1, 2, 3, 4$, so the product is $< r^5$.

So $(r-4)(r-3)(r-2)(r-1)r \leq r^5$ for all $r \geq 0$, with strict inequality for every $r \geq 1$. If $n > 1$, the sum from $r = 0$ to $n-1$ includes the term $r = 1$, so at least one inequality is strict, and

$$\sum_{r=0}^{n-1} r^5 > \tfrac{1}{6}(n-5)(n-4)(n-3)(n-2)(n-1)n.$$

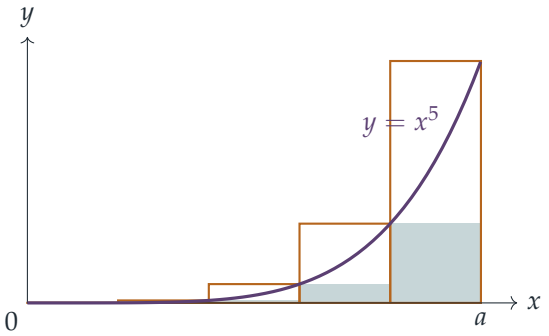
(For $n = 1$ both sides are 0, which is why the condition is needed.)

Part (iii)

Take $f(x) = x^5$, which is increasing. The two sums in the definition are

$$L_n = \sum_{r=0}^{n-1} \frac{a}{n}\Big(\frac{ra}{n}\Big)^5 = \frac{a^6}{n^6}\sum_{r=0}^{n-1} r^5, \qquad U_n = \sum_{r=1}^{n} \frac{a}{n}\Big(\frac{ra}{n}\Big)^5 = \frac{a^6}{n^6}\sum_{r=1}^{n} r^5.$$

For $a > 0$ these are the areas of the rectangles below and above the curve:



(Shaded: L_n; outlined: U_n; here $n = 5$.) The algebra below does not need $a > 0$, though: $a^6 > 0$ for any $a \neq 0$, and $a = 0$ is trivial since every sum is 0.

Using part (ii) for L_n (with $n > 1$) and part (i) for U_n,

$$L_n > \frac{a^6}{6n^6}(n-5)(n-4)(n-3)(n-2)(n-1)n = \frac{a^6}{6}\prod_{k=1}^{5}\left(1-\frac{k}{n}\right),$$

$$U_n < \frac{a^6}{6n^6}n(n+1)(n+2)(n+3)(n+4)(n+5) = \frac{a^6}{6}\prod_{k=1}^{5}\left(1+\frac{k}{n}\right).$$

Also $U_n - L_n = \dfrac{a^6}{n^6}\left(n^5 - 0^5\right) = \dfrac{a^6}{n} > 0$, so altogether

$$\frac{a^6}{6}\prod_{k=1}^{5}\left(1-\frac{k}{n}\right) < L_n < U_n < \frac{a^6}{6}\prod_{k=1}^{5}\left(1+\frac{k}{n}\right).$$

As $n \to \infty$, each bracket tends to 1, so both outer bounds tend to $\frac{1}{6}a^6$. By the squeeze (sandwich) theorem, L_n and U_n both converge, and to the same limit $\frac{1}{6}a^6$. So both limits in the definition exist and are equal, and therefore

$$\int_0^a x^5 \,\mathrm{d}x = \tfrac{1}{6}a^6.$$

Checks

- Small cases of (i): for $n = 1$, $1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 120 = \frac{1}{6} \cdot 720$; for $n = 2$, $120 + 720 = 840 = \frac{1}{6} \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7$.
- The identities in (i) and (ii), and both inequalities, hold exactly for every n up to 200 (checked by computer).
- With $a = 1$ and $n = 1000$: $L_n \approx 0.166167$ and $U_n \approx 0.167167$, either side of $\frac{1}{6} \approx 0.166667$, and $U_n - L_n = 0.001 = a^6/n$.

Common slips

- Claiming $(r-4)\cdots r < r^5$ for all $r \geq 0$: at $r = 0$ it is an equality, so the strict inequality needs the $r = 1$ term, that is, $n > 1$.
- Showing only that L_n and U_n are bounded, or only that $U_n - L_n \to 0$. The definition needs both limits to *exist*; the squeeze gives existence and the value together.

Beyond STEP

Part (iii) is a hands-on version of the Riemann (or Darboux) integral: a function is integrable when its lower and upper sums can be squeezed together, and for a monotonic function the gap is always $(b-a)\,|\mathrm{f}(b) - \mathrm{f}(a)|/n$, so every monotonic function on a closed interval is integrable. The products in (i) and (ii) are "rising" and "falling factorial" powers, which make the calculus of finite differences look just like ordinary calculus. Abbott's *Understanding Analysis* develops the integral this way.

WORKED SOLUTION

STEP III 1989, Question 11

Mechanics • Oblique Impact of Spheres

STEP III 1989, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q11

The key idea

First pin down the geometry at the moment of impact: it fixes the line of centres, and everything else follows from it. Then use the standard recipe for an oblique impact of smooth spheres when one of them is constrained. The impulse between the spheres acts along the line of centres. Newton's law of restitution applies to velocity components along that line. And instead of conservation of momentum, which fails because the string or the pivot supplies an impulse of its own, we use whichever equation the constraint leaves untouched: horizontal momentum for A with the string, angular momentum about O with the rod.

Solution

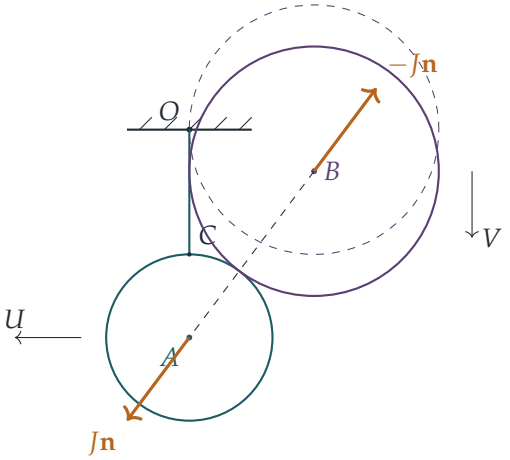
Geometry at impact

We model both spheres as smooth and uniform, and the string as light and inextensible. Before the impact the string hangs vertically. Since C lies on the line OA, we have $OA = 3a + 2a = 5a$, with A directly below O.

Sphere B starts with $OB = 3a$ horizontal and falls vertically. Let it fall a distance h. At impact the spheres touch, so $AB = 2a + 3a = 5a$. The horizontal separation of A and B is $3a$, so their vertical separation is $\sqrt{25a^2 - 9a^2} = 4a$. Hence B is $4a$ above A, or a below the level of O, and $h = a$.

So the line of centres is a 3–4–5 triangle: from B to A it goes $3a$ horizontally (towards OA) and $4a$ down. Let $\mathbf{n}$ be the unit vector from B towards A, and take the x-axis horizontal, pointing from OA towards B. Then

$$\mathbf{n} = \left(-\tfrac{3}{5}, -\tfrac{4}{5}\right).$$



The dashed sphere marks where B starts, and the solid one where it is at impact.

Impact with the string

Let J be the magnitude of the impulse between the spheres, so A receives $J\mathbf{n}$ and B receives $-J\mathbf{n}$. The spheres are smooth, so $J\mathbf{n}$ acts along the line of centres and passes through A. The string's impulse acts along OC, which also passes through A. So nothing gives sphere A any spin, and A moves as a particle on a circle of radius $5a$ about O. Immediately after the impact the string is still vertical, so A moves horizontally, with speed U in the direction of negative x.

Sphere A, horizontally. The string gives no horizontal impulse, so

$$3mU = \tfrac{3}{5}J, \qquad \text{that is,} \qquad J = 5mU.$$

(Vertically, the string supplies an upward impulse $\frac{4}{5}J$ that cancels the downward part of $J\mathbf{n}$. This is positive, so the string really does stay taut.)

Sphere B. Before the impact $\mathbf{v}_B = (0, -V)$, whose component along $\mathbf{n}$ is $\frac{4}{5}V$. The impulse $-J\mathbf{n}$ changes only this component, and it reduces it by $J/(25m)$:

$$\mathbf{v}'_B \cdot \mathbf{n} = \tfrac{4}{5}V - \frac{J}{25m} = \tfrac{4}{5}V - \tfrac{1}{5}U.$$

Ask yourself Why can't we just conserve momentum along the line of centres?

The string's impulse on A is vertical, and that has a component along $\mathbf{n}$. So the total momentum along $\mathbf{n}$ changes. The only direction in which no external impulse acts is horizontal, and that is the equation we used for A.

Restitution. Along $\mathbf{n}$, the speed of approach is $\frac{4}{5}V - 0$, since A starts at rest. The speed of separation is $\mathbf{v}'_A \cdot \mathbf{n} - \mathbf{v}'_B \cdot \mathbf{n}$, where $\mathbf{v}'_A \cdot \mathbf{n} = (-U)(-\frac{3}{5}) = \frac{3}{5}U$. So

$$\tfrac{3}{5}U - \left(\tfrac{4}{5}V - \tfrac{1}{5}U\right) = \tfrac{3}{4} \cdot \tfrac{4}{5}V \quad \Longrightarrow \quad \tfrac{4}{5}U = \tfrac{7}{5}V \quad \Longrightarrow \quad U = \tfrac{7}{4}V.$$

Impact with the rod

Now the sphere and the rod form a single rigid body turning about O. The pivot can supply an impulse in any direction, so horizontal momentum is no longer conserved. But the pivot's impulse acts at O, so we take moments about O. Its moment of inertia about O is, by the parallel axis theorem,

$$I = \tfrac{2}{5}(3m)(2a)^2 + 3m(5a)^2 = \tfrac{24}{5}ma^2 + 75ma^2 = \tfrac{399}{5}ma^2.$$

The impulse $J\mathbf{n}$ acts through A, which is a depth $5a$ below O. Only its horizontal component $\frac{3}{5}J$ has a moment about O, and that moment is $5a \cdot \frac{3}{5}J = 3aJ$. If the body's angular speed just after the impact is ω, then A moves horizontally with speed $U' = 5a\omega$, and

$$I\omega = 3aJ \quad\Longrightarrow\quad \tfrac{399}{5}ma^2 \cdot \frac{U'}{5a} = 3aJ \quad\Longrightarrow\quad J = \tfrac{133}{25}\,mU'.$$

The equations for B and for restitution are exactly as before, because they involve only the components along $\mathbf{n}$. So

$$\tfrac{3}{5}U' - \left(\tfrac{4}{5}V - \frac{J}{25m}\right) = \tfrac{3}{5}V \quad\Longrightarrow\quad \left(\tfrac{3}{5} + \tfrac{133}{625}\right)U' = \tfrac{7}{5}V \quad\Longrightarrow\quad \tfrac{508}{625}U' = \tfrac{7}{5}V,$$

so $U' = \frac{875}{508}V$. Comparing this with $U = \frac{7}{4}V$,

$$\frac{U'}{U} = \frac{875}{508} \cdot \frac{4}{7} = \frac{500}{508} = \frac{125}{127},$$

so the speed of A immediately after the impact is $125U/127$, as required.

It makes sense that the answer is a little smaller than before. With the string, A resists horizontal motion like a particle of mass $3m$. With the rod, the sphere must also start spinning, and its effective horizontal mass at A becomes $I/(5a)^2 = \frac{399}{125}m$, which is slightly more than $3m$.

Checks

- Energy: with $m = a = V = 1$, the kinetic energy is 12.5 before the impact. Afterwards it is 11.625 with the string and about 11.58 with the rod. Both are smaller, as they must be when $e < 1$.
- Elastic limit: with $e = 1$ and the string, the same steps give $\frac{4}{5}U = \frac{8}{5}V$, so $U = 2V$, and kinetic energy is conserved exactly. This confirms the signs.
- A computer solved both impacts in exact fractions, with $a = m = V = 1$. It found $U = 7/4$ and $U' = 875/508$, giving the ratio $125/127$. It also checked that B's tangential velocity is unchanged and that B falls a distance a before contact.

Common slips

- Conserving total momentum along the line of centres. The string or pivot supplies an external impulse with a component in that direction.

- Forgetting that the sphere on the rod rotates. Its moment of inertia about O needs the $\frac{2}{5}(3m)(2a)^2$ term as well as $3m(5a)^2$.
- Taking the line of centres at $45°$, or as vertical. The touching condition fixes it as a 3–4–5 triangle.

Beyond STEP

Constrained impacts are a first taste of *impulsive motion of rigid bodies*. The general rule is that an impulse at a fixed pivot drops out of the angular momentum equation about that pivot. This leads on to the idea of the *centre of percussion*: the point where you can strike a pivoted body so that the pivot feels no impulse at all, which is why a cricket bat has a "sweet spot". Kleppner and Kolenkow's *An Introduction to Mechanics* treats angular momentum and rigid-body impulses at exactly this level.

WORKED SOLUTION

STEP III 1989, Question 12

Mechanics • Rotating Frames

STEP III 1989, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q12

The key idea

The plane is smooth and the rod is light and freely pivoted, so the only horizontal force on P acts along the rod. The acceleration of P perpendicular to AP must therefore be zero. Once we have worked out that component, setting it to zero gives a pendulum equation for ϕ, and the usual "multiply by $\dot{\phi}$ and integrate" trick gives $\dot{\phi}$.

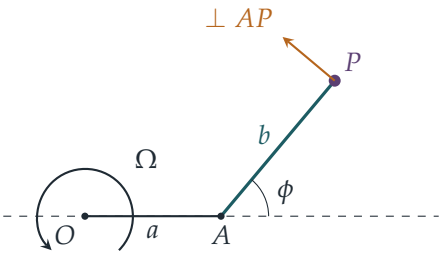
Solution

Model and set-up

Model P as a particle on a smooth plane and the rod as light and rigid, freely pivoted at A. Work in a fixed (inertial) frame. Use complex numbers for positions in the plane, with O at the origin and the real axis along OA at $t = 0$, and take $\Omega > 0$ anticlockwise. Since A is fixed in the plane, it moves on a circle of radius a, and AP makes angle ϕ with OA produced, which itself has turned through Ωt. So, writing $\theta = \Omega t + \phi$,

$$z_A = a\,\mathrm{e}^{\mathrm{i}\Omega t}, \qquad z_P = a\,\mathrm{e}^{\mathrm{i}\Omega t} + b\,\mathrm{e}^{\mathrm{i}\theta}.$$

Here $\mathrm{e}^{\mathrm{i}\theta}$ is the unit vector along AP, and $\mathrm{i}\,\mathrm{e}^{\mathrm{i}\theta}$ is the unit vector perpendicular to AP in the direction of increasing ϕ.



The perpendicular component of acceleration

Differentiating, with $\dot\theta = \Omega + \dot\phi$ and $\ddot\theta = \ddot\phi$ (as Ω is constant),

$$\dot z_P = \mathrm{i}a\Omega\,\mathrm{e}^{\mathrm{i}\Omega t} + \mathrm{i}b(\Omega+\dot\phi)\,\mathrm{e}^{\mathrm{i}\theta},$$
$$\ddot z_P = -a\Omega^2\,\mathrm{e}^{\mathrm{i}\Omega t} + b\big(\mathrm{i}\ddot\phi - (\Omega+\dot\phi)^2\big)\mathrm{e}^{\mathrm{i}\theta}.$$

To resolve along and perpendicular to AP, write $\mathrm{e}^{\mathrm{i}\Omega t} = \mathrm{e}^{\mathrm{i}\theta}\mathrm{e}^{-\mathrm{i}\phi} = \mathrm{e}^{\mathrm{i}\theta}(\cos\phi - \mathrm{i}\sin\phi)$. Then

$$\ddot z_P = \big[-a\Omega^2\cos\phi - b(\Omega+\dot\phi)^2\big]\mathrm{e}^{\mathrm{i}\theta} + \big[\,b\ddot\phi + a\Omega^2\sin\phi\,\big]\,\mathrm{i}\,\mathrm{e}^{\mathrm{i}\theta}.$$

So the component of the acceleration of P perpendicular to AP, in the direction of increasing ϕ, is

$$\boxed{\,b\,\frac{\mathrm{d}^2\phi}{\mathrm{d}t^2} + a\Omega^2\sin\phi\,}$$

(The first term is the familiar $b\ddot\phi$ of motion on a circle about A. The second comes from A's own centripetal acceleration $a\Omega^2$ towards O, resolved perpendicular to the rod.)

Finding $\mathrm{d}\phi/\mathrm{d}t$

Ask yourself Why must this component of acceleration be zero?

The forces on P are its weight and the normal reaction (both vertical, and they balance), no friction (the plane is smooth), and the force from the rod. A light rod that is freely pivoted at A can only push or pull along its own length. So there is no horizontal force perpendicular to AP, and Newton's second law gives

$$b\ddot\phi + a\Omega^2\sin\phi = 0.$$

This is the pendulum equation. Multiplying by $\dot\phi$ and integrating,

$$\tfrac12 b\dot\phi^2 = a\Omega^2\cos\phi + C.$$

Initial conditions. Initially O, A, P are in a straight line. Since $b > a$, P cannot lie between O and A, and we take P to be beyond A, so $\phi = 0$ at $t = 0$. Then both exponentials equal 1 and

$$\dot z_P(0) = \mathrm{i}\big[(a+b)\Omega + b\dot\phi(0)\big],$$

which is perpendicular to OP (the real axis). Note that $(a+b)\Omega$ is just the speed of the point of the plane underneath P, and $b\dot\phi(0)$ is P's speed relative to the plane. The printed speed is $(a+2\sqrt{ab})\Omega$, but this looks like a misprint for $(a+b+2\sqrt{ab})\Omega = (\sqrt{a}+\sqrt{b})^2\Omega$, which is the value that gives the stated result (see the remark at the end). With that value, P moves at $2\sqrt{ab}\,\Omega$ relative to the plane, so

$$\dot\phi(0) = 2\Omega\sqrt{a/b}.$$

Putting $\phi = 0$ into the energy equation gives $\tfrac12 b\cdot 4a\Omega^2/b = a\Omega^2 + C$, so $C = a\Omega^2$ and

$$\tfrac12 b\dot\phi^2 = a\Omega^2(1+\cos\phi) = 2a\Omega^2\cos^2\tfrac{\phi}{2}, \qquad \text{so} \qquad \dot\phi^2 = \frac{4a\Omega^2}{b}\cos^2\tfrac{\phi}{2}.$$

Initially $\dot\phi > 0$, and $\dot\phi$ can change sign only by passing through zero, which needs $\cos\frac{\phi}{2} = 0$, that is $\phi = \pm\pi$. So as long as $-\pi < \phi < \pi$ we take the positive root:

$$\boxed{\frac{\mathrm{d}\phi}{\mathrm{d}t} = 2\Omega\sqrt{\frac{a}{b}}\cos\frac{\phi}{2}}$$

P never crosses OA

Ask yourself Couldn't ϕ just creep up to π and go past it?

P is on the line OA only when ϕ is a multiple of π (for $\phi = \pi$ it lies a distance $b - a$ beyond O). Since $\dot\phi > 0$ while $0 \le \phi < \pi$, ϕ increases from 0, and the only way back onto the line is to reach $\phi = \pi$. We show that this never happens by solving the equation. Separating the variables, with $k = \sqrt{a/b}$,

$$\int_0^{\phi} \sec\tfrac{u}{2}\,\mathrm{d}u = 2k\Omega t \quad\Longrightarrow\quad \ln\Big(\sec\tfrac{\phi}{2} + \tan\tfrac{\phi}{2}\Big) = k\Omega t.$$

With $s = \sin\frac{\phi}{2}$ we have $\sec\frac{\phi}{2} + \tan\frac{\phi}{2} = \frac{1+s}{\sqrt{1-s^2}} = \sqrt{\frac{1+s}{1-s}}$, so the left-hand side is $\frac{1}{2}\ln\frac{1+s}{1-s} = \operatorname{artanh} s$. Hence

$$\sin\frac{\phi}{2} = \tanh\Big(\Omega t\sqrt{\tfrac{a}{b}}\Big).$$

For every $t > 0$ the right-hand side lies strictly between 0 and 1, so $0 < \phi < \pi$ for all $t > 0$. Then P always stays strictly on one side of the line OA. It approaches the line (at the point beyond O) as $t \to \infty$ but never reaches it, so P never crosses OA. ■

(Without solving: $\phi = \pi$, $\dot\phi = 0$ is an equilibrium of the pendulum equation, and near it $\psi = \pi - \phi$ satisfies $\dot\psi \approx -k\Omega\psi$, so ψ decays exponentially and is never zero.)

Remark on the printed speed. If we take the speed literally as $(a + 2\sqrt{ab})\Omega$, then $\dot\phi(0) = (2k-1)\Omega$, and the same integration gives $\dot\phi^2 = \Omega^2\big[(2k-1)^2 - 4k^2\sin^2\frac{\phi}{2}\big]$. This is not a neat expression. P then oscillates about $\phi = 0$ (crossing OA produced every swing) if $b < 16a$, and goes right round, crossing OA, if $b > 16a$. So the printed value cannot be what was intended.

Checks

- Setting $a = 0$ puts A at O, and then $\dot\phi \equiv 0$: P simply rides round with the plane at speed $b\Omega$. That is right, because the initial speed becomes $b\Omega$.
- Units: $\Omega\sqrt{a/b}$ is a rate, so it has units of s^{-1}, as $\dot\phi$ should.
- We stepped $b\ddot\phi = -a\Omega^2\sin\phi$ numerically (RK4) with $a = 1$, $b = 3$, $\Omega = 1.3$. At $t = 2$ it gave $\phi \approx 2.2644$ and $\dot\phi \approx 0.6375$, agreeing with $2\arcsin\tanh(\Omega t\sqrt{a/b})$ and $2\Omega\sqrt{a/b}\cos\frac{\phi}{2}$ to 10 significant figures. At $t = 10$, $\phi \approx 3.1394 < \pi$.

Common slips

- Forgetting the frame's own contribution $(a+b)\Omega$ to P's initial velocity, and treating the given speed as speed relative to the plane.
- Writing the acceleration of P as just that of circular motion about A. The pivot A is itself accelerating, which is where the $a\Omega^2 \sin\phi$ comes from.
- Getting the sign of $a\Omega^2 \sin\phi$ wrong. With the wrong sign, $\phi = 0$ would be unstable, which is unphysical because the "centrifugal" pull is away from O.

Beyond STEP

Viewed from the rotating plane, P is a pendulum hanging in a uniform "gravity" $a\Omega^2$ pointing along OA produced. The Coriolis force acts along the rod, so it does no work and drops out. The initial condition puts the motion exactly on the *separatrix*: the borderline between swinging and going over the top, where the pendulum takes infinite time to reach the inverted position. Rotating frames, fictitious forces and phase-plane pictures like this are covered in Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP III 1989, Question 13

Mechanics • Simple Harmonic Motion

STEP III 1989, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q13

The key idea

Each string runs from O (under the table) up through its hole and across the table to P. The piece under the table always has length exactly a, the natural length, so the extension of each string is simply the distance from P to its hole. That makes each tension proportional to a vector, and with position vectors the five forces add up in one line, because the five holes are symmetrically placed and their position vectors sum to zero.

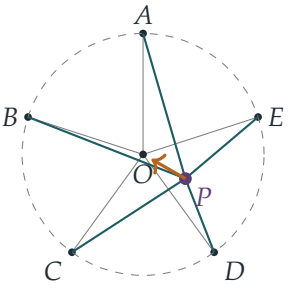
Solution

Setting up

Model P as a particle on a smooth table, the strings as light elastic strings and the holes as smooth, so the tension is the same all along each string. Take O as origin on the table, let $\mathbf{r}$ be the position vector of P and let $\mathbf{a}_1, \ldots, \mathbf{a}_5$ be the position vectors of $A, \ldots, E$, each of length a.

The string through hole i consists of a vertical piece of length a from O (under the table, directly below the centre) to the hole, and a piece on the table of length $|\mathbf{a}_i - \mathbf{r}|$. Its total length is $a + |\mathbf{a}_i - \mathbf{r}|$, so its extension is $|\mathbf{a}_i - \mathbf{r}|$. This is never negative, so every string is taut (or just slack, when P is at the hole) wherever P is: no string ever goes slack and changes the equations. By Hooke's law the tension is $\frac{\lambda}{a}|\mathbf{a}_i - \mathbf{r}|$, and on the table it pulls P towards the hole, in the direction of the unit vector $\frac{\mathbf{a}_i - \mathbf{r}}{|\mathbf{a}_i - \mathbf{r}|}$. So the force on P from string i is

$$\frac{\lambda}{a}|\mathbf{a}_i - \mathbf{r}| \cdot \frac{\mathbf{a}_i - \mathbf{r}}{|\mathbf{a}_i - \mathbf{r}|} = \frac{\lambda}{a}(\mathbf{a}_i - \mathbf{r}).$$



The teal lines are the five strings on the table, seen from above; the amber arrow shows the resultant force on P, which points straight at O.

The first part

The vertical forces on P (weight and normal reaction) balance. Adding the five horizontal forces,

$$\mathbf{F} = \frac{\lambda}{a}\sum_{i=1}^{5}(\mathbf{a}_i - \mathbf{r}) = \frac{\lambda}{a}\Big(\sum_{i=1}^{5}\mathbf{a}_i\Big) - \frac{5\lambda}{a}\mathbf{r}.$$

Ask yourself Why is $\mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3 + \mathbf{a}_4 + \mathbf{a}_5 = \mathbf{0}$?

Rotating the whole figure through 72° about O permutes the five holes, so it leaves the sum $\mathbf{s} = \sum \mathbf{a}_i$ unchanged. But the only vector unchanged by a rotation through 72° is the zero vector, so $\mathbf{s} = \mathbf{0}$. (Equivalently, the fifth roots of unity sum to zero.)

Hence Newton's second law gives

$$m\ddot{\mathbf{r}} = -\frac{5\lambda}{a}\mathbf{r}, \qquad \text{that is} \qquad \ddot{\mathbf{r}} = -\omega^2\mathbf{r} \quad \text{with} \quad \omega^2 = \frac{5\lambda}{am}.$$

P starts from rest at F, so $\mathbf{r}(0) = \overrightarrow{OF}$ and $\dot{\mathbf{r}}(0) = \mathbf{0}$. The solution is $\mathbf{r} = \cos(\omega t)\,\overrightarrow{OF}$, so P moves to and fro along the straight line FO (to the reflection of F in O and back), and its displacement from O satisfies the simple harmonic equation. The period is

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{am}{5\lambda}},$$

as required. (Note that this works for *any* starting point F, not just for small displacements: the force law is exactly linear.)

The final part

Now string PAO has modulus $k\lambda$, so its force on P is $\frac{k\lambda}{a}(\mathbf{a}_1 - \mathbf{r})$. Using $\mathbf{a}_2 + \mathbf{a}_3 + \mathbf{a}_4 + \mathbf{a}_5 = -\mathbf{a}_1$,

$$\begin{aligned}\mathbf{F} &= \frac{k\lambda}{a}(\mathbf{a}_1 - \mathbf{r}) + \frac{\lambda}{a}\sum_{i=2}^{5}(\mathbf{a}_i - \mathbf{r}) \\ &= \frac{k\lambda}{a}(\mathbf{a}_1 - \mathbf{r}) + \frac{\lambda}{a}(-\mathbf{a}_1 - 4\mathbf{r}) \\ &= \frac{\lambda}{a}\big[(k-1)\mathbf{a}_1 - (k+4)\mathbf{r}\big] = -\frac{(k+4)\lambda}{a}\,(\mathbf{r} - \mathbf{r}_0)\,, \qquad \mathbf{r}_0 = \frac{k-1}{k+4}\,\mathbf{a}_1.\end{aligned}$$

So the equilibrium position is the point $\mathbf{r}_0$ on OA (where $\mathbf{F} = \mathbf{0}$), and

$$\ddot{\mathbf{r}} = -\frac{(k+4)\lambda}{am}(\mathbf{r} - \mathbf{r}_0).$$

If P is displaced along OA from $\mathbf{r}_0$ and released from rest, then $\mathbf{r} - \mathbf{r}_0$ starts parallel to $\mathbf{a}_1$ and the acceleration is always parallel to $\mathbf{r} - \mathbf{r}_0$, with zero initial velocity. So, exactly as before, $\mathbf{r} - \mathbf{r}_0 = \cos(\omega' t)(\mathbf{r}(0) - \mathbf{r}_0)$: P moves in a straight line along OA, with simple harmonic motion about $\mathbf{r}_0$ of angular frequency ω', where

$$\omega'^2 = \frac{(k+4)\lambda}{am}.$$

The period is $T/2$ exactly when $\omega' = 2\omega$, that is, $\omega'^2 = 4\omega^2$:

$$\frac{(k+4)\lambda}{am} = \frac{20\lambda}{am} \quad\Longrightarrow\quad k + 4 = 20 \quad\Longrightarrow\quad k = 16.$$

With $k = 16$ the equilibrium position is $\mathbf{r}_0 = \frac{15}{20}\mathbf{a}_1$, three-quarters of the way from O to A. (For the model to hold, the amplitude must be less than $\frac{1}{4}a$, so that P does not reach the hole at A.)

Checks

- Setting $k = 1$ recovers the first part: $\mathbf{r}_0 = \mathbf{0}$ and $\omega'^2 = 5\lambda/(am)$.
- Units: λ is a force, so $\lambda/(am)$ has units $\mathrm{N}/(\mathrm{m\,kg}) = \mathrm{s}^{-2}$, and T is a time.
- We summed the five tension vectors at 1000 random positions (with $a = 1.3$, $m = 0.7$, $\lambda = 2.1$) and got exactly $-5\lambda\mathbf{r}/a$, and for $k = 16$ exactly $-20\lambda(\mathbf{r} - \mathbf{r}_0)/a$. Stepping the motion numerically from $(0.4, -0.3)$ returned P to its start after time $T \approx 1.85$, staying on the line through O; from $\mathbf{r}_0 + 0.1a\,\hat{\mathbf{a}}_1$ with $k = 16$ it returned after $T/2$.

Common slips

- Taking the extension to be $|\mathbf{a}_i - \mathbf{r}| - a$ or $|\mathbf{r}|$: the natural length a is used up exactly by the piece under the table.

- Resolving the five tensions with angles and trigonometry. It can be done, but vectors make the cancellation immediate.
- Assuming the new oscillation is still centred at O: the stiffer string drags the equilibrium position towards A.

Beyond STEP

A force $\mathbf{F} = -c\,\mathbf{r}$ in two dimensions gives the isotropic harmonic oscillator: the general motion (with a sideways initial velocity) is an ellipse centred at the equilibrium point, and every orbit has the same period. Together with the inverse-square law, it is one of only two central forces for which every bounded orbit closes (Bertrand's theorem). Kleppner and Kolenkow, *An Introduction to Mechanics*, treats oscillators and central forces in this vector style.

WORKED SOLUTION

STEP III 1989, Question 14

Mechanics • Pulleys and Impulse

STEP III 1989, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q14

The key idea

For part (i), cut the disc into thin rings: each ring has all its mass at one distance from the centre, so both the mass and the moment of inertia are single integrals in r. For part (ii), the whole system has one degree of freedom (the depth of D), so one equation of motion gives the acceleration. When B is suddenly stopped, the string constraint tells us at once how fast D can then move, and the impulse is just the change in its momentum.

Solution

Part (i)

(The axis in part (i) is the one through the centre O; the "C" in the paper is a misprint.) Let the density be $\rho = kr$ at distance r from O, with k constant. The thin ring between radii r and $r + \mathrm{d}r$ has area $2\pi r\,\mathrm{d}r$, so mass $2\pi kr^2\,\mathrm{d}r$, and every point of it is at distance r from the axis. Hence

$$m = \int_0^a 2\pi kr^2\,\mathrm{d}r = \frac{2\pi ka^3}{3}, \qquad I = \int_0^a r^2 \cdot 2\pi kr^2\,\mathrm{d}r = \frac{2\pi ka^5}{5}.$$

Dividing, $I/m = \frac{3}{5}a^2$, so

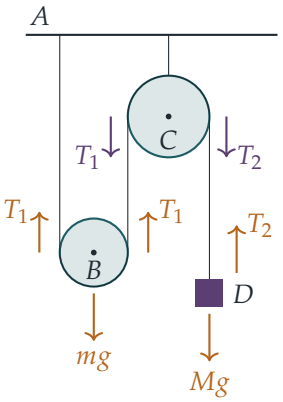
$$I = \tfrac{3}{5}ma^2,$$

as required. (It is bigger than the uniform disc's $\frac{1}{2}ma^2$, as it should be: more of the mass is near the rim.)

Part (ii): the motion before B is stopped

Model: B is a smooth pulley of mass m, and C is a disc of the type in (i) with mass m and radius a, so its moment of inertia about its fixed axis is $\frac{3}{5}ma^2$. The string is light and inextensible and does not slip on C. Because B is smooth, the tension T_1 is the

same in the two strands AB and BC. Call the tension in the strand CD T_2; since C is rough, T_2 need not equal T_1.



Let D move down with acceleration f. The two strands supporting B each shorten by half of the amount by which CD lengthens, so B rises with acceleration $\frac{1}{2}f$; and since the string does not slip on C, the angular acceleration of C is f/a. The equations of motion are:

$$\begin{aligned} D\ (\text{down}): &\quad Mg - T_2 = Mf, \\ B\ (\text{up}): &\quad 2T_1 - mg = \tfrac{1}{2}mf, \\ C\ (\text{moments about its axis}): &\quad (T_2 - T_1)a = \tfrac{3}{5}ma^2 \cdot \frac{f}{a}. \end{aligned}$$

Substituting $T_2 = Mg - Mf$ and $T_1 = \frac{1}{2}mg + \frac{1}{4}mf$ into the last equation,

$$Mg - Mf - \tfrac{1}{2}mg - \tfrac{1}{4}mf = \tfrac{3}{5}mf \quad\Longrightarrow\quad f\left(M + \tfrac{17}{20}m\right) = g\left(M - \tfrac{1}{2}m\right),$$

so

$$f = \frac{10(2M - m)g}{20M + 17m}.$$

This is constant, and positive precisely because $m < 2M$: that is why the condition is there. It tells us D really does descend from rest. So after t seconds D is moving down with speed

$$v = ft = \frac{10(2M - m)gt}{20M + 17m},$$

B is rising at $\frac{1}{2}v$ and C is turning with angular speed v/a.

(An energy check gives the same f: the kinetic energy is $\frac{1}{2}\left(M + \frac{1}{4}m + \frac{3}{5}m\right)\dot{x}^2$ when D has fallen x, and the potential energy lost is $\left(M - \frac{1}{2}m\right)gx$.)

Part (ii): the impulse

Ask yourself After B is stopped, can D still move?

While the string is taut, the speed of D is always twice the speed of B, because A is fixed and AB and BC each change length by the same amount. Once B is at rest, both of these lengths are fixed, so no string can pass round C and D must stop too, as must C. The string does stay taut: D is moving away from C, so it pulls CD tight, and C, still turning, draws string up from the BC side, so that strand is pulled tight as well. (In fact the impulsive tensions come out as Mv in CD and $(M+\frac{3}{5}m)v$ in BC, both positive, which confirms this.)

So the velocity of D drops from v downwards to zero, and the impulse on D (from the string) equals its change of momentum:

$$\text{impulse on } D = Mv = \frac{10M(2M-m)gt}{20M+17m} \quad \text{vertically upwards.}$$

Checks

- If $m \to 0$ (light pulleys) then $f \to g$ and the impulse is Mgt: D simply falls freely, as expected.
- If $m = 2M$ the impulse is zero: the system is in equilibrium and never starts moving.
- With $M = 3$, $m = 2.5$, $g = 9.8$, $t = 1.3$, stepping the motion numerically gives $Mv \approx 13.0507$, matching the formula, and the kinetic energy gained matches the potential energy lost. The moment of inertia ratio from numerical integration is 0.6000.

Common slips

- Taking the same tension on both sides of C. The pulley C has mass and is rough, so the tension difference is exactly what turns it.
- Forgetting that B moves at half the speed of D, which gives $\frac{1}{4}m$, not m, in the effective mass.
- Giving the impulse without its direction: it acts upwards on D.

Beyond STEP

Writing the kinetic energy as $\frac{1}{2}M_{\text{eff}}\dot{x}^2$ in a single coordinate is the starting point of Lagrangian mechanics, where constraints like the pulley geometry are built in from the beginning and the tensions never appear. Sudden constraints, like stopping B, are handled the same way with impulsive equations. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers rigid-body rotation and impulse in this style.

WORKED SOLUTION

STEP III 1989, Question 15

Probability and Statistics • Distribution of the Median

STEP III 1989, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q15

The key idea

The median of $2n+1$ values lies close to z exactly when one value falls in a tiny interval at z, n values fall below it and n above it. Parts (i) and (ii) give the probabilities of the "all below" and "all above" pieces, and counting the ways to choose which values play which role gives the density. After that, the fact that a density integrates to 1 evaluates the integral for free, and the same trick, used with $n+1$ in place of n, gives the variance.

Solution

Throughout, X has density $\frac{1}{2c}$ on $[-c, c]$, so for $-c \le k \le c$

$$\mathrm{P}(X > k) = \frac{c-k}{2c}, \qquad \mathrm{P}(X < k) = \frac{c+k}{2c}.$$

Part (i)

The n values are independent, so we multiply:

$$\mathrm{P}(\text{all } n \text{ values exceed } k) = \left(\frac{c-k}{2c}\right)^n.$$

Part (ii)

In the same way,

$$\mathrm{P}(\text{all } n \text{ values are less than } k) = \left(\frac{c+k}{2c}\right)^n.$$

The density of the median

Let $X_1, \ldots, X_{2n+1}$ be the sample, and Z its median, the $(n+1)$th smallest value. Since X is continuous, ties happen with probability 0, so we ignore them. Every X_i lies in

$[-c, c]$, so Z does too.

Take z in $(-c, c)$ and a small $\delta z > 0$. Up to terms of order $(\delta z)^2$, the event $z < Z \leq z + \delta z$ happens exactly when

- one of the values lies in $(z, z + \delta z]$,
- n of the other values are less than z, and
- the remaining n values are greater than $z + \delta z$.

(Arrangements with two or more values in the little interval have probability of order $(\delta z)^2$, which doesn't matter in the limit.)

Ask yourself How many ways are there to assign the $2n + 1$ values to these three roles?

There are $2n + 1$ choices for the value in the little interval, and then $\binom{2n}{n}$ ways to choose which n of the rest go below z. That gives

$$(2n+1)\binom{2n}{n} = \frac{(2n+1)!}{(n!)^2}$$

assignments. They are mutually exclusive, and each has the same probability. For a fixed assignment, independence and parts (i) and (ii) with $k = z$ give the probability

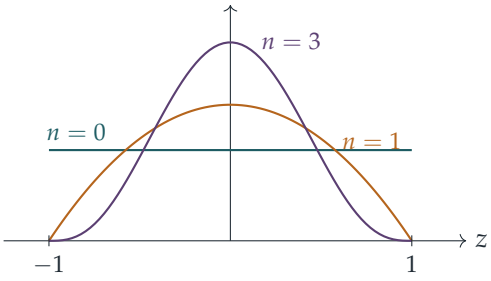
$$\frac{\delta z}{2c}\left(\frac{c+z}{2c}\right)^n \left(\frac{c-z-\delta z}{2c}\right)^n.$$

Adding over the assignments, dividing by δz and letting $\delta z \to 0$, the density of Z is

$$\begin{aligned} g(z) &= \frac{(2n+1)!}{(n!)^2} \cdot \frac{1}{2c} \cdot \frac{(c+z)^n (c-z)^n}{(2c)^{2n}} \\ &= \frac{(2n+1)!}{(n!)^2 (2c)^{2n+1}} (c^2 - z^2)^n, \qquad -c \leq z \leq c, \end{aligned}$$

using $(c + z)(c - z) = c^2 - z^2$. This is the required result.

The picture below shows the density for $c = 1$. When $n = 0$ the "median" is a single value, which is uniform. As n grows, the median concentrates near 0.



The integral

A probability density integrates to 1 over its range, so

$$\frac{(2n+1)!}{(n!)^2(2c)^{2n+1}}\int_{-c}^{c}(c^2-z^2)^n\,\mathrm{d}z = 1,$$

and therefore

$$\int_{-c}^{c}(c^2-z^2)^n\,\mathrm{d}z = \frac{(n!)^2(2c)^{2n+1}}{(2n+1)!}.$$

This holds for every integer $n \geq 0$, and we use that in a moment.

Mean and variance

The density g is an even function of z, so $zg(z)$ is odd and its integral over the symmetric interval $[-c, c]$ is zero:

$$\mathrm{E}(Z) = 0.$$

Hence $\mathrm{var}(Z) = \mathrm{E}(Z^2)$. Write $K_n = \dfrac{(2n+1)!}{(n!)^2(2c)^{2n+1}}$ and $I_n = \displaystyle\int_{-c}^{c}(c^2-z^2)^n\,\mathrm{d}z = \frac{1}{K_n}$.

Ask yourself How can we integrate $z^2(c^2-z^2)^n$ without expanding it?

Write $z^2 = c^2 - (c^2 - z^2)$. Then

$$\begin{aligned}\mathrm{E}(Z^2) &= K_n\int_{-c}^{c}\big(c^2-(c^2-z^2)\big)(c^2-z^2)^n\,\mathrm{d}z \\ &= K_n\big(c^2 I_n - I_{n+1}\big) = c^2 - \frac{K_n}{K_{n+1}},\end{aligned}$$

using $K_n I_n = 1$ and $I_{n+1} = 1/K_{n+1}$, which is the integral result with $n+1$ in place of n. Now

$$\frac{K_n}{K_{n+1}} = \frac{(2n+1)!}{(2n+3)!}\cdot\frac{((n+1)!)^2}{(n!)^2}\cdot\frac{(2c)^{2n+3}}{(2c)^{2n+1}} = \frac{(n+1)^2\cdot 4c^2}{(2n+2)(2n+3)} = \frac{2(n+1)c^2}{2n+3}.$$

Therefore

$$\mathrm{var}(Z) = c^2\left(1 - \frac{2n+2}{2n+3}\right) = \frac{c^2}{2n+3}.$$

Checks

- When $n = 0$ the median is just one value of X. The density becomes $\frac{1}{2c}$, the integral is $2c$, and the variance is $\frac{c^2}{3}$, which is the variance of the uniform distribution on $[-c, c]$.
- When $n = 1$ and $c = 1$, the integral formula gives $\int_{-1}^{1}(1-z^2)\,\mathrm{d}z = \frac{1\cdot 8}{6} = \frac{4}{3}$, which is correct.
- A simulation of 10^5 samples of size 5 ($n = 2$) with $c = 1.7$ gave a median variance of 0.415, against $c^2/7 \approx 0.413$.

Common slips

- Forgetting the factor $2n+1$ for the choice of the middle value, and using only $\binom{2n}{n}$.
- Missing the factor $\frac{1}{2c}$ for the middle value's own density, which leaves the power $(2c)^{2n}$ instead of $(2c)^{2n+1}$.
- Expanding $(c^2-z^2)^n$ binomially to find $\mathrm{E}(Z^2)$. It can be done, but the sum is hard to simplify.

Beyond STEP

The median is one of the *order statistics* of a sample. The same counting argument gives the density of the rth smallest of m values from any continuous distribution, as $\frac{m!}{(r-1)!\,(m-r)!}F^{r-1}(1-F)^{m-r}f$. For the uniform distribution these are *beta distributions*, and the integral you deduced is a case of the beta function $B(a,b)=\frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$. The variance $c^2/(2n+3)$ shrinks like $1/n$, which is the first step towards the asymptotic normality of the sample median. Blitzstein and Hwang's *Introduction to Probability* covers order statistics and the beta distribution well.

WORKED SOLUTION

STEP III 1989, Question 16

Probability and Statistics • Probability and Hypothesis Testing

STEP III 1989, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q16

The key idea

Three yes/no characteristics split the population into eight cells, and the eight statements are just linear equations for the proportions in those cells. Fill in the green-eyed cells first (the conditions there are the most specific), then the hazel-eyed ones. The statistical parts then rest on one number, the probability that a Ruritanian is narrow-headed, and ask whether the observations are surprising under that value.

Solution

The printed paper has three slips of the pen. We read "arrow-headed" in (iv) as *narrow-headed*, "fair-headed" in (vi) as *fair-haired*, and "hazel-haired" in (viii) as *hazel-eyed*. These are the only sensible readings, since the three characteristics are hair (fair or dark), eyes (green or hazel) and head (narrow or broad).

The proportion of narrow-heads

Choose a Ruritanian at random. Write F, D for fair- and dark-haired, G, H for green- and hazel-eyed, and N, B for narrow- and broad-headed. We are told $\mathrm{P}(F) = \frac{1}{4}$ and $\mathrm{P}(G) = \frac{1}{5}$.

Green-eyed cells. By (v), $\mathrm{P}(N \mid G) = \mathrm{P}(B \mid G) = \frac{1}{2}$, so

$$\mathrm{P}(G \cap N) = \mathrm{P}(G \cap B) = \tfrac{1}{10}.$$

By (vi), the green-eyed broad-heads split equally between fair and dark hair:

$$\mathrm{P}(F \cap G \cap B) = \mathrm{P}(D \cap G \cap B) = \tfrac{1}{20}.$$

By (iv), $\mathrm{P}(F \cap G \cap N) = 0$, so $\mathrm{P}(D \cap G \cap N) = \frac{1}{10}$.

Hazel-eyed cells. Let

$$a = \mathrm{P}(F \cap H \cap B), \quad b = \mathrm{P}(F \cap H \cap N), \quad c = \mathrm{P}(D \cap H \cap B), \quad d = \mathrm{P}(D \cap H \cap N).$$

- Fair hair: the green-eyed fair-haired make up $0 + \frac{1}{20}$, so $a + b = \frac{1}{4} - \frac{1}{20} = \frac{1}{5}$.
- Hazel eyes: $a + b + c + d = \frac{4}{5}$, so $c + d = \frac{3}{5}$.
- Statement (vii): $\mathrm{P}(B \cap D) = \frac{1}{20} + c = \frac{1}{2}$, so $c = \frac{9}{20}$ and hence $d = \frac{3}{20}$.
- Statement (viii): $\mathrm{P}(F \cap B \mid H) = \mathrm{P}(D \cap N \mid H)$; multiplying by $\mathrm{P}(H)$ gives $a = d = \frac{3}{20}$, and so $b = \frac{1}{20}$.

All eight proportions, in twentieths, are:

	$G \cap N$	$G \cap B$	$H \cap N$	$H \cap B$
F	0	1	1	3
D	2	1	3	9

They are all non-negative and add to 20, so the model is consistent. Adding the narrow-headed columns,

$$\mathrm{P}(N) = \frac{0+2+1+3}{20} = \frac{3}{10}.$$

So the model says that 30% of Ruritanians are narrow-headed.

Six broad-headed acquaintances

If my six acquaintances were a random sample, independent of one another, the chance that all six are broad-headed would be

$$\left(\tfrac{7}{10}\right)^6 = 0.1176\ldots \approx 0.12.$$

Ask yourself Is a 12% event good evidence against the model?

No. Something with probability about 1 in 8 is not unusual; it would not be significant at any conventional level such as 5%. The data are also weak in a second way: one's acquaintances are not a random sample. They may share a family, region or social group, so they need not be independent, and they need not represent the whole population. The observation gives at most very slight grounds for doubt.

A random sample of 200

Let X be the number of narrow-heads in a random sample of 200. The population is large, so the draws are effectively independent and, under the model, $X \sim \mathrm{B}(200, \frac{3}{10})$, with

$$\mathrm{E}(X) = 60, \qquad \mathrm{Var}(X) = 200 \times 0.3 \times 0.7 = 42, \qquad \sqrt{42} \approx 6.481.$$

Since np and $n(1-p)$ are both large, we use the normal approximation $X \approx \mathrm{N}(60, 42)$, with a continuity correction:

$$\mathrm{P}(X \le 50) \approx \Phi\left(\frac{50.5 - 60}{\sqrt{42}}\right) = \Phi(-1.466) \approx 0.071.$$

So under the model, 50 or fewer narrow-heads would happen about 7% of the time.

Ask yourself Should we now reject the model?

Not at the 5% level: a one-tailed test of $p = 0.3$ against $p < 0.3$ gives a p-value of about 0.07. The result does cast some doubt on the model, though. The sample proportion $\frac{50}{200} = 0.25$ is about 1.5 standard deviations below the predicted value. It also points the same way as my acquaintances, who had fewer narrow-heads than expected. It would be sensible to collect more data before trusting the model's figure of 30%. (If the test were two-tailed, the p-value would be about 0.14, and the evidence weaker still.)

Checks

- Each of the eight conditions can be read straight off the table, for example (vii): $1 + 9 = 10$ twentieths $= \frac{1}{2}$.
- A computer check confirms that the cell proportions are the unique solution, that $\mathrm{P}(N) = \frac{3}{10}$ and that $(0.7)^6 = 0.117649$. The exact binomial probability $\mathrm{P}(X \leq 50) = 0.0695$ is close to the normal approximation 0.0713.

Common slips

- Reading (v) or (vi) as statements about the whole population rather than conditional ones: "as likely as" within the green-eyed group means equal halves of $\frac{1}{5}$.
- Forgetting the continuity correction. Using 50 instead of 50.5 gives $\Phi(-1.543) \approx 0.061$.
- Treating acquaintances as a random sample without comment.

Beyond STEP

The last part is a hypothesis test, and the two pieces of evidence raise a natural question: how should several weak pieces of evidence be combined? Undergraduate statistics answers this with likelihood (and, for the Bayesian, posterior probabilities), which measure how well each value of p explains *all* the data together. Grimmett and Welsh, *Probability: An Introduction*, gives the probability background, and Blitzstein and Hwang, *Introduction to Probability*, has an approachable account of the Bayesian view.

Beyond STEP

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